

Foundations of Computer Science

Discrete Structures and Logic

Study Guide with Solved Previous-Year Questions

What is inside

- Ch. 1–3 Propositional logic, predicate logic, rules of inference
- Ch. 4 Proof techniques and mathematical induction
- Ch. 5 Sets
- Ch. 6–7 Relations; partial orders, Hasse diagrams and lattices
- Ch. 8 Functions
- App. A–B Corrections to the class notes; one-page revision sheet

Built from your handwritten class notes (18 Aug – 15 Sep 2026), ten NIT Silchar (PG, CSE) question papers, 2015–2024, and selected exercises from Rosen's *Discrete Mathematics*. Every definition is followed by an example; **27 previous-year questions** and **88 practice questions** are solved in full.

About this guide

How each chapter is organised

Every chapter follows the same pattern: **definitions, each with an example**, then **worked examples from your notes**, then a section of **solved previous-year questions (PYQs)** on that chapter's topics, and finally **practice questions from Rosen** in the same style. The coloured boxes mean:

DEFINITION	the formal statement to memorise
EXAMPLE	a worked instance of the definition just given
REMEMBER	a rule, shortcut or checklist
WATCH OUT	a common mistake
PYQ	a question from a past paper (source, question number, marks), followed by its solution
PRACTICE	a question from Kenneth Rosen's <i>Discrete Mathematics</i> (section and exercise number, and the skill it tests), followed by its solution

A superscript † marks material that is *not in your handwritten notes* but is a standard prerequisite or is needed to answer a PYQ (for example free and bound variables, symmetric difference, the closures of a relation, inverse images of a function, and the formula for counting onto functions).

Which questions are included

The course covers **propositional logic, predicate logic, rules of inference, proofs, sets, relations and functions**. Partial orders, Hasse diagrams and lattices are treated as part of relations (Chapters 6 and 7). Your handwritten notes also contain a short section on algebraic structures (groups, rings). It is **not part of the course**, so it has been removed, together with the one PYQ on it (M23 Q1(c), an abelian-group proof). The first page of the notes also lists Languages and Automata, Computability and Computational Complexity, but no notes on these have been written yet.

So **every PYQ whose topic is outside the course or missing from the notes has been left out**: the 2015, 2016, 2017 and 2018 papers (compiler design, automata, Turing machines, P/NP, ...) and the corresponding parts of the December 2024 paper. The tables on the next pages show exactly what was included (with page numbers) and what was dropped and why.

Rosen practice questions. Only questions that add something new were taken from the trimmed Rosen text: a different skill, a common trap, or a proof that the notes do not show. Routine repeats of an example already in a chapter were skipped. They are phrased like the PYQs (find, decide with a proof or counterexample, prove), each with a full solution. The last index in this front matter lists them all.

Contents

About this guide	i
PYQ coverage index	v
1 Propositional Logic	1
1.1 Propositions and the basic connectives	1
1.2 The conditional and the biconditional	2
1.3 Translating English into logic	3
1.4 Knights and knaves	3
1.5 Tautology, contradiction, contingency, equivalence	4
1.6 Standard logical equivalences	5
1.7 Proving an equivalence by a chain of laws	5
1.8 Satisfiability	6
1.9 Conditionals in programs and bit strings [†]	6
1.10 Solved previous-year questions	7
1.11 Practice questions from Rosen	10
2 Predicate Logic	15
2.1 Predicates and propositional functions	15
2.2 Quantifiers	15
2.3 Negation and De Morgan's laws for quantifiers	16
2.4 Translating English into predicate logic	17
2.5 Nested quantifiers	18
2.6 Free and bound variables [†]	19
2.7 Domains inside the quantifier, and truth sets [†]	20
2.8 The uniqueness quantifier [†]	20
2.9 Solved previous-year questions	20
2.10 Practice questions from Rosen	23
3 Rules of Inference	24
3.1 Arguments and validity	24
3.2 Rules of inference for propositional logic	24
3.3 Building a proof from premises	25
3.4 Fallacies	26
3.5 Rules of inference for quantified statements	27
3.6 Solved previous-year questions	28
3.7 Practice questions from Rosen	28
4 Proof Techniques	33
4.1 Vocabulary [†]	33
4.2 Direct proof	33
4.3 Indirect proof (proof by contraposition)	34
4.4 Vacuous and trivial proofs	34
4.5 Proof by contradiction	34

4.6	Mistakes in proofs	35
4.7	Exhaustive proof and proof by cases	35
4.8	Proof by cases, continued [†]	36
4.9	Proving “if and only if” and lists of equivalent statements [†]	37
4.10	Counterexamples [†]	37
4.11	Existence and uniqueness proofs [†]	37
4.12	Mathematical induction	38
4.13	Solved previous-year questions	39
4.14	Practice questions from Rosen	40
5	Sets	45
5.1	Sets and elements	45
5.2	Singleton set	45
5.3	Subset and proper subset	46
5.4	Empty (null) set	46
5.5	Equal sets	46
5.6	Cardinality [†]	47
5.7	Power set	47
5.8	Cartesian product	47
5.9	Set operations	48
5.10	Set identities [†]	48
5.11	Generalised union and intersection [†]	50
5.12	Symmetric difference [†]	50
5.13	Sets as bit strings [†]	50
5.14	Practice questions from Rosen	51
6	Relations	57
6.1	Relations and ordered pairs	57
6.2	Properties of relations	57
6.3	Representing relations	59
6.4	Combining relations [†]	60
6.5	Closures [†]	61
6.6	Equivalence relations	62
6.7	Partial orders	63
6.8	n -ary relations and databases [†]	63
6.9	Solved previous-year questions	64
6.10	Practice questions from Rosen	65
7	Partial Orders, Hasse Diagrams and Lattices	73
7.1	Partially ordered sets	73
7.2	Hasse diagrams	73
7.3	Maximal and minimal elements	74
7.4	Upper and lower bounds, LUB and GLB	75
7.5	Lattices	76
7.6	Dual of a poset [†]	77
7.7	Lexicographic order [†]	77
7.8	Compatible total orders (topological sorting) [†]	78
7.9	Solved previous-year questions	78
7.10	Practice questions from Rosen	79
8	Functions	84
8.1	Functions: domain, codomain, image, range	84

8.2	One-to-one, onto, bijection	84
8.3	Inverse function	86
8.4	Composition of functions	86
8.5	Floor and ceiling functions	86
8.6	Is a rule a function? [†]	87
8.7	Image of a set and inverse image [†]	87
8.8	Old functions, new functions [†]	87
8.9	Composition, inverses and one-to-one/onto [†]	88
8.10	Floor and ceiling: more properties [†]	88
8.11	Partial functions [†]	88
8.12	Solved previous-year questions	89
8.13	Practice questions from Rosen	90
A	Corrections to the class notes	95
B	One-page revision sheet	96

PYQ coverage index

Paper codes used in this guide.

M23	Mid-Semester (PG) Examination, October 2023 — CS 5101, 30 marks
MU	Mid-Semester paper, CS 5101 (date not visible on the scan), 30 marks
M24	Mid-Semester (PG) Examination, September 2024 — CS 5101, 30 marks
E24	End-Semester Examination (PG), December 2024 — CS 5101, 50 marks

Previous-year questions included and solved (27)

Paper	Q	Topic	Marks	Chapter	Page
M23	1(a)	Truth table of $(P \vee Q) \rightarrow (P \wedge Q)$	3	1 Propositional Logic	7
M23	1(b)	Predicates, quantifiers, free and bound variables	3	2 Predicate Logic	20
M23	2(a)	Composition of functions; $f \circ g$	2	8 Functions	89
M23	2(b)	Function $\mathbb{N} \rightarrow \mathbb{N}$ onto but not one-to-one	2	8 Functions	89
M23	2(c)	Relation $x \neq y$: R, R^{-1} , transitivity	3	6 Relations	64
M23	2(d)	Hasse diagram of divisors poset $\{1, 2, 3, 4, 6, 8, 12\}$	3	7 Partial Orders	78
M23	3(a)	True/false statements on relations (R^{-1} , σ , symmetric/antisymmetric)	4	6 Relations	64
M23	3(b)	Predicate-logic translations with P, E, L	6	2 Predicate Logic	22
MU	1(a)	Which sentences are propositions	4	1 Propositional Logic	7
MU	1(b)	Satisfiability of an 8-clause CNF	2	1 Propositional Logic	8
MU	1(c)	Quantifiers: Jerry loves exactly two; Lydia	4	2 Predicate Logic	21
MU	2(a)	Knights and knaves (Anita, Boris, Carmen)	4	1 Propositional Logic	9
MU	2(b)	Rules of inference in the Socrates argument	2	3 Rules of Inference	28
MU	2(c)	Negations with De Morgan's laws	4	2 Predicate Logic	21

Paper	Q	Topic	Marks	Chapter	Page
MU	3(a)	Proof by contradiction: irrational + rational	4	4 Proof Techniques	39
MU	3(b)	Induction: $H_{2^n} \geq 1 + n/2$	6	4 Proof Techniques	40
M24	1(a)	n odd $\iff 5n + 6$ odd	2	4 Proof Techniques	39
M24	1(b)	Knights and knaves (A, B) with propositional logic	3	1 Propositional Logic	8
M24	1(c)	Quantifier translations over the integers	5	2 Predicate Logic	21
M24	2(a)	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$ by equivalences	3	1 Propositional Logic	9
M24	2(b)	Master's-degree requirement in predicate logic	3	2 Predicate Logic	22
M24	2(c)	Satisfiability of $(p \rightarrow q) \wedge (p \rightarrow \neg q) \wedge \dots$	4	1 Propositional Logic	10
M24	3(a)	Are these arguments valid?	4	3 Rules of Inference	28
M24	3(b)	Induction: $\sum i^2 = n(n+1)(2n+1)/6$	6	4 Proof Techniques	40
E24	1(b)	$f \circ g$ and $g \circ f$ for $3x + 2, x^2 - 4$	5	8 Functions	89
E24	2(b)	Hasse diagram of divisors of 60; maximal/minimal, LUB, GLB	7	7 Partial Orders	79
E24	4	Number of onto functions; $4 \rightarrow 3$	10	8 Functions	89

Questions left out (topic not in your notes)

Paper	Questions left out	Topic area
Mid-Sem Oct 2023 (CS 5101)	Q1(c): $(S, *)$ with $a * b = a + b + ab$ is an abelian group	Algebraic structures (removed from the course)
End-Sem Nov 2015 (CS-1501)	Q1–Q8: predictive parsing, SDT, code optimisation and generation, global register allocation, CFG vs AST, type checking, grammar design and ambiguity, LR error recovery	Compiler design
Mid-Sem Sep 2015 (CS-1501)	Q1–Q5: Church–Turing thesis, halting problem, P vs NP, intractable problems, algorithm efficiency	Computability, complexity
End-Sem Nov 2016 (CS-1501; scanned twice)	Q1–Q6, D1–D2: ambiguity, shift-reduce parsing, register allocation on an AST, LR error recovery, priority-based graph colouring, type checking, data-flow and liveness analysis, global register allocation	Compiler design
End-Sem Nov–Dec 2017 (CS-1501)	Q1–Q6: P and NP, recursively enumerable languages, Turing machines, regular expressions, ambiguous grammar, predictive and CLR parsing, DFA/NFA (Thompson)	Automata, complexity, compilers
Mid-Sem Sep 2018 (CS-1501)	Q1–Q3: universal TM, palindrome TM, PCP, r.e. union, asymptotic complexity, Hamiltonian cycle in NP, CNF to 3SAT, NFA to DFA	Automata, computability, complexity
End-Sem Nov 2018 (CS-1501)	Q1–Q6: halting, TMs for $0^m 1^m$ and $m \times n$, r.e. and recursive languages, undecidability, 3SAT, tractability, minimised DFA, DAG of a basic block, optimisation sources, error recovery, LALR parsing	Automata, complexity, compilers
End-Sem Dec 2024 (CS 5101)	Q1(a) DFA for counts of a's and b's; Q2(a) regex with no consecutive a's; Q3(a) PDA for $a^n b^m c^{n+m}$; Q3(b) TM for ww^R ; Q5(a) TM and valid C programs; Q5(b) TM for unary product; Q6(a) TSP is NP-complete; Q6(b) reductions and NP-completeness	Automata, computability, complexity

Rosen practice questions (88)

Source	Skill	Chapter	Page
Rosen §1.1 Ex. 6	truth values from given data	1 Propositional Logic	10
Rosen §1.1 Ex. 3–5	negating English propositions	1 Propositional Logic	10
Rosen §1.1 Ex. 19	inclusive or exclusive?	1 Propositional Logic	11
Rosen §1.1 Ex. 16–17	truth values with false facts	1 Propositional Logic	11
Rosen §1.1 Ex. 15	translating with “necessary but not sufficient”	1 Propositional Logic	11
Rosen §1.1 Ex. 22–24	rewriting as “if p , then q ”	1 Propositional Logic	12
Rosen §1.1 Ex. 27	converse, contrapositive, inverse	1 Propositional Logic	12
Rosen §1.1 Ex. 29–30	number of rows of a truth table	1 Propositional Logic	13
Rosen §1.1 Ex. 33 and 35	truth tables, two variables	1 Propositional Logic	13
Rosen §1.1 Ex. 37(d)	truth table, three variables	1 Propositional Logic	14
Rosen §2.1 Ex. 41–42	quantified statements over $\mathbb{R}$ and $\mathbb{Z}$	2 Predicate Logic	23
Rosen §2.1 Ex. 43	truth sets over the integers	2 Predicate Logic	23
Rosen §1.8 Ex. 15	uniqueness quantifier	2 Predicate Logic	23
Rosen §1.6 Ex. 20	valid or not?	3 Rules of Inference	29
Rosen §1.6 Ex. 23	find the error (existential instantiation)	3 Rules of Inference	29
Rosen §1.6 Ex. 24	find the error (universal generalization)	3 Rules of Inference	29
Rosen §1.6 Ex. 25–26	universal modus tollens and universal transitivity	3 Rules of Inference	30
Rosen §1.6 Ex. 27–28	derivations with $\forall$	3 Rules of Inference	30
Rosen §1.6 Ex. 29	derivation with $\exists$	3 Rules of Inference	31
Rosen §1.6 Ex. 30–31	resolution	3 Rules of Inference	31
Rosen §1.6 Ex. 33	resolution and satisfiability	3 Rules of Inference	31
Rosen §1.6 Ex. 34	translate, then test validity	3 Rules of Inference	32
Rosen §1.6 Ex. 35	a long argument in words	3 Rules of Inference	32
Rosen §1.7 Ex. 5 and 7	direct proofs	4 Proof Techniques	41
Rosen §1.7 Ex. 8	contradiction with squares	4 Proof Techniques	41
Rosen §1.7 Ex. 11–13	prove or disprove (rational, irrational)	4 Proof Techniques	41
Rosen §1.7 Ex. 15–16	contraposition	4 Proof Techniques	41
Rosen §1.7 Ex. 17	contraposition and contradiction, same statement	4 Proof Techniques	42
Rosen §1.7 Ex. 23	contradiction (pigeonhole)	4 Proof Techniques	42
Rosen §1.7 Ex. 25	contradiction with parity cases	4 Proof Techniques	42
Rosen §1.7 Ex. 31	several equivalent statements	4 Proof Techniques	43
Rosen §1.7 Ex. 34–35	is this reasoning correct?	4 Proof Techniques	43
Rosen §1.7 Ex. 38	counterexample	4 Proof Techniques	43
Rosen §1.8 Ex. 27	conjecture, then proof by cases	4 Proof Techniques	43
Rosen §1.8 Ex. 29–30	exhaustive proofs	4 Proof Techniques	44
Rosen §1.8 Ex. 22	forward reasoning	4 Proof Techniques	44

Source	Skill	Chapter	Page
Rosen §1.8 Ex. 9–10	constructive and non-constructive existence	4 Proof Techniques	44
Rosen §1.8 Ex. 17	existence and uniqueness	4 Proof Techniques	44
Rosen §2.1 Ex. 9–10	true or false with $\emptyset$	5 Sets	51
Rosen §2.1 Ex. 7, 8 and 18	$\in$ versus $\subseteq$	5 Sets	51
Rosen §2.1 Ex. 20, 21, 23, 24	cardinality and power sets	5 Sets	52
Rosen §2.1 Ex. 22 and 25	proofs about power sets	5 Sets	52
Rosen §2.1 Ex. 26, 31–38	Cartesian products	5 Sets	52
Rosen §2.2 Ex. 14 and 25	compute set expressions; recover the sets	5 Sets	53
Rosen §2.2 Ex. 16, 18, 24	prove a set identity	5 Sets	54
Rosen §2.2 Ex. 17	membership table	5 Sets	54
Rosen §2.2 Ex. 29 and 30	what can you conclude?	5 Sets	54
Rosen §2.2 Ex. 32, 35–41	symmetric difference	5 Sets	55
Rosen §2.2 Ex. 46	inclusion–exclusion for three sets	5 Sets	55
Rosen §2.2 Ex. 50 and 51	indexed unions and intersections	5 Sets	55
Rosen §2.2 Ex. 52–56	sets as bit strings	5 Sets	56
Rosen §9.1 Ex. 3	test the six properties	6 Relations	65
Rosen §9.1 Ex. 6 and 7	relations on $\mathbb{R}$ and $\mathbb{Z}$	6 Relations	65
Rosen §9.1 Ex. 8–10, 15, 22	the empty relation; properties that co-occur	6 Relations	66
Rosen §9.1 Ex. 26, 27, 40, 41	inverse, complement, set operations on relations	6 Relations	67
Rosen §9.1 Ex. 32, 33, 36	composition and powers	6 Relations	67
Rosen §9.1 Ex. 25, 42–46	counting relations	6 Relations	67
Rosen §9.4 Ex. 1–3	closures	6 Relations	68
Rosen §9.4 Ex. 29	smallest relation with given properties	6 Relations	68
Rosen §9.3 Ex. 3 and 7	from a matrix to properties	6 Relations	69
Rosen §9.3 Ex. 14 and 15	matrix operations and powers	6 Relations	69
Rosen §9.5 Ex. 1, 24 and 26	which are equivalence relations?	6 Relations	70
Rosen §9.5 Ex. 9, 15, 16, 25, 29, 39, 40	$f(x) = f(y)$ gives an equivalence relation	6 Relations	70

Source	Skill	Chapter	Page
Rosen §9.5 Ex. 35–37, 49	congruence classes and refinement	6 Relations	71
Rosen §9.5 Ex. 41, 44, 47	partitions and their relations	6 Relations	71
Rosen §9.5 Ex. 55 and 56	smallest equivalence relation; unions and intersections	6 Relations	72
Rosen §9.5 Ex. 61–63	how many equivalence relations?	6 Relations	72
Rosen §9.6 Ex. 1 and 7	which relations are partial orders?	7 Partial Orders	79
Rosen §9.6 Ex. 13–15	duals, comparable elements	7 Partial Orders	80
Rosen §9.6 Ex. 16–19	lexicographic order	7 Partial Orders	80
Rosen §9.6 Ex. 22, 23, 29	draw the Hasse diagram	7 Partial Orders	81
Rosen §9.6 Ex. 33 and 35	maximal, minimal, greatest, least, bounds	7 Partial Orders	82
Rosen §9.6 Ex. 36, 40, 50, 60	existence and uniqueness	7 Partial Orders	82
Rosen §9.6 Ex. 44 and 46	is it a lattice?	7 Partial Orders	82
Rosen §9.6 Ex. 62 and 63	compatible total orders	7 Partial Orders	83
Rosen §2.3 Ex. 1–3	is it a function?	8 Functions	90
Rosen §2.3 Ex. 12–14	one-to-one and onto, with counterexamples	8 Functions	90
Rosen §2.3 Ex. 20	example functions $\mathbb{N} \rightarrow \mathbb{N}$	8 Functions	91
Rosen §2.3 Ex. 22, 23, 39	bijections $\mathbb{R} \rightarrow \mathbb{R}$ and inverses	8 Functions	91
Rosen §2.3 Ex. 33–35	composition and one-to-one, onto	8 Functions	92
Rosen §2.3 Ex. 38 and 70	when do $f \circ g$ and $g \circ f$ agree; inverse of a composition	8 Functions	92
Rosen §2.3 Ex. 40 and 41	image of a union and intersection	8 Functions	92
Rosen §2.3 Ex. 42–45	inverse images	8 Functions	93
Rosen §2.3 Ex. 54 and 73	prove or disprove: floor and ceiling	8 Functions	93
Rosen §2.3 Ex. 56–59, 61	counting with floor and ceiling	8 Functions	93
Rosen §2.3 Ex. 71	characteristic functions	8 Functions	94
Rosen §2.3 Ex. 72	finite sets of equal size	8 Functions	94
Rosen §2.3 Ex. 77	partial functions	8 Functions	94

Propositional Logic

1.1 Propositions and the basic connectives

DEFINITION — Proposition

A *proposition* is a declarative sentence that is either **true** or **false**, but not both. Its truth value is written **T** or **F**. We name propositions with variables $p, q, r, s, \dots$

EXAMPLE

Sentence	Verdict
"Boston is the capital of Massachusetts."	proposition (true)
"The moon is made of green cheese."	proposition (false)
" $2 + 3 = 7$ "	proposition (false)
"What time is it?"	not a proposition (question)
"Answer this question."	not a proposition (command)
" $x + 2 = 11$ "	not a proposition (depends on x)
"This statement is false."	not a proposition (paradox)

A proposition that cannot be split further is *atomic*; propositions built from others using connectives are *compound*. The connectives are:

DEFINITION — Negation, conjunction, disjunction

- $\neg p$ (*not p*) is true exactly when p is false.
- $p \wedge q$ (*p and q*) is true exactly when *both* are true.
- $p \vee q$ (*p or q*) is true when *at least one* is true (inclusive or).
- $p \oplus q$ (exclusive or)[†] is true when *exactly one* is true.

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \oplus q$
T	T	F	T	T	F
T	F	F	F	T	T
F	T	T	F	T	T
F	F	T	F	F	F

EXAMPLE

Let p : "It is raining" and q : "It is cold". Then $p \wedge q$ = "It is raining and cold", $p \vee q$ = "It is raining or cold (or both)", $\neg p$ = "It is not raining", $p \oplus q$ = "Either it is raining or it is cold, but not both".

REMEMBER — precedence

Evaluate in the order $\neg$, then $\wedge$, then $\vee$, then $\rightarrow$, then $\leftrightarrow$. When in doubt, use brackets: $p \vee q \wedge r$ means $p \vee (q \wedge r)$.

1.2 The conditional and the biconditional

DEFINITION — Conditional statement $p \rightarrow q$

“If p then q ”. Here p is the *hypothesis* (antecedent) and q the *conclusion* (consequent). It is false only when p is true and q is false; in every other row it is true.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

EXAMPLE — the “promise” reading

A teacher promises: “If you score 100%, I will give you a prize.” Only one situation breaks the promise: you score 100% and get no prize (row **TF**). If you do not score 100%, the teacher has not broken the promise whatever happens, so the promise stays true (rows **FT** and **FF**).

Many ways to say $p \rightarrow q$: if p then q ; q if p ; p only if q ; q whenever p ; p is sufficient for q ; q is necessary for p ; q follows from p .

DEFINITION — Converse, inverse, contrapositive

For $p \rightarrow q$:

converse	$q \rightarrow p$
inverse	$\neg p \rightarrow \neg q$
contrapositive	$\neg q \rightarrow \neg p$

p	q	$p \rightarrow q$	$q \rightarrow p$	$\neg q \rightarrow \neg p$	$\neg p \rightarrow \neg q$
T	T	T	T	T	T
T	F	F	T	F	T
F	T	T	F	T	F
F	F	T	T	T	T

So $p \rightarrow q \equiv \neg q \rightarrow \neg p$ (a statement and its contrapositive always agree), and the converse is equivalent to the inverse, but *neither is equivalent to the original*.

EXAMPLE

$p \rightarrow q$: “If it rains, the ground is wet.” Converse: “If the ground is wet, it rains” (false: a sprinkler could be on). Contrapositive: “If the ground is not wet, it did not rain” (true, and says the same thing as the original).

DEFINITION — Biconditional $p \leftrightarrow q$

“ p if and only if q ” (iff). It is true exactly when p and q have the *same* truth value, and $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

EXAMPLE

“You can take the flight if and only if you have a ticket”: if you have a ticket you may fly, and if you may fly then you have a ticket.

1.3 Translating English into logic

Method. (1) Pick a letter for each atomic statement. (2) Mark the connective words (*and, or, not, if, only if, unless*). (3) Put the sentence together, using brackets to show grouping. Useful dictionary: “ A unless B ” $\equiv \neg B \rightarrow A$; “ A only if B ” $\equiv A \rightarrow B$; “ A but B ” $\equiv A \wedge B$.

EXAMPLE — Roller coaster

“You cannot ride the roller coaster if you are under 4 feet tall unless you are older than 16 years old.”

Let p : you can ride; q : you are under 4 feet tall; r : you are older than 16.

$$(q \wedge \neg r) \rightarrow \neg p.$$

EXAMPLE — Network access

“If the computer is within the local network, or it is not within the local network but has a valid login ID, or it is under the use of the administrators, then the internet is accessible to the computer.”

Let p : within the local network; q : has a valid login ID; r : used by administrators; s : internet accessible. Mark the three blocks joined by “or” (*make sure the sentence is broken into blocks*):

$$(p \vee (\neg p \wedge q) \vee r) \rightarrow s.$$

1.4 Knights and knaves

An island has two kinds of inhabitants: *knights*, who always tell the truth, and *knaves*, who always lie. To solve such puzzles with propositional logic:

1. Let a letter mean “that person is a knight” (so its negation means “knave”).
2. If person X says statement S , then X is a knight $\leftrightarrow S$ is true. (A knight’s statement is true, a knave’s is false.)
3. Find the assignment(s) satisfying all such biconditionals (truth table or case analysis).

EXAMPLE — from the notes

A says “B is a knight”. B says “the two of us are opposite types.” Let p : A is a knight, q : B is a knight. The conditions are $p \leftrightarrow q$ (from A) and $q \leftrightarrow (p \oplus q)$ (from B). The complete solution is worked out in the PYQ section below (both are knaves).

1.5 Tautology, contradiction, contingency, equivalence

DEFINITION — Tautology, contradiction, contingency

A compound proposition is a

- **tautology** if it is *always true* (every row of its truth table is **T**), e.g. $p \vee \neg p$;
- **contradiction** if it is *always false*, e.g. $p \wedge \neg p$;
- **contingency** if it can be true or false depending on the truth values of its variables, e.g. $p \rightarrow q$ or $p \wedge q$.

EXAMPLE

p	$p \vee \neg p$	$p \wedge \neg p$
T	T	F
F	T	F

Here $p \vee \neg p$ is a tautology (all **T**) and $p \wedge \neg p$ a contradiction (all **F**).

DEFINITION — Logical equivalence

$p \equiv q$ means p and q have the same truth value in every row; equivalently, $p \leftrightarrow q$ is a tautology.

EXAMPLE — De Morgan and the conditional as $\neg p \vee q$

p	q	$\neg(p \vee q)$	$\neg p \wedge \neg q$	$p \rightarrow q$	$\neg p \vee q$
T	T	F	F	T	T
T	F	F	F	F	F
F	T	F	F	T	T
F	F	T	T	T	T

The 3rd and 4th columns agree, so $\neg(p \vee q) \equiv \neg p \wedge \neg q$. The last two agree, so $p \rightarrow q \equiv \neg p \vee q$ — an **important result** used in almost every derivation.

1.6 Standard logical equivalences

Law	Equivalences	
Identity	$p \wedge \mathbf{T} \equiv p$	$p \vee \mathbf{F} \equiv p$
Domination	$p \vee \mathbf{T} \equiv \mathbf{T}$	$p \wedge \mathbf{F} \equiv \mathbf{F}$
Idempotent	$p \vee p \equiv p$	$p \wedge p \equiv p$
Double negation	$\neg(\neg p) \equiv p$	
Commutative	$p \vee q \equiv q \vee p$	$p \wedge q \equiv q \wedge p$
Associative	$(p \vee q) \vee r \equiv p \vee (q \vee r)$	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
Distributive	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
De Morgan	$\neg(p \vee q) \equiv \neg p \wedge \neg q$	$\neg(p \wedge q) \equiv \neg p \vee \neg q$
Absorption	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
Negation	$p \vee \neg p \equiv \mathbf{T}$	$p \wedge \neg p \equiv \mathbf{F}$

Equivalences involving $\rightarrow$ and $\leftrightarrow$:

$$\begin{aligned}
 p \rightarrow q &\equiv \neg p \vee q & \neg(p \rightarrow q) &\equiv p \wedge \neg q \\
 p \rightarrow q &\equiv \neg q \rightarrow \neg p & p \vee q &\equiv \neg p \rightarrow q \\
 p \leftrightarrow q &\equiv (p \rightarrow q) \wedge (q \rightarrow p) & p \leftrightarrow q &\equiv (p \wedge q) \vee (\neg p \wedge \neg q)
 \end{aligned}$$

1.7 Proving an equivalence by a chain of laws

To show $A \equiv B$ without a truth table, start from one side and rewrite step by step, naming the law used at each step.

EXAMPLE — $\neg(p \rightarrow q) \equiv p \wedge \neg q$

$$\begin{aligned}
 \neg(p \rightarrow q) &\equiv \neg(\neg p \vee q) && \text{(conditional as disjunction)} \\
 &\equiv \neg\neg p \wedge \neg q && \text{(De Morgan)} \\
 &\equiv p \wedge \neg q && \text{(double negation)}
 \end{aligned}$$

EXAMPLE — $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$

$$\begin{aligned}
 \neg(p \vee (\neg p \wedge q)) &\equiv \neg((p \vee \neg p) \wedge (p \vee q)) && \text{(distributive)} \\
 &\equiv \neg(\mathbf{T} \wedge (p \vee q)) && \text{(negation law)} \\
 &\equiv \neg(p \vee q) && \text{(identity)} \\
 &\equiv \neg p \wedge \neg q && \text{(De Morgan)}
 \end{aligned}$$

EXAMPLE — $(p \wedge q) \rightarrow (p \vee q)$ is a tautology

$$\begin{aligned}(p \wedge q) \rightarrow (p \vee q) &\equiv \neg(p \wedge q) \vee (p \vee q) && \text{(conditional as disjunction)} \\ &\equiv (\neg p \vee \neg q) \vee (p \vee q) && \text{(De Morgan)} \\ &\equiv (\neg p \vee p) \vee (\neg q \vee q) && \text{(commutative, associative)} \\ &\equiv \mathbf{T} \vee \mathbf{T} \equiv \mathbf{T} && \text{(negation, idempotent)}\end{aligned}$$

1.8 Satisfiability

DEFINITION — Satisfiable / unsatisfiable

A compound proposition is *satisfiable* if **at least one** assignment of truth values to its variables makes it true (such an assignment is a *satisfying assignment*). If **no** assignment makes it true it is *unsatisfiable*.

Relations to earlier ideas: a tautology is satisfiable (it is true under every assignment); a contradiction is unsatisfiable; and A is a tautology exactly when $\neg A$ is unsatisfiable.

How to test. *Satisfiable*: find one assignment and check it in every clause (a conjunction is true only if every part is true). *Unsatisfiable*: show that every assignment fails, either with the full truth table or by forced deductions.

EXAMPLE — satisfiable

$(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$. Try $p = q = r = \mathbf{T}$: the three clauses are $\mathbf{T} \vee \mathbf{F}$, $\mathbf{T} \vee \mathbf{F}$, $\mathbf{T} \vee \mathbf{F}$, all true. So it is satisfiable (so is $p = q = r = \mathbf{F}$). Note that $p = \mathbf{T}, q = \mathbf{T}, r = \mathbf{F}$ is *not* satisfying: the last clause becomes $\mathbf{F} \vee \mathbf{F}$. One failed try proves nothing — try again.

EXAMPLE — unsatisfiable

$(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \wedge \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$.

The clause $r \wedge \neg p$ forces $r = \mathbf{T}, p = \mathbf{F}$. Then $(p \vee \neg q)$ forces $q = \mathbf{F}$. But then $(q \vee \neg r) = \mathbf{F} \vee \mathbf{F} = \mathbf{F}$. The conjunction cannot be true: **not satisfiable**.

REMEMBER

A formula written as an AND of clauses, each clause an OR of literals (a CNF^+), is satisfiable iff some single assignment makes *every* clause contain at least one true literal.

1.9 Conditionals in programs and bit strings[†]

EXAMPLE — a conditional inside a program

What is the value of x after the statement `if  $2 + 2 = 4$  then  $x := x + 1$  if  $x = 0$` before it? The hypothesis $2 + 2 = 4$ is true, so the assignment is carried out and x becomes 1. If the hypothesis had been false, nothing would happen and x would stay 0. This is exactly the truth table of $p \rightarrow q$: when p is true, the action q is required.

DEFINITION — Bit string and bitwise operations

A *bit string* is a sequence of 0s and 1s; its *length* is the number of bits. Using $1 = \text{T}$ and $0 = \text{F}$, the connectives $\vee, \wedge, \oplus$ extend position by position to two strings of the same length; these are the *bitwise OR, AND and XOR*.

EXAMPLE

For $a = 01\ 1011\ 0110$ and $b = 11\ 0001\ 1101$:

a	01 1011 0110
b	11 0001 1101
$a \text{ OR } b$	11 1011 1111
$a \text{ AND } b$	01 0001 0100
$a \text{ XOR } b$	10 1010 1011

(Bits are grouped in fours only to make them readable.) Bit strings are used again in Chapter 5 to represent sets.

1.10 Solved previous-year questions**PYQ • Mid-Sem Oct 2023 Q1(a) • 3 marks**

Construct the truth table $(P \vee Q) \rightarrow (P \wedge Q)$.

Solution. Build the table column by column: first $P \vee Q$ and $P \wedge Q$, then the conditional.

P	Q	$P \vee Q$	$P \wedge Q$	$(P \vee Q) \rightarrow (P \wedge Q)$
T	T	T	T	T
T	F	T	F	F
F	T	T	F	F
F	F	F	F	T

The conditional is false exactly when the hypothesis $P \vee Q$ is true and the conclusion $P \wedge Q$ is false (rows 2 and 3). The final column contains both **T** and **F**, so the statement is a **contingency**. Its column is **T, F, F, T**, which is the column of $P \leftrightarrow Q$, so $(P \vee Q) \rightarrow (P \wedge Q) \equiv P \leftrightarrow Q$.

PYQ • Mid-Sem (undated, CS5101) Q1(a) • 4 marks

Which of these sentences are propositions? What are the truth values of those that are propositions?

- (i) Boston is the capital of Massachusetts (ii) Answer this question (iii) $x + 2 = 11$ (iv) The moon is made of green cheese.

Solution. A proposition is a declarative sentence that is exactly one of true/false.

Verdict	Reason
(i) Proposition, true	Boston is the capital of Massachusetts.
(ii) Not a proposition	It is a command; it cannot be true or false.
(iii) Not a proposition	Contains the free variable x ; true for $x = 9$, false otherwise. It becomes a proposition only after x is given a value or quantified.
(iv) Proposition, false	A declarative sentence with a definite (false) truth value.

PYQ • Mid-Sem (undated, CS5101) Q1(b) • 2 marks

Determine whether the given compound proposition is satisfiable:

$$(p \vee q \vee r) \wedge (p \vee \neg q \vee \neg s) \wedge (q \vee \neg r \vee s) \wedge (\neg p \vee r \vee s) \wedge (\neg p \vee q \vee \neg s) \wedge (p \vee \neg q \vee \neg r) \wedge (\neg p \vee \neg q \vee s) \wedge (\neg p \vee \neg r \vee \neg s)$$

Solution. It suffices to exhibit one satisfying assignment. Try $p = \text{F}$, $q = \text{T}$, $r = \text{F}$, $s = \text{F}$ and check every clause:

Clause	Substitution	Value
$p \vee q \vee r$	$\text{F} \vee \text{T} \vee \text{F}$	T
$p \vee \neg q \vee \neg s$	$\text{F} \vee \text{F} \vee \text{T}$	T
$q \vee \neg r \vee s$	$\text{T} \vee \text{T} \vee \text{F}$	T
$\neg p \vee r \vee s$	$\text{T} \vee \text{F} \vee \text{F}$	T
$\neg p \vee q \vee \neg s$	$\text{T} \vee \text{T} \vee \text{T}$	T
$p \vee \neg q \vee \neg r$	$\text{F} \vee \text{F} \vee \text{T}$	T
$\neg p \vee \neg q \vee s$	$\text{T} \vee \text{F} \vee \text{F}$	T
$\neg p \vee \neg r \vee \neg s$	$\text{T} \vee \text{T} \vee \text{T}$	T

All eight clauses are true, so the conjunction is true. Hence the proposition is **satisfiable**. (Checking all 16 assignments shows exactly three satisfy it: $(p, q, r, s) = (\text{T}, \text{T}, \text{F}, \text{T}), (\text{F}, \text{T}, \text{F}, \text{F})$ and $(\text{F}, \text{F}, \text{T}, \text{T})$.)

PYQ • Mid-Sem Sep 2024 Q1(b) • 3 marks

On an island every inhabitant is a knight (always tells the truth) or a knave (always lies). You meet two people A and B . A says “ B is a knight” and B says “The two of us are opposite types.” What are A and B ? Use propositional logic.

Solution. Let p : “ A is a knight” and q : “ B is a knight”. A person is a knight exactly when what they say is true, so:

- A ’s statement is q , giving the condition $p \leftrightarrow q$;

- B 's statement is "opposite types" $\equiv p \oplus q \equiv \neg(p \leftrightarrow q)$, giving $q \leftrightarrow \neg(p \leftrightarrow q)$.

p	q	$p \leftrightarrow q$	$p \oplus q$	$q \leftrightarrow (p \oplus q)$	both conditions hold?
T	T	T	F	F	no
T	F	F	T	F	no
F	T	F	T	T	no (A 's fails)
F	F	T	F	T	yes

Only $p = \text{F}$, $q = \text{F}$ works. **Both A and B are knaves.**

Check in words: if A were a knight then B would be a knight, and B 's statement "we are opposite types" would be false — impossible for a knight. So A is a knave; his statement " B is a knight" is then false, so B is a knave. B 's statement "we are opposite types" is indeed false (both knaves), consistent with a knave lying.

PYQ • Mid-Sem (undated, CS 5101) Q2(a) • 4 marks

Anita, Boris and Carmen are inhabitants of an island whose inhabitants are knights (always truthful) or knaves (always lie). What are Anita, Boris and Carmen if Anita says "I am a knave and Boris is a knight" and Boris says "Exactly one of the three of us is a knight"?

Solution. Let a, b, c mean "Anita / Boris / Carmen is a knight".

Anita. Her statement is $\neg a \wedge b$, so $a \leftrightarrow (\neg a \wedge b)$. If $a = \text{T}$, the right side is $\text{F} \wedge b = \text{F}$, so $\text{T} \leftrightarrow \text{F}$ is false. Hence $a = \text{F}$: **Anita is a knave.** A knave's statement is false; with $a = \text{F}$ the first part " $\neg a$ " is true, so the statement can be false only if $b = \text{F}$. Hence **Boris is a knave.**

Boris. Being a knave, his statement "exactly one of us is a knight" must be false. Anita and Boris are already knaves, so Carmen would be the only possible knight; if $c = \text{T}$ the number of knights would be exactly one and Boris's statement would be true — contradiction. So $c = \text{F}$: **Carmen is a knave.**

Check. Anita: "I am a knave" is true but "Boris is a knight" is false, so the conjunction is false (a knave lies) ✓. Boris: the number of knights is $0 \neq 1$, so his statement is false ✓.

Answer: all three are knaves. (It is the only consistent assignment; a truth table over (a, b, c) confirms it.)

PYQ • Mid-Sem Sep 2024 Q2(a) • 3 marks

Check whether $p \leftrightarrow q$ and $(p \rightarrow q) \wedge (q \rightarrow p)$ are logically equivalent, by developing a series of logical equivalences.

Solution. By the truth table of $\leftrightarrow$, $p \leftrightarrow q$ is true exactly when p, q are both true or both false, i.e. $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$. Now start from the other side:

$$\begin{aligned}
 (p \rightarrow q) \wedge (q \rightarrow p) &\equiv (\neg p \vee q) \wedge (\neg q \vee p) && \text{(conditional as disjunction)} \\
 &\equiv [(\neg p \vee q) \wedge \neg q] \vee [(\neg p \vee q) \wedge p] && \text{(distributive)} \\
 &\equiv (\neg p \wedge \neg q) \vee (q \wedge \neg q) \vee (\neg p \wedge p) \vee (q \wedge p) && \text{(distributive, twice)} \\
 &\equiv (\neg p \wedge \neg q) \vee \mathbf{F} \vee \mathbf{F} \vee (p \wedge q) && \text{(negation law)} \\
 &\equiv (p \wedge q) \vee (\neg p \wedge \neg q) && \text{(identity, commutative)} \\
 &\equiv p \leftrightarrow q.
 \end{aligned}$$

Hence the two propositions are **logically equivalent**. (A four-row truth table gives the same

column **T, F, F, T** for both.)

PYQ • Mid-Sem Sep 2024 Q2(c) • 4 marks

Determine whether the compound proposition is satisfiable: $(p \rightarrow q) \wedge (p \rightarrow \neg q) \wedge (\neg p \rightarrow q) \wedge (\neg p \rightarrow \neg q)$.

Solution. Method 1 (laws). Combine the first two conjuncts and the last two:

$$\begin{aligned}(p \rightarrow q) \wedge (p \rightarrow \neg q) &\equiv p \rightarrow (q \wedge \neg q) \equiv p \rightarrow \mathbf{F} \equiv \neg p, \\ (\neg p \rightarrow q) \wedge (\neg p \rightarrow \neg q) &\equiv \neg p \rightarrow (q \wedge \neg q) \equiv \neg p \rightarrow \mathbf{F} \equiv p.\end{aligned}$$

So the whole proposition is $\neg p \wedge p \equiv \mathbf{F}$, a contradiction.

Method 2 (cases). If $p = \mathbf{T}$: $p \rightarrow q$ needs $q = \mathbf{T}$, while $p \rightarrow \neg q$ needs $q = \mathbf{F}$ — impossible. If $p = \mathbf{F}$: $\neg p \rightarrow q$ needs $q = \mathbf{T}$ while $\neg p \rightarrow \neg q$ needs $q = \mathbf{F}$ — impossible.

No assignment makes it true, so the proposition is **not satisfiable** (it is a contradiction).

1.11 Practice questions from Rosen

Rosen's exercises for Section 1.1 mostly repeat the skills above. The ten below are the ones that add something new (a trap in the wording, or a different type of table), in the style of the mid-semester questions.

PRACTICE • Rosen §1.1 Ex. 6 • truth values from given data

Smartphone A has 256 MB RAM, 32 GB ROM and an 8 MP camera; B has 288 MB, 64 GB and 4 MP; C has 128 MB, 32 GB and 5 MP. Determine the truth value of:

- (a) B has the most RAM of the three. (b) C has more ROM *or* a higher-resolution camera than B.
- (c) B has more RAM, more ROM *and* a higher-resolution camera than A.
- (d) If B has more RAM and more ROM than C, then it also has a higher-resolution camera (than C).
- (e) A has more RAM than B if and only if B has more RAM than A.

Solution. Evaluate each simple statement from the data, then combine.

- (a) $288 > 256$ and $288 > 128$ **T**
- (b) ROM: $32 > 64$ is **F**; camera: $5 > 4$ is **T**; **F** $\vee$ **T** **T**
- (c) RAM $288 > 256$ **T**, ROM $64 > 32$ **T**, camera $4 > 8$ **F**; **F**
T $\wedge$ **T** $\wedge$ **F**
- (d) hypothesis: $288 > 128$ **T** and $64 > 32$ **T**, so **T**; conclusion: **F**
 $4 > 5$ is **F**; **T** $\rightarrow$ **F**
- (e) $256 > 288$ is **F**; $288 > 256$ is **T**; **F** $\leftrightarrow$ **T** **F**

PRACTICE • Rosen §1.1 Ex. 3–5 • negating English propositions

Write the negation of each: (a) The summer in Maine is hot and sunny. (b) Abby sent more than 100 text messages every day. (c) Steve has more than 100 GB free disk space on his laptop. (d) Zach blocks e-mails and texts from Jennifer.

Solution. (a) $\neg(p \wedge q) \equiv \neg p \vee \neg q$ (De Morgan): “The summer in Maine is not hot *or* it is not sunny.” (Not “not hot and not sunny”.)

(b) “Every day” is a $\forall$ statement, so the negation is a $\exists$ statement: “There is a day on which Abby sent *at most* 100 text messages.” (Not “Abby sent at most 100 messages every day”.)

(c) The negation of “more than 100” ($>$) is “at most 100” ($\leq$): “Steve has at most 100 GB of free disk space.” (Not “less than 100”; exactly 100 also makes the original false.)

(d) De Morgan again: “Zach does not block e-mails from Jennifer *or* he does not block texts from her.”

PRACTICE • Rosen §1.1 Ex. 19 • inclusive or exclusive?

Decide whether an inclusive or exclusive “or” is intended, and explain: (a) Coffee or tea comes with dinner. (b) A password must have at least three digits or be at least eight characters long. (c) The prerequisite for the course is a course in number theory or a course in cryptography. (d) You can pay using U.S. dollars or euros.

Solution. The two readings differ only in the row where *both* statements are true: $p \vee q$ is **T** there, $p \oplus q$ is **F**. So ask: is the case “both” allowed?

(a) **Exclusive**: dinner comes with one of the two drinks, not both.

(b) **Inclusive**: a password with three digits *and* eight characters certainly satisfies the rule.

(c) **Inclusive**: having taken both courses is fine.

(d) Usually **exclusive** (one payment, in one currency), but the sentence does not forbid paying part in each, so either reading is defensible. In exercises of this kind, state the reading you choose and why.

PRACTICE • Rosen §1.1 Ex. 16–17 • truth values with false facts

Determine whether each is true or false. (i) $2 + 2 = 4 \iff 1 + 1 = 2$. (ii) $1 + 1 = 2 \iff 2 + 3 = 4$. (iii) $1 + 1 = 3 \iff$ monkeys can fly. (iv) $0 > 1 \iff 2 > 1$. (v) If $1 + 1 = 2$ then $2 + 2 = 5$. (vi) If $1 + 1 = 3$ then $2 + 2 = 4$. (vii) If $1 + 1 = 3$ then $2 + 2 = 5$. (viii) If monkeys can fly then $1 + 1 = 3$.

Solution. Find the truth value of each part, then use the tables of $\leftrightarrow$ and $\rightarrow$.

- (i) **T** $\leftrightarrow$ **T** **T** same values
- (ii) **T** $\leftrightarrow$ **F** **F** different values
- (iii) **F** $\leftrightarrow$ **F** **T** both false
- (iv) **F** $\leftrightarrow$ **T** **F** different values
- (v) **T** $\rightarrow$ **F** **F** the only false row of $\rightarrow$
- (vi) **F** $\rightarrow$ **T** **T** false hypothesis
- (vii) **F** $\rightarrow$ **F** **T** false hypothesis
- (viii) **F** $\rightarrow$ **F** **T** false hypothesis

PRACTICE • Rosen §1.1 Ex. 15 • translating with “necessary but not sufficient”

Let p : grizzly bears have been seen in the area; q : hiking is safe on the trail; r : berries are ripe along the trail. Write in symbols: (a) Berries are ripe along the trail, but grizzly bears have not been seen. (b) Grizzly bears have not been seen and hiking is safe, but berries are ripe. (c) If

berries are ripe, hiking is safe if and only if grizzly bears have not been seen. (d) It is not safe to hike, but grizzly bears have not been seen and the berries are ripe. (e) For hiking to be safe, it is necessary but not sufficient that berries not be ripe and grizzly bears not have been seen. (f) Hiking is not safe whenever grizzly bears have been seen and berries are ripe.

Solution. “But” means $\wedge$. “X is necessary for Y” is $Y \rightarrow X$; “X is sufficient for Y” is $X \rightarrow Y$.

(a) $r \wedge \neg p$

(b) $\neg p \wedge q \wedge r$

(c) $r \rightarrow (q \leftrightarrow \neg p)$

(d) $\neg q \wedge \neg p \wedge r$

(e) Let $s = \neg r \wedge \neg p$. Necessary: $q \rightarrow s$. Not sufficient: $\neg(s \rightarrow q)$. So $(q \rightarrow s) \wedge \neg(s \rightarrow q)$; equivalently $(q \rightarrow s) \wedge s \wedge \neg q$.

(f) “A whenever B” is $B \rightarrow A$: $(p \wedge r) \rightarrow \neg q$.

PRACTICE • Rosen §1.1 Ex. 22–24 • rewriting as “if p, then q”

Rewrite in the form “if p, then q”: (a) It is necessary to wash the boss’s car to get promoted. (b) A sufficient condition for the warranty to be good is that you bought the computer less than a year ago. (c) Willy gets caught whenever he cheats. (d) You can access the website only if you pay a subscription fee. (e) Getting elected follows from knowing the right people. (f) Jan will go swimming unless the water is too cold. (g) I will remember to send you the address only if you send me an e-mail message.

Solution. Key patterns: “p only if q” and “q is necessary for p” both give $p \rightarrow q$; “p is sufficient for q” and “q whenever p” give $p \rightarrow q$; “q unless r” gives $\neg r \rightarrow q$.

(a) If you get promoted, then you (have) washed the boss’s car.

(b) If you bought the computer less than a year ago, then the warranty is good.

(c) If Willy cheats, then he gets caught.

(d) If you can access the website, then you pay a subscription fee.

(e) If you know the right people, then you get elected.

(f) If the water is not too cold, then Jan will go swimming.

(g) If I remember to send you the address, then you sent me an e-mail message.

Note the reversals in (a), (d), (g): the “only if” or “necessary” part is the *conclusion*.

PRACTICE • Rosen §1.1 Ex. 27 • converse, contrapositive, inverse

State the converse, contrapositive and inverse of: (a) I come to class whenever there is going to be a quiz. (b) A positive integer is a prime only if it has no divisors other than 1 and itself.

Solution. (a) p : there is going to be a quiz; q : I come to class. The statement is $p \rightarrow q$.

- Converse $q \rightarrow p$: If I come to class, then there is going to be a quiz.

- Contrapositive $\neg q \rightarrow \neg p$: If I do not come to class, then there is not going to be a quiz.

- Inverse $\neg p \rightarrow \neg q$: If there is not going to be a quiz, then I do not come to class.

(b) p : n is prime; q : n has no divisors other than 1 and itself. “p only if q” is $p \rightarrow q$.

- Converse: If n has no divisors other than 1 and itself, then n is prime.

- Contrapositive: If n has a divisor other than 1 and itself, then n is not prime.

- Inverse: If n is not prime, then n has a divisor other than 1 and itself.

The contrapositive is equivalent to the original. The converse and inverse are equivalent to each other but not to the original (here they fail for $n = 1$, which has no other divisors yet is not

prime).

PRACTICE • Rosen §1.1 Ex. 29–30 • number of rows of a truth table

How many rows does the truth table of each proposition have? (a) $p \rightarrow \neg p$ (b) $(p \vee \neg r) \wedge (q \vee \neg s)$ (c) $q \vee p \vee \neg s \vee \neg r \vee \neg t \vee u$ (d) $(p \wedge r \wedge s) \vee (q \wedge t) \vee (r \wedge \neg t)$

Solution. Rows = 2^n where n is the number of *distinct* propositional variables (repeats do not count).

	variables	n	rows
(a)	p	1	2
(b)	p, q, r, s	4	16
(c)	p, q, r, s, t, u	6	64
(d)	p, q, r, s, t	5	32

PRACTICE • Rosen §1.1 Ex. 33 and 35 • truth tables, two variables

Construct truth tables for (a) $(p \vee q) \rightarrow (p \oplus q)$; (b) $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow q)$; (c) $(p \rightarrow q) \vee (\neg p \rightarrow q)$; (d) $(p \rightarrow q) \wedge (\neg p \rightarrow q)$. What do (b), (c), (d) turn out to be?

Solution.

p	q	$p \vee q$	$p \oplus q$	(a)	$p \leftrightarrow q$	$\neg p \leftrightarrow q$	(b)	$p \rightarrow q$	$\neg p \rightarrow q$	(c)	(d)
T	T	T	F	F	T	F	T	T	T	T	T
T	F	T	T	T	F	T	T	F	T	T	F
F	T	T	T	T	F	T	T	T	T	T	T
F	F	F	F	T	T	F	T	T	F	T	F

(a) is false only when p, q are both true: it is equivalent to $\neg(p \wedge q)$. (b) and (c) are **tautologies**. (d) has the same column as q , so $(p \rightarrow q) \wedge (\neg p \rightarrow q) \equiv q$. (Algebraically: $(\neg p \vee q) \wedge (p \vee q) \equiv q \vee (\neg p \wedge p) \equiv q$.)

PRACTICE • Rosen §1.1 Ex. 37(d) • truth table, three variables

Construct the truth table of $(p \rightarrow q) \wedge (\neg p \rightarrow r)$ and describe what it computes.

Solution. Three variables give $2^3 = 8$ rows.

p	q	r	$p \rightarrow q$	$\neg p \rightarrow r$	$(p \rightarrow q) \wedge (\neg p \rightarrow r)$
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	F	T	F
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	F	F
F	F	T	T	T	T
F	F	F	T	F	F

The last column equals q when p is true and r when p is false: the formula means “if p then q , else r ”.

Predicate Logic

Propositional logic cannot express “every student...” or “there is a number such that...”. Predicate logic adds *variables*, *predicates* and *quantifiers*.

2.1 Predicates and propositional functions

DEFINITION — Predicate / propositional function

A *predicate* is a statement containing one or more variables, such as $P(x)$: “ $x > 3$ ”. It is not true or false by itself — it does not “answer yes or no” — but becomes a proposition once every variable is given a value (or quantified). Because of this, $P(x)$ is also called a *propositional function*. A predicate with n variables is an n -place predicate, written $Q(x, y)$, $R(x, y, z)$, ...

DEFINITION — Domain (universe of discourse)

The *domain* of a variable is the set of values it may take. The truth of a quantified statement depends on the domain, so always state it.

EXAMPLE

$P(x)$: “ $x > 3$ ”. Then $P(5)$ is true and $P(2)$ is false.

$Q(x, y)$: “ $x + y = 5$ ”. Then $Q(2, 3)$ is true and $Q(1, 1)$ is false.

$R(x, y, z)$: “ $x + y = z$ ” is a 3-place predicate; $R(1, 2, 3)$ is true.

2.2 Quantifiers

DEFINITION — Universal quantifier $\forall$

$\forall x P(x)$ reads “for all x , $P(x)$ ”. It is **true** if $P(x)$ is true for *every* x in the domain and **false** if there is at least one x (a *counterexample*) for which $P(x)$ is false.

EXAMPLE

Domain $\mathbb{R}$. Let $P(x)$: $x + 1 > x$. Then $\forall x (x + 1 > x)$ is **true** (it holds for all real x).

Let $S(x)$: $x^2 > x$. Then $\forall x S(x)$ is **false**: $x = \frac{1}{2}$ gives $\frac{1}{4} > \frac{1}{2}$, which is false — a counterexample.

DEFINITION — Existential quantifier $\exists$

$\exists x P(x)$ reads “there exists an x such that $P(x)$ ” (“for some x ”, “for at least one x ”). It is **true** if $P(x)$ is true for *at least one* x in the domain and **false** if $P(x)$ is false for every x .

EXAMPLE

Domain $\mathbb{R}$. Let $Q(x): x \geq 3$. Then $\exists x (x \geq 3)$ is true ($x = 4$ works). But $\exists x (x^2 = -1)$ is false over $\mathbb{R}$: no real number squares to -1 .

WATCH OUT

“True for some” does *not* mean “true for some but not all”. $\exists x P(x)$ is also true when $P(x)$ holds for every x . It is false only when P fails for *all* x .

Statement	True when	False when
$\forall x P(x)$	$P(x)$ true for every x	there is an x with $P(x)$ false
$\exists x P(x)$	there is an x with $P(x)$ true	$P(x)$ false for every x

REMEMBER — finite domains

[†]If the domain is $\{1, 2, 3\}$ then $\forall x P(x) \equiv P(1) \wedge P(2) \wedge P(3)$ and $\exists x P(x) \equiv P(1) \vee P(2) \vee P(3)$. This explains why the De Morgan laws below look like the propositional ones.

2.3 Negation and De Morgan’s laws for quantifiers

DEFINITION — De Morgan’s laws for quantifiers

$$\neg \forall x P(x) \equiv \exists x \neg P(x), \quad \neg \exists x P(x) \equiv \forall x \neg P(x).$$

To negate a quantified statement, push the negation inside, **swapping** $\forall \leftrightarrow \exists$, and negate the predicate.

EXAMPLE

Domain: the students in this class; $P(x)$: “ x has taken a course in Deep Learning (DL)”.

- $\forall x P(x)$: “Every student has taken a DL course.”
 $\neg \forall x P(x) \equiv \exists x \neg P(x)$: “There is at least one student who has *not* taken a DL course.”
- $\exists x P(x)$: “Some student has taken a DL course.”
 $\neg \exists x P(x) \equiv \forall x \neg P(x)$: “No student has taken a DL course.”

WATCH OUT

The negation of “every student took DL” is *not* “no student took DL”. $\neg \forall x P(x)$ says “not all”, which is $\exists x \neg P(x)$, not $\forall x \neg P(x)$.

EXAMPLE — $\neg \forall x (P(x) \rightarrow Q(x)) \equiv \exists x (P(x) \wedge \neg Q(x))$

$$\begin{aligned} \neg \forall x (P(x) \rightarrow Q(x)) &\equiv \exists x \neg (P(x) \rightarrow Q(x)) && \text{(quantifier De Morgan)} \\ &\equiv \exists x \neg (\neg P(x) \vee Q(x)) && \text{(conditional as disjunction)} \\ &\equiv \exists x (P(x) \wedge \neg Q(x)) && \text{(De Morgan, double negation)} \end{aligned}$$

Meaning: “not every P is a Q ” $\equiv$ “some P is not a Q ”.

2.4 Translating English into predicate logic

DEFINITION — Restricting the domain: the $\forall \rightarrow / \exists \wedge$ rule

When the domain is bigger than the group the sentence talks about (say the domain is *all people* but the sentence is about students, with $S(x)$: “ x is a student”):

- “Every S is P ” is written $\forall x (S(x) \rightarrow P(x))$ — use **implication** with $\forall$;
- “Some S is P ” is written $\exists x (S(x) \wedge P(x))$ — use **conjunction** with $\exists$.

If instead the domain is *only* the students, $S(x)$ is not needed: $\forall x P(x)$ and $\exists x P(x)$.

EXAMPLE — Mexico

$P(x)$: “ x has visited Mexico”.

	Domain: students of this class	Domain: all people ($S(x)$: student of this class)
Some student has visited Mexico	$\exists x P(x)$	$\exists x (S(x) \wedge P(x))$
Every student has visited Mexico	$\forall x P(x)$	$\forall x (S(x) \rightarrow P(x))$

WATCH OUT

$\exists x (S(x) \rightarrow P(x))$ is wrong for “some student...” (it is true as soon as one non-student exists, since $\mathbf{F} \rightarrow \cdot$ is true). $\forall x (S(x) \wedge P(x))$ is wrong for “every student...” (it claims *everyone* is a student who visited Mexico).

EXAMPLE — Lions

Domain: all creatures. $L(x)$: x is a lion; $F(x)$: x is fierce; $C(x)$: x drinks coffee.

- All lions are fierce: $\forall x (L(x) \rightarrow F(x))$.
- Some lions do not drink coffee: $\exists x (L(x) \wedge \neg C(x))$.
- Some fierce creatures do not drink coffee: $\exists x (F(x) \wedge \neg C(x))$.

EXAMPLE — Birds

Domain: birds. $H(x)$: hummingbird; $C(x)$: richly coloured; $L(x)$: lives on honey; $S(x)$: small. (“Large” is written $\neg S(x)$.)

- (a) All hummingbirds are richly coloured: $\forall x (H(x) \rightarrow C(x))$.
- (b) No large bird lives on honey: $\forall x (\neg S(x) \rightarrow \neg L(x))$.
- (c) Birds that do not live on honey are dull in colour: $\forall x (\neg L(x) \rightarrow \neg C(x))$.
- (d) Hummingbirds are small: $\forall x (H(x) \rightarrow S(x))$.

2.5 Nested quantifiers

DEFINITION — Nested quantifiers

A *nested* quantified statement has one quantifier inside the scope of another, e.g. $\forall x \exists y (x + y = 5)$. Read left to right: “for every x there is a y (which may depend on x) such that...”. **The order of different quantifiers matters.**

EXAMPLE — order matters

Domain $\mathbb{R}$, $P(x, y)$: $x + y = 5$.

Statement	Meaning	Value
$\forall x \forall y (x + y = 5)$	every pair sums to 5	false ($x = y = 0$)
$\exists x \exists y (x + y = 5)$	some pair sums to 5	true ($2 + 3$)
$\forall x \exists y (x + y = 5)$	each x has a partner y	true ($y = 5 - x$)
$\exists y \forall x (x + y = 5)$	one y works for all x	false

Swapping $\forall x \exists y$ to $\exists y \forall x$ turned a true statement into a false one.

REMEMBER — negating nested quantifiers

Push $\neg$ inward one quantifier at a time, swapping $\forall \leftrightarrow \exists$ at each step, and finally negate the predicate: $\neg \forall x \exists y P \equiv \exists x \forall y \neg P$.

EXAMPLE — Negating

$\neg \forall x \exists y (xy = 1) \equiv \exists x \neg \exists y (xy = 1) \equiv \exists x \forall y \neg (xy = 1) \equiv \exists x \forall y (xy \neq 1)$: “some x has no multiplicative inverse”.

EXAMPLE — Translations from the notes

1. Domain $\mathbb{Z}$: “The sum of two positive integers is always positive”: $\forall x \forall y ((x > 0 \wedge y > 0) \rightarrow x + y > 0)$.
2. Domain $\mathbb{R}$: “Every real number except zero has a multiplicative inverse”: $\forall x (x \neq 0 \rightarrow \exists y (xy = 1))$ (equivalently $\forall x \exists y (x \neq 0 \rightarrow xy = 1)$).
3. Domain: students of this class. $C(x)$: x has a computer; $F(x, y)$: x and y are friends.
 $\forall x (C(x) \vee \exists y (C(y) \wedge F(x, y)))$: “Every student has a computer or has a friend who has a computer.”
4. Same domain: $\exists x \forall y \forall z ((F(x, y) \wedge F(x, z) \wedge y \neq z) \rightarrow \neg F(y, z))$: “There is a student none of whose friends are friends with each other.” (Any two *distinct* friends of this student are not friends.)
5. “There does not exist a woman who has taken a flight on every airline in the world.”
 $P(w, f)$: woman w has taken flight f ; $Q(f, a)$: flight f is on airline a .

$$\begin{aligned} \neg \exists w \forall a \exists f (P(w, f) \wedge Q(f, a)) &\equiv \forall w \exists a \forall f \neg (P(w, f) \wedge Q(f, a)) \\ &\equiv \forall w \exists a \forall f (P(w, f) \rightarrow \neg Q(f, a)). \end{aligned}$$

“For every woman there is an airline on which she has taken no flight.”

2.6 Free and bound variables[†]

DEFINITION — Scope, bound and free variables

The *scope* of a quantifier $\forall x$ or $\exists x$ is the formula it applies to (usually the bracket right after it). An occurrence of a variable is **bound** if it lies in the scope of a quantifier that names it (or is the quantifier’s own variable); otherwise it is **free**. A formula with *no free variables* is a proposition; a formula with free variables is still a predicate.

EXAMPLE

Formula	Bound / free	Proposition?
$P(x)$	x free	no
$\forall x (x + y > 3)$	x bound, y free	no (truth depends on y)
$\forall x \exists y (x + y = 0)$ over $\mathbb{Z}$	x, y bound	yes (true)
$(\exists x P(x)) \wedge Q(x)$	first x bound, second x free	no

Bound variables are dummies: $\forall x P(x)$ and $\forall y P(y)$ mean the same.

2.7 Domains inside the quantifier, and truth sets[†]

DEFINITION — Restricted quantifiers

Writing the domain inside the quantifier is shorthand:

$$\forall x \in D P(x) \text{ means } \forall x (x \in D \rightarrow P(x)), \quad \exists x \in D P(x) \text{ means } \exists x (x \in D \wedge P(x)).$$

(This is the same $\forall \rightarrow$ and $\exists \wedge$ pattern as in the translation rule of Section 2.4.)

EXAMPLE

$\forall x \in \mathbb{R} (x^2 \geq 0)$ says “the square of every real number is non-negative” (true). $\exists x \in \mathbb{Z} (x^2 = 1)$ says “some integer has square 1” (true, $x = \pm 1$).

DEFINITION — Truth set

The *truth set* of $P(x)$ over the domain D is $\{x \in D \mid P(x) \text{ is true}\}$. Then $\forall x P(x)$ is true exactly when the truth set is all of D , and $\exists x P(x)$ is true exactly when the truth set is non-empty.

EXAMPLE

Domain $\mathbb{Z}$. $P(x): |x| = 1$ has truth set $\{-1, 1\}$. $Q(x): x^2 = 2$ has truth set $\emptyset$ (no integer has square 2), so $\exists x Q(x)$ is false. $R(x): |x| = x$ has truth set $\{0, 1, 2, \dots\}$, so $\exists x R(x)$ is true but $\forall x R(x)$ is false ($x = -3$).

2.8 The uniqueness quantifier[†]

DEFINITION — $\exists!$

$\exists! x P(x)$ means “there is *exactly one* x with $P(x)$ ”. It can be written

$$\exists x (P(x) \wedge \forall y (P(y) \rightarrow y = x)) :$$

some x satisfies P , and anything satisfying P is equal to it. In terms of truth sets: the truth set has exactly one element.

EXAMPLE

Over $\mathbb{R}$, $\exists! x (2x + 1 = 7)$ is true ($x = 3$ only). Over $\mathbb{R}$, $\exists! x (x^2 = 4)$ is false (both 2 and -2 work), but over the positive integers it is true. Over $\mathbb{R}$, $\exists! x (x^2 = -1)$ is false: no x at all.

2.9 Solved previous-year questions

PYQ • Mid-Sem Oct 2023 Q1(b) • 3 marks

Define predicates, quantifiers, free and bound variables. Give suitable examples.

Solution. Predicate. A statement with variables, e.g. $P(x): “x > 3”$, that becomes a proposition when the variables are given values. $P(5)$ is true and $P(2)$ is false. (It is also called a propositional function; the set of allowed values of x is its domain.)

Quantifiers. They turn a predicate into a proposition by saying *how many* values of x satisfy it.

- Universal $\forall x P(x)$: $P(x)$ holds for *every* x in the domain. E.g. over $\mathbb{R}$, $\forall x (x^2 \geq 0)$ is true.
- Existential $\exists x P(x)$: $P(x)$ holds for *at least one* x . E.g. over $\mathbb{Z}$, $\exists x (x^2 = 9)$ is true ($x = 3$).

Bound and free variables. A variable occurrence in the scope of a quantifier that names it is *bound*; otherwise it is *free*. In $\forall x (x + y > 3)$ the variable x is bound and y is free, so this is still a predicate in y . In $\forall x \exists y (x + y = 0)$ every variable is bound, so it is a proposition. In $(\exists x P(x)) \wedge Q(x)$ the first x is bound and the second is free.

PYQ • Mid-Sem (undated, CS5101) Q1(c) • 4 marks

Let $L(x, y)$ be “ x loves y ”, the domain for both variables being all people. Use quantifiers to express: (i) There are exactly two people whom Jerry loves. (ii) There is somebody whom Lydia does not love.

Solution. (i) “Exactly two” means *at least* two distinct people and *no one else*:

$$\exists x \exists y (x \neq y \wedge L(\text{Jerry}, x) \wedge L(\text{Jerry}, y) \wedge \forall z (L(\text{Jerry}, z) \rightarrow (z = x \vee z = y))).$$

The first three conjuncts give two different people Jerry loves; the last says anyone Jerry loves is one of them.

(ii) $\exists x \neg L(\text{Lydia}, x)$, which is the same as $\neg \forall x L(\text{Lydia}, x)$ (“it is not the case that Lydia loves everybody”).

PYQ • Mid-Sem (undated, CS5101) Q2(c) • 4 marks

Find the negations of these statements using De Morgan’s laws. (i) If it snows today, then I will go skiing tomorrow. (ii) In every mathematics class there is some student who falls asleep during lectures.

Solution. (i) Let p : “it snows today”, q : “I will go skiing tomorrow”. The statement is $p \rightarrow q \equiv \neg p \vee q$. Then

$$\neg(p \rightarrow q) \equiv \neg(\neg p \vee q) \equiv \neg\neg p \wedge \neg q \equiv p \wedge \neg q.$$

Negation: “It snows today **and** I will **not** go skiing tomorrow.”

(ii) Let $M(c)$: “ c is a mathematics class”; $T(s, c)$: “ s is a student in class c ”; $A(s)$: “ s falls asleep during lectures”. The statement is

$$\forall c (M(c) \rightarrow \exists s (T(s, c) \wedge A(s))).$$

Negate, swapping quantifiers:

$$\begin{aligned} \neg \forall c (M(c) \rightarrow \exists s (T(s, c) \wedge A(s))) &\equiv \exists c (M(c) \wedge \neg \exists s (T(s, c) \wedge A(s))) \\ &\equiv \exists c (M(c) \wedge \forall s (\neg T(s, c) \vee \neg A(s))) \\ &\equiv \exists c (M(c) \wedge \forall s (T(s, c) \rightarrow \neg A(s))). \end{aligned}$$

Negation: “There is a mathematics class in which **no** student falls asleep during lectures.”

PYQ • Mid-Sem Sep 2024 Q1(c) • 5 marks

Express each statement using mathematical and logical operators, predicates and quantifiers, where the domain consists of all integers. (i) “The difference of two positive integers is not

necessarily positive.” (ii) “The sum of the squares of two integers is greater than or equal to the square of their sum.”

Solution. (i) The full claim “the difference of two positive integers is positive” is $\forall x \forall y ((x > 0 \wedge y > 0) \rightarrow x - y > 0)$. “Not necessarily” negates it:

$$\neg \forall x \forall y ((x > 0 \wedge y > 0) \rightarrow x - y > 0) \equiv \exists x \exists y ((x > 0 \wedge y > 0) \wedge x - y \leq 0).$$

Witness: $x = 1, y = 2$ gives $x - y = -1 \leq 0$.

(ii) $\forall x \forall y (x^2 + y^2 \geq (x + y)^2)$.

Remark. The question only asks for the translation, but note the statement is false: $(x + y)^2 - (x^2 + y^2) = 2xy$, so it holds only when $xy \leq 0$; $x = y = 1$ gives $2 \geq 4$, which is false.

PYQ • Mid-Sem Sep 2024 Q2(b) • 3 marks

The domain consists of persons. Express in terms of propositional functions, quantifiers and logical connectives: “A student must take at least 60 course hours, or at least 45 course hours and write a master’s thesis, and receive a grade no lower than a B in all required courses, to receive a master’s degree.”

Solution. Define predicates on a person x :

$S(x)$: x is a student; $D(x)$: x receives a master’s degree; $H_{60}(x)$: x takes at least 60 course hours; $H_{45}(x)$: x takes at least 45 course hours; $T(x)$: x writes a master’s thesis; $B(x)$: x receives a grade no lower than B in all required courses.

The wording “must ... to receive a degree” states *necessary conditions*: receiving the degree implies the requirements. The requirements are $[H_{60} \vee (H_{45} \wedge T)] \wedge B$. Hence

$$\forall x ((S(x) \wedge D(x)) \rightarrow [H_{60}(x) \vee (H_{45}(x) \wedge T(x))] \wedge B(x)).$$

PYQ • Mid-Sem Oct 2023 Q3(b) • 6 marks

Using the universe of integers for all quantifiers, express the English sentences in predicate logic using $P(x)$: “ x is prime”, $E(x)$: “ x is even” and $L(x, y)$: $x < y$. Replace $*$ with the appropriate entry.

(i) “No prime greater than 2 is even” is $\forall x (* \wedge *) \rightarrow *$. (ii) “For every number, there is a larger prime number” is $\forall x (* \wedge *)$. (iii) Translate $\forall x ((E(x) \wedge P(x)) \rightarrow L(x, 100))$ into English.

Solution. (i) “ x is a prime greater than 2” is $P(x) \wedge L(2, x)$ (since $L(2, x)$ means $2 < x$). The conclusion “is not even” is $\neg E(x)$:

$$\forall x (P(x) \wedge L(2, x)) \rightarrow \neg E(x) \quad \text{i.e. } \forall x ((P(x) \wedge L(2, x)) \rightarrow \neg E(x)).$$

Entries: $*$ = $P(x), L(2, x), \neg E(x)$.

(ii) For each x there *exists* a y that is prime and larger than x :

$$\forall x \exists y (P(y) \wedge L(x, y)).$$

Entries: $*$ = $\exists y, P(y), L(x, y)$.

(iii) $E(x) \wedge P(x)$ means “ x is an even prime”; $L(x, 100)$ means $x < 100$. So the expression says: “Every even prime number is less than 100.” (True, because the only even prime is 2.)

2.10 Practice questions from Rosen

The trimmed Rosen text has only a few quantifier exercises (Sections 2.1 and 1.8); they are extra practice.

PRACTICE • Rosen §2.1 Ex. 41–42 • quantified statements over $\mathbb{R}$ and $\mathbb{Z}$

Translate into English and determine the truth value: (a) $\forall x \in \mathbb{R} (x^2 = -1)$ (b) $\exists x \in \mathbb{Z} (x^2 = 2)$ (c) $\forall x \in \mathbb{Z} (x^2 > 0)$ (d) $\exists x \in \mathbb{R} (x^2 = x)$ (e) $\exists x \in \mathbb{R} (x^3 = -1)$ (f) $\forall x \in \mathbb{Z} (x - 1 \in \mathbb{Z})$.

Solution. A $\forall$ statement is refuted by one counterexample; an $\exists$ statement is proved by one witness.

	In English		Reason
(a)	The square of every real number is -1 .	F	$0^2 = 0 \neq -1$ (indeed $x^2 \geq 0$)
(b)	Some integer has square 2.	F	only $\pm\sqrt{2}$, not integers
(c)	The square of every integer is positive.	F	counterexample $x = 0$
(d)	Some real number equals its own square.	T	witnesses $x = 0$ and $x = 1$
(e)	Some real number has cube -1 .	T	witness $x = -1$
(f)	Subtracting 1 from any integer gives an integer.	T	$\mathbb{Z}$ is closed under subtraction

PRACTICE • Rosen §2.1 Ex. 43 • truth sets over the integers

Find the truth set of each predicate when the domain is $\mathbb{Z}$: (a) $P(x): x^2 < 3$ (b) $Q(x): x^2 > x$ (c) $R(x): 2x + 1 = 0$.

Solution. (a) $x^2 < 3$ means $|x| < \sqrt{3} \approx 1.73$, so $x \in \{-1, 0, 1\}$. Truth set $\{-1, 0, 1\}$.
 (b) $x^2 > x \iff x(x-1) > 0 \iff x < 0$ or $x > 1$. Over $\mathbb{Z}$: $\{\dots, -3, -2, -1\} \cup \{2, 3, 4, \dots\}$.
 (Check: $x = 0$ and $x = 1$ fail because $0 > 0$ and $1 > 1$ are false.)
 (c) $2x + 1 = 0$ gives $x = -\frac{1}{2} \notin \mathbb{Z}$. Truth set $\emptyset$; so $\exists x R(x)$ is false over $\mathbb{Z}$ (though true over $\mathbb{R}$).

PRACTICE • Rosen §1.8 Ex. 15 • uniqueness quantifier

Show that each of these expresses “there is a unique x such that $P(x)$ ”:

- (a) $\exists x \forall y (P(y) \leftrightarrow x = y)$
- (b) $\exists x P(x) \wedge \forall x \forall y ((P(x) \wedge P(y)) \rightarrow x = y)$
- (c) $\exists x (P(x) \wedge \forall y (P(y) \rightarrow x = y))$

Solution. Let T be the truth set of P ; “unique” means $|T| = 1$. Each formula says exactly that:
 (c) Some x_0 has $P(x_0)$ and every y with $P(y)$ equals x_0 , so $T = \{x_0\}$; and $T = \{x_0\}$ makes (c) true.
 (b) The first conjunct says $T \neq \emptyset$ (at least one), the second that any two members of T are equal (at most one).
 (a) It says $P(y)$ holds exactly when $y = x_0$, i.e. $T = \{x_0\}$, and conversely.

Rules of Inference

3.1 Arguments and validity

DEFINITION — Argument, premises, conclusion

An *argument* is a list of propositions $p_1, p_2, \dots, p_n$ called the *premises* (hypotheses), followed by a proposition q called the *conclusion*. It is written

$$\begin{array}{c} p_1 \\ p_2 \\ \vdots \\ p_n \\ \hline \therefore q \end{array}$$

DEFINITION — Valid argument

An argument is *valid* if the conclusion is true **whenever all the premises are true**; equivalently, $(p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow q$ is a **tautology**. Validity is a property of the *form* of the argument, not of whether the sentences happen to be true.

EXAMPLE — from the notes

"If you have the current password then you can log on to the network." $p \rightarrow q$

"You have the current password." p

"Therefore you can log on to the network." $\therefore q$

Validity: $((p \rightarrow q) \wedge p) \rightarrow q$ is a tautology. Only one row makes both premises true ($p = q = \text{T}$) and there the conclusion q is true.

DEFINITION — Sound argument

[†]An argument is *sound* if it is valid **and** all its premises are actually true. A valid argument can still have a false conclusion if a premise is false.

EXAMPLE — valid but not sound

"If $\sqrt{2} > \frac{3}{2}$ then $(\sqrt{2})^2 > (\frac{3}{2})^2$. $\sqrt{2} > \frac{3}{2}$. Therefore $2 > \frac{9}{4}$." The form is modus ponens (valid), but the second premise is false ($\sqrt{2} \approx 1.414 < 1.5$), and the conclusion $2 > 2.25$ is false.

3.2 Rules of inference for propositional logic

Each rule is a valid argument form (its associated conditional is a tautology). Rules marked [†] are standard but not written in your notes.

Rule	Form	Tautology
Modus ponens	$p, p \rightarrow q \therefore q$	$(p \wedge (p \rightarrow q)) \rightarrow q$
Modus tollens	$\neg q, p \rightarrow q \therefore \neg p$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$
Hypothetical syllogism	$p \rightarrow q, q \rightarrow r \therefore p \rightarrow r$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$
Addition	$p \therefore p \vee q$	$p \rightarrow (p \vee q)$
Simplification	$p \wedge q \therefore p$	$(p \wedge q) \rightarrow p$
Conjunction [†]	$p, q \therefore p \wedge q$	$((p) \wedge (q)) \rightarrow (p \wedge q)$
Disjunctive syllogism [†]	$p \vee q, \neg p \therefore q$	$((p \vee q) \wedge \neg p) \rightarrow q$
Resolution [†]	$p \vee q, \neg p \vee r \therefore q \vee r$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$

EXAMPLE — one example per rule

- **Modus ponens.** “If it snows today we will go skiing. It is snowing today. $\therefore$ We will go skiing.”
- **Modus tollens.** “If it snows today we will go skiing. We did not go skiing. $\therefore$ It did not snow today.”
- **Hypothetical syllogism.** “If it rains today then we will not have a barbecue today [$p \rightarrow \neg q$]. If we do not have a barbecue today then we will have one tomorrow [$\neg q \rightarrow r$]. $\therefore$ If it rains today then we will have a barbecue tomorrow [$p \rightarrow r$].”
- **Addition.** “It is below freezing now. $\therefore$ It is below freezing or raining now.”
- **Simplification.** “It is cold and it is raining. $\therefore$ It is cold.”
- **Disjunctive syllogism.**[†] “The prize is in box A or box B. It is not in box A. $\therefore$ It is in box B.”

3.3 Building a proof from premises

A *derivation* lists numbered steps; each step is a premise or follows from earlier steps by a named rule.

EXAMPLE — from the notes: swimming and canoeing

Premises: “If it is not sunny this afternoon *and* it is colder than yesterday” ($\neg s \wedge c$); “we will go swimming only if it is sunny” ($r \rightarrow s$); “if we do not go swimming, then we will take a canoe trip” ($\neg r \rightarrow t$); “if we take a canoe trip, then we will be home by sunset” ($t \rightarrow h$). Conclusion: “we will be home by sunset” (h).

#	Step	Reason
1	$\neg s \wedge c$	premise
2	$\neg s$	simplification, 1
3	$r \rightarrow s$	premise
4	$\neg r$	modus tollens, 2 and 3
5	$\neg r \rightarrow t$	premise
6	t	modus ponens, 4 and 5
7	$t \rightarrow h$	premise
8	h	modus ponens, 6 and 7

So the argument is valid.

3.4 Fallacies

DEFINITION — Fallacy

A *fallacy* is an incorrect argument that looks like a valid rule but whose associated conditional is *not* a tautology.

DEFINITION — Fallacy of affirming the conclusion (converse error)

$$\frac{p \rightarrow q \quad q}{\therefore p}$$

The associated conditional $((p \rightarrow q) \wedge q) \rightarrow p$ is **not** a tautology: take $p = \text{F}$, $q = \text{T}$. Then $p \rightarrow q = \text{T}$ and $q = \text{T}$ (both premises true) but the conclusion $p = \text{F}$.

EXAMPLE — from the notes

“If you do every problem in the book, then you will learn discrete math. You learned discrete math. $\therefore$ You did every problem in this book.” This is a fallacy: you may have learned the subject another way ($p = \text{F}, q = \text{T}$).

DEFINITION — Fallacy of denying the hypothesis (inverse error)

†

$$\frac{p \rightarrow q \quad \neg p}{\therefore \neg q}$$

Invalid: $p = \text{F}$, $q = \text{T}$ makes both premises true and the conclusion $\neg q$ false.

3.5 Rules of inference for quantified statements

Rule	Form	Remark
Universal instantiation (UI)	$\forall x P(x) \therefore P(c)$	c is any element of the domain
Universal generalization (UG)	$P(c)$ for arbitrary $c \therefore \forall x P(x)$	c must be <i>arbitrary</i> , not a specially chosen element
Existential instantiation (EI)	$\exists x P(x) \therefore P(c)$ for some element c	c is a <i>new</i> name, not used earlier
Existential generalization (EG)	$P(c)$ for some element $c \therefore \exists x P(x)$	

EXAMPLE — from the notes: Sita

“Everyone in the class CS 5101 has taken a course in computer science. Sita is a student of this class. $\therefore$ Sita has taken a course in computer science.”

Let $P(x)$: x is in CS 5101; $Q(x)$: x has taken a CS course.

#	Step	Reason
1	$\forall x (P(x) \rightarrow Q(x))$	premise
2	$P(\text{Sita}) \rightarrow Q(\text{Sita})$	universal instantiation, 1
3	$P(\text{Sita})$	premise
4	$Q(\text{Sita})$	modus ponens, 2 and 3

EXAMPLE — from the notes: the book and the exam

Premises: “A student in this class has not read the book” and “Every student in this class passed the exam”. Conclusion: “Someone who passed the exam has not read the book.”

$P(x)$: x is in this class; $Q(x)$: x has read the book; $R(x)$: x passed the exam.

#	Step	Reason
1	$\exists x (P(x) \wedge \neg Q(x))$	premise
2	$P(c) \wedge \neg Q(c)$	existential instantiation, 1
3	$P(c)$	simplification, 2
4	$\neg Q(c)$	simplification, 2
5	$\forall x (P(x) \rightarrow R(x))$	premise
6	$P(c) \rightarrow R(c)$	universal instantiation, 5
7	$R(c)$	modus ponens, 3 and 6
8	$R(c) \wedge \neg Q(c)$	conjunction, 7 and 4
9	$\exists x (R(x) \wedge \neg Q(x))$	existential generalization, 8

Do EI (step 2) *before* UI, so that c is the specific element that the existential premise supplies.

3.6 Solved previous-year questions

PYQ • Mid-Sem (undated, CS 5101) Q2(b) • 2 marks

What rules of inference are used in this famous argument? “All men are mortal. Socrates is a man. Therefore, Socrates is mortal.”

Solution. Let $M(x)$: x is a man; $D(x)$: x is mortal.

#	Step	Reason
1	$\forall x (M(x) \rightarrow D(x))$	premise (“All men are mortal”)
2	$M(\text{Socrates}) \rightarrow D(\text{Socrates})$	universal instantiation , 1
3	$M(\text{Socrates})$	premise (“Socrates is a man”)
4	$D(\text{Socrates})$	modus ponens , 2 and 3

The rules used are **universal instantiation** and **modus ponens**.

PYQ • Mid-Sem Sep 2024 Q3(a) • 4 marks

Determine whether these are valid arguments.

(i) If x is a positive real number, then x^2 is a positive real number. Therefore, if a^2 is positive, where a is a real number, then a is a positive real number.

(ii) If $x^2 \neq 0$, where x is a real number, then $x = 0$. Let a be a real number with $a^2 = 0$; then $a = 0$.

Solution. (i) Not valid. Let $p(x)$: x is a positive real; $q(x)$: x^2 is a positive real. The premise is $\forall x (p(x) \rightarrow q(x))$; the conclusion is $\forall a (q(a) \rightarrow p(a))$, the *converse*. This is the fallacy of **affirming the conclusion**: $p \rightarrow q \therefore q \rightarrow p$.

Counterexample: $a = -2$. Then $a^2 = 4$ is positive (the hypothesis of the conclusion is true) but a is not positive, while the premise “if $x > 0$ then $x^2 > 0$ ” is true. So the premise is true and the conclusion is false.

(ii) Not valid, as printed. Let $P(x)$: $x^2 \neq 0$ and $Q(x)$: $x = 0$. The premise is $\forall x (P(x) \rightarrow Q(x))$. The second statement $a^2 = 0$ says $\neg P(a)$, and the conclusion is $Q(a)$: $a = 0$. The pattern

$$P(a) \rightarrow Q(a), \quad \neg P(a) \therefore Q(a)$$

is the fallacy of **denying the hypothesis**: with $P(a) = Q(a) = \text{F}$ both premises are true but the conclusion is false. (Also the first premise is false for real numbers: $x = 2$ has $x^2 \neq 0$ but $x \neq 0$; the conclusion $a = 0$ happens to be true, but it does not follow from these premises by a valid rule.)

Note. If the intended statement was the textbook version “If $x^2 \neq 0$ then $x \neq 0$. Let a have $a^2 \neq 0$; then $a \neq 0$ ”, it is **valid**: universal instantiation gives $a^2 \neq 0 \rightarrow a \neq 0$, and modus ponens with $a^2 \neq 0$ gives $a \neq 0$.

3.7 Practice questions from Rosen

These come from the exercises on rules of inference in Rosen (Section 1.6). They extend the previous-year questions with the two things that are asked most often: *spotting the error in a bad derivation* and *building a derivation with the quantifier rules*.

PRACTICE • Rosen §1.6 Ex. 20 • valid or not?

Determine whether these arguments are valid. (a) If x is a positive real number, then x^2 is a positive real number. Therefore, if a^2 is positive, where a is a real number, then a is a positive real number. (b) If $x^2 = 0$, where x is a real number, then $x = 0$. Let a be a real number with $a^2 = 0$; then $a = 0$.

Solution. (a) Invalid. With $P(x)$: “ x is positive” and $Q(x)$: “ x^2 is positive”, the premise is $\forall x (P(x) \rightarrow Q(x))$ and the conclusion is $Q(a) \rightarrow P(a)$, the *converse*. This is the fallacy of affirming the conclusion. Counterexample: $a = -2$; $a^2 = 4 > 0$ but a is not positive.

(b) Valid. Premises $\forall x (x^2 = 0 \rightarrow x = 0)$ and $a^2 = 0$.

- | | | |
|---|---|--------------------------------|
| 1 | $\forall x (x^2 = 0 \rightarrow x = 0)$ | premise |
| 2 | $a^2 = 0 \rightarrow a = 0$ | universal instantiation from 1 |
| 3 | $a^2 = 0$ | premise |
| 4 | $a = 0$ | modus ponens from 2, 3 |

PRACTICE • Rosen §1.6 Ex. 23 • find the error (existential instantiation)

Identify the error in this “proof” that $\exists x P(x) \wedge \exists x Q(x)$ implies $\exists x (P(x) \wedge Q(x))$.

1. $\exists x P(x) \wedge \exists x Q(x)$ premise; 2. $\exists x P(x)$ simplification from 1; 3. $P(c)$ existential instantiation from 2; 4. $\exists x Q(x)$ simplification from 1; 5. $Q(c)$ existential instantiation from 4; 6. $P(c) \wedge Q(c)$ conjunction from 3, 5; 7. $\exists x (P(x) \wedge Q(x))$ existential generalization.

Solution. The error is **step 5**. Existential instantiation must introduce a *new* name that has not been used before. In step 3, c was already chosen as an element with $P(c)$; nothing guarantees that the same element also satisfies Q . Step 5 should use a different name d , and then step 6 would give $P(c) \wedge Q(d)$, from which no $\exists x (P(x) \wedge Q(x))$ follows.

The conclusion is in fact not a valid consequence: over the integers take $P(x)$: “ x is even” and $Q(x)$: “ x is odd”. Then $\exists x P(x) \wedge \exists x Q(x)$ is true but $\exists x (P(x) \wedge Q(x))$ is false.

PRACTICE • Rosen §1.6 Ex. 24 • find the error (universal generalization)

Identify the errors in this “proof” that $\forall x (P(x) \vee Q(x))$ implies $\forall x P(x) \vee \forall x Q(x)$.

1. $\forall x (P(x) \vee Q(x))$ premise; 2. $P(c) \vee Q(c)$ universal instantiation; 3. $P(c)$ simplification from 2; 4. $\forall x P(x)$ universal generalization from 3; 5. $Q(c)$ simplification from 2; 6. $\forall x Q(x)$ universal generalization from 5; 7. $\forall x P(x) \wedge \forall x Q(x)$ conjunction from 4, 6.

Solution.

- **Steps 3 and 5:** simplification applies to a *conjunction* $p \wedge q$. Step 2 is a *disjunction*; from $P(c) \vee Q(c)$ we cannot conclude $P(c)$, nor $Q(c)$.
- **Steps 4 and 6:** universal generalization needs a statement that holds for an *arbitrary* element; step 3 was not derived for arbitrary c .
- **Step 7:** even if the steps were right they would prove $\forall x P(x) \wedge \forall x Q(x)$, not the claimed $\vee$.

The claim itself is false: over the integers with $P(x)$: “ x even” and $Q(x)$: “ x odd”, $\forall x (P(x) \vee Q(x))$ is true while $\forall x P(x)$ and $\forall x Q(x)$ are both false.

PRACTICE • Rosen §1.6 Ex. 25–26 • universal modus tollens and universal transitivity

(a) Show that $\forall x (P(x) \rightarrow Q(x))$ and $\neg Q(a)$ (for a particular a in the domain) imply $\neg P(a)$. (b) Show that $\forall x (P(x) \rightarrow Q(x))$ and $\forall x (Q(x) \rightarrow R(x))$ imply $\forall x (P(x) \rightarrow R(x))$.

Solution. (a)

- | | | |
|---|-------------------------------------|--------------------------------|
| 1 | $\forall x (P(x) \rightarrow Q(x))$ | premise |
| 2 | $P(a) \rightarrow Q(a)$ | universal instantiation from 1 |
| 3 | $\neg Q(a)$ | premise |
| 4 | $\neg P(a)$ | modus tollens from 2, 3 |

(b) Let a be an *arbitrary* element of the domain.

- | | | |
|---|-------------------------------------|--|
| 1 | $\forall x (P(x) \rightarrow Q(x))$ | premise |
| 2 | $\forall x (Q(x) \rightarrow R(x))$ | premise |
| 3 | $P(a) \rightarrow Q(a)$ | universal instantiation from 1 |
| 4 | $Q(a) \rightarrow R(a)$ | universal instantiation from 2 |
| 5 | $P(a) \rightarrow R(a)$ | hypothetical syllogism from 3, 4 |
| 6 | $\forall x (P(x) \rightarrow R(x))$ | universal generalization from 5 (a arbitrary) |

PRACTICE • Rosen §1.6 Ex. 27–28 • derivations with $\forall$

Use rules of inference to show: (a) if $\forall x (P(x) \rightarrow (Q(x) \wedge S(x)))$ and $\forall x (P(x) \wedge R(x))$ are true, then $\forall x (R(x) \wedge S(x))$ is true. (b) if $\forall x (P(x) \vee Q(x))$ and $\forall x ((\neg P(x) \wedge Q(x)) \rightarrow R(x))$ are true, then $\forall x (\neg R(x) \rightarrow P(x))$ is true.

Solution. (a) Let a be arbitrary.

- | | | |
|----|---|---------------------------------|
| 1 | $\forall x (P(x) \wedge R(x))$ | premise |
| 2 | $P(a) \wedge R(a)$ | universal instantiation from 1 |
| 3 | $P(a)$ | simplification from 2 |
| 4 | $R(a)$ | simplification from 2 |
| 5 | $\forall x (P(x) \rightarrow (Q(x) \wedge S(x)))$ | premise |
| 6 | $P(a) \rightarrow (Q(a) \wedge S(a))$ | universal instantiation from 5 |
| 7 | $Q(a) \wedge S(a)$ | modus ponens from 3, 6 |
| 8 | $S(a)$ | simplification from 7 |
| 9 | $R(a) \wedge S(a)$ | conjunction from 4, 8 |
| 10 | $\forall x (R(x) \wedge S(x))$ | universal generalization from 9 |

(b) Let a be arbitrary. We need $\neg R(a) \rightarrow P(a)$, which is equivalent to $R(a) \vee P(a)$.

- | | | |
|---|--|--|
| 1 | $P(a) \vee Q(a)$ | universal instantiation from the first premise |
| 2 | $(\neg P(a) \wedge Q(a)) \rightarrow R(a)$ | universal instantiation from the second premise |
| 3 | $P(a) \vee \neg Q(a) \vee R(a)$ | from 2: $p \rightarrow r \equiv \neg p \vee r$ and De Morgan |
| 4 | $P(a) \vee R(a)$ | resolution from 1, 3 (cancel $Q(a), \neg Q(a)$) |
| 5 | $\neg R(a) \rightarrow P(a)$ | equivalent to 4 |
| 6 | $\forall x (\neg R(x) \rightarrow P(x))$ | universal generalization from 5 |

PRACTICE • Rosen §1.6 Ex. 29 • derivation with $\exists$

Show that if $\forall x (P(x) \vee Q(x))$, $\forall x (\neg Q(x) \vee S(x))$, $\forall x (R(x) \rightarrow \neg S(x))$ and $\exists x \neg P(x)$ are true, then $\exists x \neg R(x)$ is true.

Solution. Apply existential instantiation *first*, so that the name c is new.

1	$\exists x \neg P(x)$	premise
2	$\neg P(c)$	existential instantiation from 1
3	$\forall x (P(x) \vee Q(x))$	premise
4	$P(c) \vee Q(c)$	universal instantiation from 3
5	$Q(c)$	disjunctive syllogism from 2, 4
6	$\forall x (\neg Q(x) \vee S(x))$	premise
7	$\neg Q(c) \vee S(c)$	universal instantiation from 6
8	$S(c)$	disjunctive syllogism from 5, 7
9	$\forall x (R(x) \rightarrow \neg S(x))$	premise
10	$R(c) \rightarrow \neg S(c)$	universal instantiation from 9
11	$\neg R(c)$	modus tollens from 10, 8
12	$\exists x \neg R(x)$	existential generalization from 11

PRACTICE • Rosen §1.6 Ex. 30–31 • resolution

Use resolution to show: (a) “Allen is a bad boy or Hillary is a good girl” and “Allen is a good boy or David is happy” imply “Hillary is a good girl or David is happy”. (b) “It is not raining or Yvette has her umbrella”, “Yvette does not have her umbrella or she does not get wet” and “It is raining or Yvette does not get wet” imply “Yvette does not get wet”.

Solution. Resolution: from $p \vee q$ and $\neg p \vee r$ conclude $q \vee r$.

(a) p : Allen is a bad boy; q : Hillary is a good girl; r : David is happy. (“Allen is a good boy” is $\neg p$.) The premises are $p \vee q$ and $\neg p \vee r$. Resolving on p : $q \vee r$.

(b) p : it is raining; q : Yvette has her umbrella; r : Yvette gets wet. Premises: (1) $\neg p \vee q$, (2) $\neg q \vee \neg r$, (3) $p \vee \neg r$.

- Resolve (1) and (2) on q : $\neg p \vee \neg r$.
- Resolve this with (3) on p : $\neg r \vee \neg r = \neg r$: Yvette does not get wet. ■

PRACTICE • Rosen §1.6 Ex. 33 • resolution and satisfiability

Use resolution to show that $(p \vee q) \wedge (\neg p \vee q) \wedge (p \vee \neg q) \wedge (\neg p \vee \neg q)$ is not satisfiable.

Solution.

- Resolve $p \vee q$ and $\neg p \vee q$ on p : q .
- Resolve $p \vee \neg q$ and $\neg p \vee \neg q$ on p : $\neg q$.
- Resolve q and $\neg q$: the empty clause, i.e. a contradiction **F**.

Each resolution step is valid, so whenever the four clauses are all true, **F** would be true, which is impossible. Hence no assignment satisfies the formula. (Check by truth table: each clause is false in exactly one of the four rows (p, q) , and the four clauses rule out four different rows.)

PRACTICE • Rosen §1.6 Ex. 34 • translate, then test validity

Assumptions: (1) “Logic is difficult or not many students like logic.” (2) “If mathematics is easy, then logic is not difficult.” Which of these are valid conclusions? (a) Mathematics is not easy, if many students like logic. (b) Not many students like logic, if mathematics is not easy. (c) Mathematics is not easy or logic is difficult. (d) Logic is not difficult or mathematics is not easy. (e) If not many students like logic, then either mathematics is not easy or logic is not difficult.

Solution. Let L : logic is difficult; S : many students like logic; M : mathematics is easy. Premises: (1) $L \vee \neg S$, (2) $M \rightarrow \neg L$ (equivalently $\neg M \vee \neg L$).

(a) $S \rightarrow \neg M$: **valid**. If S holds, (1) forces L (disjunctive syllogism); with (2) and L , modus tollens gives $\neg M$.

(b) $\neg M \rightarrow \neg S$: **invalid**. Take $M = \text{F}$, $S = \text{T}$, $L = \text{T}$: (1) $\text{T} \vee \text{F} = \text{T}$, (2) $\text{F} \rightarrow \text{F} = \text{T}$, but the conclusion is $\text{T} \rightarrow \text{F} = \text{F}$.

(c) $\neg M \vee L$: **invalid**. Take $M = \text{T}$, $L = \text{F}$, $S = \text{F}$: (1) $\text{F} \vee \text{T} = \text{T}$, (2) $\text{T} \rightarrow \text{T} = \text{T}$, but $\neg M \vee L = \text{F} \vee \text{F} = \text{F}$.

(d) $\neg L \vee \neg M$: **valid**. It is premise (2) rewritten.

(e) $\neg S \rightarrow (\neg M \vee \neg L)$: **valid**. The consequent is premise (2) again, so the implication is true whatever $\neg S$ is.

PRACTICE • Rosen §1.6 Ex. 35 • a long argument in words

Is this argument valid? “If Superman were able and willing to prevent evil, he would do so. If Superman were unable to prevent evil, he would be impotent; if he were unwilling to prevent evil, he would be malevolent. Superman does not prevent evil. If Superman exists, he is neither impotent nor malevolent. Therefore, Superman does not exist.”

Solution. Let A : able; W : willing; P : prevents evil; I : impotent; M : malevolent; E : exists. Premises:

$$(A \wedge W) \rightarrow P, \quad (\neg A \rightarrow I) \wedge (\neg W \rightarrow M), \quad \neg P, \quad E \rightarrow (\neg I \wedge \neg M).$$

- | | | |
|---|--|--|
| 1 | $\neg(A \wedge W)$, i.e. $\neg A \vee \neg W$ | modus tollens from $(A \wedge W) \rightarrow P$ and $\neg P$ |
| 2 | $I \vee M$ | if $\neg A$ then I ; if $\neg W$ then M ; one of them holds by 1 |
| 3 | $\neg(\neg I \wedge \neg M)$ | equivalent to 2 (De Morgan) |
| 4 | $\neg E$ | modus tollens from $E \rightarrow (\neg I \wedge \neg M)$ and 3 |

The argument is **valid** (a brute-force check over all 2^6 truth assignments agrees). Valid does not mean the premises are true; it means that if they were, Superman could not exist.

Proof Techniques

4.1 Vocabulary[†]

A *theorem* is a statement proved true; a *lemma* is a helper theorem; a *corollary* follows quickly from a theorem; an *axiom* is assumed without proof; a *conjecture* is a statement believed true but not yet proved. Throughout, an integer n is **even** if $n = 2k$ for some integer k , and **odd** if $n = 2k + 1$ for some integer k . A real number is **rational** if it equals a/b with integers a, b , $b \neq 0$; otherwise it is **irrational**.

Most theorems have the form $p \rightarrow q$. The techniques below differ in what they assume and what they aim for.

4.2 Direct proof

DEFINITION — Direct proof of $p \rightarrow q$

Assume p is true. Using definitions, axioms, previously proved results and rules of inference, show step by step that q must be true.

EXAMPLE — If n is odd, then n^2 is odd

Let n be odd, so $n = 2k + 1$ for some integer k . Then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1 = 2p + 1, \quad p = 2k^2 + 2k \in \mathbb{Z}.$$

So n^2 is odd.

EXAMPLE — If m and n are perfect squares, then mn is a perfect square

Let $m = k^2$ and $n = \ell^2$ with k, ℓ integers. Then $mn = k^2\ell^2 = (k\ell)^2$, and $k\ell$ is an integer. So mn is a perfect square.

WATCH OUT — when a direct proof gets stuck

Try to prove directly: *if $3n + 2$ is odd then n is odd*. Assume $3n + 2 = 2k + 1$; then $n = \frac{2k-1}{3}$, and nothing tells us this is an odd integer. When the hypothesis gives no useful handle, switch to the contrapositive (next section).

4.3 Indirect proof (proof by contraposition)

DEFINITION — Proof by contraposition

Since $p \rightarrow q \equiv \neg q \rightarrow \neg p$, prove $p \rightarrow q$ by assuming $\neg q$ and proving $\neg p$ (a direct proof of the contrapositive).

EXAMPLE — If $3n + 2$ is odd, then n is odd

Contrapositive: if n is even then $3n + 2$ is even. Let $n = 2k$. Then $3n + 2 = 6k + 2 = 2(3k + 1)$, which is even. The contrapositive holds, hence so does the original statement.

EXAMPLE — If $n = ab$ with a, b positive integers, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$

Contrapositive: if $a > \sqrt{n}$ and $b > \sqrt{n}$, then $n \neq ab$. Indeed $ab > \sqrt{n} \cdot \sqrt{n} = n$, so $ab \neq n$.

4.4 Vacuous and trivial proofs

DEFINITION — Vacuous proof

If the hypothesis p of $p \rightarrow q$ is false, then $p \rightarrow q$ is true whatever q is (row **FT** and **FF** of the truth table). Showing p is false is a *vacuous proof*.

EXAMPLE

Let $P(n)$: “if $n > 1$ then $n^2 > n$ ” with domain the integers. $P(0)$ is “if $0 > 1$ then $0^2 > 0$ ”. The hypothesis $0 > 1$ is false, so $P(0)$ is true (vacuously), even though $0^2 > 0$ itself is false.

DEFINITION — Trivial proof

If the conclusion q of $p \rightarrow q$ is true, then $p \rightarrow q$ is true whatever p is (rows **TT** and **FT**). Showing q is true is a *trivial proof*.

EXAMPLE

Let $P(n)$: “if a and b are positive integers with $a \geq b$, then $a^n \geq b^n$ ”, domain the non-negative integers. For $n = 0$ the conclusion is $a^0 \geq b^0$, i.e. $1 \geq 1$, which is true. So $P(0)$ holds trivially, without using $a \geq b$.

4.5 Proof by contradiction

DEFINITION — Proof by contradiction

To prove a statement S , assume $\neg S$ and derive a contradiction (something false, such as $r \wedge \neg r$). Then $\neg S$ is impossible, so S is true. For $p \rightarrow q$: assume the hypothesis is true *and* the conclusion is false ($p \wedge \neg q$); reaching a contradiction shows this case cannot occur, because **T** $\rightarrow$ **F** is the only false row of $p \rightarrow q$.

EXAMPLE — If $3n + 2$ is odd then n is odd

Assume $3n + 2$ is odd *and* n is even. Then $n = 2k$, so $3n + 2 = 2(3k + 1)$ is even. This contradicts “ $3n + 2$ is odd”. Hence n is odd.

EXAMPLE — $\sqrt{2}$ is irrational

Suppose $\sqrt{2}$ is rational. Then $\sqrt{2} = a/b$ with integers a, b , $b \neq 0$, and a/b in lowest terms (a, b have no common factor except 1). Squaring, $2 = a^2/b^2$, i.e. $2b^2 = a^2$. So a^2 is even, and therefore a is even (an odd number has an odd square). Write $a = 2c$. Then $2b^2 = 4c^2$, so $b^2 = 2c^2$, so b^2 is even and hence b is even. Now a and b are both even, so 2 is a common factor — contradicting lowest terms. Therefore $\sqrt{2}$ is irrational.

4.6 Mistakes in proofs

EXAMPLE — the “proof” that $1 = 2$

- | | | |
|---|-----------------------------|--------------------------------|
| 1 | Let $a = b$ | |
| 2 | $a^2 = ab$ | multiply by a |
| 3 | $a^2 - b^2 = ab - b^2$ | subtract b^2 |
| 4 | $(a + b)(a - b) = b(a - b)$ | factor |
| 5 | $a + b = b$ | divide both sides by $(a - b)$ |
| 6 | $2b = b$ | since $a = b$ |
| 7 | $2 = 1$ | divide by b |

Error: step 5 divides by $a - b$, but $a = b$ means $a - b = 0$, and division by zero is not allowed.

Other common mistakes: affirming the conclusion or denying the hypothesis (Chapter 3); *proof by example* (one example never proves a $\forall$ statement — though one *counterexample* disproves it); assuming what you are trying to prove (circular reasoning)[†].

EXAMPLE — two more faulty “proofs”[†]

“Theorem”: if n^2 is positive then n is positive. “Proof”: if n is positive then n^2 is positive; n^2 is positive; therefore n is positive. This affirms the conclusion of $p \rightarrow q$ (Chapter 3). The theorem is false: $n = -2$.

“Proof” that n^2 even $\Rightarrow n$ even: “Suppose $n^2 = 2k$. Let $n = 2\ell$.” Here $n = 2\ell$ is the claim “ n is even”, so the argument assumes what it must prove (circular reasoning, also called begging the question).

4.7 Exhaustive proof and proof by cases

DEFINITION — Exhaustive proof / proof by cases

When a statement involves only finitely many possibilities, verify each one (*exhaustive proof*). More generally, split the situation into cases that together cover every possibility and prove the statement in each case (*proof by cases*).

EXAMPLE — $(n + 1)^3 \geq 3^n$ for positive integers $n \leq 4$

n	$(n + 1)^3$	3^n	holds?
1	8	3	yes
2	27	9	yes
3	64	27	yes
4	125	81	yes

All four cases hold, so the statement is proved.

EXAMPLE — The only consecutive positive perfect powers not exceeding 100 are 8 and 9

List all perfect powers ≤ 100 :

squares	1, 4, 9, 16, 25, 36, 49, 64, 81, 100
cubes	1, 8, 27, 64
fourth powers	1, 16, 81
fifth powers	1, 32
sixth powers	1, 64

(Higher powers give nothing new below 100.) Merged: 1, 4, 8, 9, 16, 25, 27, 32, 36, 49, 64, 81, 100. Checking neighbours, the only pair of consecutive integers is (8, 9).

EXAMPLE — proof by cases: $n^2 + n$ is even for every integer n

Case 1: n even. $n = 2k$, so $n^2 + n = 4k^2 + 2k = 2(2k^2 + k)$: even.

Case 2: n odd. $n = 2k + 1$, so $n^2 + n = (4k^2 + 4k + 1) + (2k + 1) = 2(2k^2 + 3k + 1)$: even.

4.8 Proof by cases, continued[†]

EXAMPLE — $n^2 \geq n$ for every integer n

Split into three cases that cover all integers.

Case $n = 0$: $0^2 = 0 \geq 0$.

Case $n \geq 1$: multiplying $n \geq 1$ by the positive number n gives $n^2 \geq n$.

Case $n \leq -1$: $n^2 \geq 0 > n$.

DEFINITION — Without loss of generality (WLOG)

If a proof has several cases that are the same argument with the roles of variables swapped, we may treat one case and say “without loss of generality”: the other cases follow by symmetry.

EXAMPLE — if xy and $x + y$ are both even, then x and y are both even

Contrapositive: if x and y are not both even, then xy is odd or $x + y$ is odd. Not both even means x is odd or y is odd; the statement is symmetric in x and y , so WLOG x is odd.

Case y even: $x + y$ is odd.

Case y odd: xy is odd (a product of odd numbers).

Either way, “ xy even and $x + y$ even” fails. Hence the original statement holds.

4.9 Proving “if and only if” and lists of equivalent statements[†]

DEFINITION — Proving $p \leftrightarrow q$ and $p_1 \leftrightarrow \dots \leftrightarrow p_n$

Since $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$, prove *both* directions (each by any method). To show that $p_1, \dots, p_n$ are all equivalent, it is enough to prove the cycle $p_1 \rightarrow p_2 \rightarrow \dots \rightarrow p_n \rightarrow p_1$: this needs n implications instead of $n(n - 1)$.

EXAMPLE — n is odd $\iff n^2$ is odd

($\Rightarrow$) Direct: $n = 2k + 1$ gives $n^2 = 2(2k^2 + 2k) + 1$, odd.

($\Leftarrow$) Contraposition: if n is even, $n = 2k$, then $n^2 = 2(2k^2)$ is even.

EXAMPLE — n even $\iff n - 1$ odd $\iff n^2$ even

Let p_1 : n is even; p_2 : $n - 1$ is odd; p_3 : n^2 is even. Prove $p_1 \rightarrow p_2 \rightarrow p_3 \rightarrow p_1$.

$p_1 \rightarrow p_2$: $n = 2k$ gives $n - 1 = 2(k - 1) + 1$, odd.

$p_2 \rightarrow p_3$: $n - 1 = 2m + 1$ gives $n = 2(m + 1)$, so $n^2 = 2 \cdot 2(m + 1)^2$, even.

$p_3 \rightarrow p_1$: by contraposition, if n is odd then n^2 is odd (shown above), so n^2 even forces n even.

4.10 Counterexamples[†]

DEFINITION — Counterexample

To disprove $\forall x P(x)$, exhibit one element c of the domain with $P(c)$ false. A single counterexample is enough; no number of examples can prove a $\forall$ statement.

EXAMPLE

“Every positive integer is the sum of the squares of two integers” is false: 3 is a counterexample. The squares up to 3 are 0 and 1, giving sums 0, 1, 2 only. “Every prime is odd” is false: 2 is a counterexample.

4.11 Existence and uniqueness proofs[†]

DEFINITION — Existence proofs

A *constructive* proof of $\exists x P(x)$ exhibits a specific x with $P(x)$. A *non-constructive* proof shows that such an x must exist without saying which one it is.

EXAMPLE — constructive

Some positive integer is the sum of two positive cubes in two different ways: $1729 = 10^3 + 9^3 = 12^3 + 1^3$.

EXAMPLE — non-constructive: irrational x, y with x^y rational

$\sqrt{2}$ is irrational. Consider $\sqrt{2}^{\sqrt{2}}$. If it is rational, take $x = y = \sqrt{2}$. If it is irrational, take $x = \sqrt{2}^{\sqrt{2}}$ and $y = \sqrt{2}$: then $x^y = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^2 = 2$, which is rational. In either case suitable numbers exist, but the proof does not tell us which case occurs.

DEFINITION — Uniqueness proofs

To prove that *exactly one* x has property P , prove **existence** (some x works) and **uniqueness** (if x and y both work then $x = y$).

EXAMPLE

If $a \neq 0$, the equation $ar + b = 0$ has a unique real solution. *Existence*: $r = -b/a$ works, since $a(-b/a) + b = 0$. *Uniqueness*: if $ar + b = 0$ and $as + b = 0$ then $a(r - s) = 0$, so $r = s$ because $a \neq 0$.

4.12 Mathematical induction

DEFINITION — Principle of mathematical induction

To prove $P(n)$ for all integers $n \geq n_0$:

1. **Basis step**: show $P(n_0)$ is true.
2. **Inductive step**: for an arbitrary $k \geq n_0$, assume $P(k)$ (the *inductive hypothesis*) and prove $P(k + 1)$.

Conclusion: $P(n)$ holds for every $n \geq n_0$. Think of dominoes: the basis knocks over the first, and the inductive step shows each falling domino knocks over the next.

EXAMPLE — $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$

Basis ($n = 1$): LHS = 1, RHS = $\frac{1 \cdot 2}{2} = 1$.

Hypothesis: $1 + 2 + \cdots + k = \frac{k(k+1)}{2}$.

Step:

$$1 + 2 + \cdots + k + (k + 1) = \frac{k(k+1)}{2} + (k + 1) = \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2},$$

which is the formula for $n = k + 1$.

EXAMPLE — Conjecture, then prove: the sum of the first n odd integers

Compute small cases: $1 = 1$, $1 + 3 = 4$, $1 + 3 + 5 = 9$, $1 + 3 + 5 + 7 = 16$, $1 + 3 + 5 + 7 + 9 = 25$.

Conjecture: $1 + 3 + 5 + \cdots + (2n - 1) = n^2$.

Basis ($n = 1$): $2(1) - 1 = 1 = 1^2$.

Hypothesis: $1 + 3 + \cdots + (2k - 1) = k^2$.

Step: add the next odd number $2(k + 1) - 1 = 2k + 1$:

$$k^2 + (2k + 1) = (k + 1)^2.$$

EXAMPLE — $1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$ for all $n \geq 0$

Basis ($n = 0$): LHS = 1, RHS = $2^1 - 1 = 1$.

Hypothesis: $1 + 2 + \cdots + 2^k = 2^{k+1} - 1$.

Step:

$$(1 + 2 + \cdots + 2^k) + 2^{k+1} = 2^{k+1} - 1 + 2^{k+1} = 2 \cdot 2^{k+1} - 1 = 2^{k+2} - 1 = 2^{(k+1)+1} - 1.$$

4.13 Solved previous-year questions

PYQ • Mid-Sem Sep 2024 Q1(a) • 2 marks

Prove that if n is a positive integer, then n is odd if and only if $5n + 6$ is odd.

Solution. An “if and only if” needs both directions.

($\Rightarrow$) **Direct proof.** Let n be odd, $n = 2k + 1$. Then $5n + 6 = 10k + 5 + 6 = 10k + 11 = 2(5k + 5) + 1$, which is odd.

($\Leftarrow$) **Contraposition.** We show: if n is not odd then $5n + 6$ is not odd. Let n be even, $n = 2k$. Then $5n + 6 = 10k + 6 = 2(5k + 3)$, which is even.

Hence n is odd $\iff 5n + 6$ is odd. ■

(Shortcut: $5n + 6 = n + 2(2n + 3)$, so $5n + 6$ and n always have the same parity.)

PYQ • Mid-Sem (undated, CS 5101) Q3(a) • 4 marks

Use a proof by contradiction to prove that the sum of an irrational number and a rational number is irrational.

Solution. Let x be irrational and r rational. **Assume the opposite:** $x + r$ is rational, say $x + r = s$ with s rational.

Then $x = s - r$. Write $s = \frac{a}{b}$ and $r = \frac{c}{d}$ with integers a, b, c, d and $b, d \neq 0$. Then

$$x = \frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd},$$

a quotient of integers with $bd \neq 0$. So x is rational. This contradicts the assumption that x is irrational.

Hence the assumption was false: $x + r$ is irrational. ■

PYQ • Mid-Sem Sep 2024 Q3(b) • 6 marks

For all $n \geq 1$, using mathematical induction prove that $1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$.

Solution. Let $P(n)$: $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$.

Basis ($n = 1$): LHS = $1^2 = 1$; RHS = $\frac{1 \cdot 2 \cdot 3}{6} = 1$. So $P(1)$ is true.

Inductive hypothesis: assume $P(k)$ for some $k \geq 1$: $1^2 + \cdots + k^2 = \frac{k(k+1)(2k+1)}{6}$.

Inductive step: we must show $1^2 + \cdots + k^2 + (k+1)^2 = \frac{(k+1)(k+2)(2k+3)}{6}$. Using the hypothesis,

$$\begin{aligned} 1^2 + \cdots + k^2 + (k+1)^2 &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 = \frac{(k+1)[k(2k+1) + 6(k+1)]}{6} \\ &= \frac{(k+1)(2k^2 + 7k + 6)}{6} = \frac{(k+1)(k+2)(2k+3)}{6}, \end{aligned}$$

since $2k^2 + 7k + 6 = (k+2)(2k+3)$. This is $P(k+1)$.

By the principle of mathematical induction, $P(n)$ is true for all $n \geq 1$. ■

PYQ • Mid-Sem (undated, CS 5101) Q3(b) • 6 marks

The harmonic numbers H_j , $j = 1, 2, 3, \dots$ are defined by $H_j = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{j}$. For instance $H_4 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{25}{12}$. Use mathematical induction to show that $H_{2^n} \geq 1 + \frac{n}{2}$ whenever n is a non-negative integer.

Solution. Let $P(n)$: $H_{2^n} \geq 1 + \frac{n}{2}$.

Basis ($n = 0$): $H_{2^0} = H_1 = 1 \geq 1 + 0 = 1$. True. (Check $n = 2$ too: $H_4 = \frac{25}{12} \approx 2.08 \geq 1 + \frac{2}{2} = 2$.)

Inductive hypothesis: assume $H_{2^k} \geq 1 + \frac{k}{2}$ for some $k \geq 0$.

Inductive step: split $H_{2^{k+1}}$ into H_{2^k} and the next 2^k terms:

$$H_{2^{k+1}} = H_{2^k} + \underbrace{\frac{1}{2^k+1} + \frac{1}{2^k+2} + \cdots + \frac{1}{2^{k+1}}}_{2^k \text{ terms}}.$$

Each of the 2^k terms is at least the smallest one, $\frac{1}{2^{k+1}}$, so their sum is at least $2^k \cdot \frac{1}{2^{k+1}} = \frac{1}{2}$. With the hypothesis,

$$H_{2^{k+1}} \geq \left(1 + \frac{k}{2}\right) + \frac{1}{2} = 1 + \frac{k+1}{2},$$

which is $P(k+1)$.

By induction, $H_{2^n} \geq 1 + \frac{n}{2}$ for all $n \geq 0$. ■

4.14 Practice questions from Rosen

These are chosen from the exercises of Rosen §1.7–1.8 to cover each proof method once or twice: direct, contraposition, contradiction, cases, exhaustion, equivalence, counterexample, existence and uniqueness. (Induction questions are not in the trimmed Rosen text; see the PYQs above.)

PRACTICE • Rosen §1.7 Ex. 5 and 7 • direct proofs

(a) Prove that if $m + n$ and $n + p$ are even integers, then $m + p$ is even. What kind of proof did you use? (b) Show that every odd integer is the difference of two squares.

Solution. (a) Direct proof. Let $m + n = 2a$ and $n + p = 2b$ for integers a, b . Then

$$m + p = (m + n) + (n + p) - 2n = 2a + 2b - 2n = 2(a + b - n),$$

so $m + p$ is even.

(b) An odd integer is $2k + 1$. Since $(k + 1)^2 - k^2 = 2k + 1$, every odd integer equals $(k + 1)^2 - k^2$. For example $7 = 4^2 - 3^2$ ($k = 3$).

PRACTICE • Rosen §1.7 Ex. 8 • contradiction with squares

Prove that if n is a perfect square, then $n + 2$ is not a perfect square.

Solution. Let $n = a^2$ with $a \geq 0$ an integer (replace a by $-a$ if needed). **Assume** $n + 2 = b^2$ with $b \geq 0$ an integer. Then $b^2 - a^2 = 2 > 0$, so $b > a$, hence $b \geq a + 1$ and

$$2 = b^2 - a^2 \geq (a + 1)^2 - a^2 = 2a + 1.$$

So $2a + 1 \leq 2$, which forces $a = 0$. Then $b^2 = 2$, but 2 is not a perfect square: contradiction. Hence $n + 2$ is not a perfect square. ■

PRACTICE • Rosen §1.7 Ex. 11–13 • prove or disprove (rational, irrational)

(a) Prove or disprove: the product of two irrational numbers is irrational. (b) Prove that the product of a non-zero rational number and an irrational number is irrational. (c) Prove that if x is irrational then $1/x$ is irrational.

Solution. (a) False. Counterexample: $\sqrt{2}$ is irrational but $\sqrt{2} \cdot \sqrt{2} = 2$ is rational.

(b) True, by contradiction. Let $q \neq 0$ be rational and x irrational, and **assume** $qx = r$ is rational. Then $x = r/q$ is a quotient of rationals with $q \neq 0$, hence rational: contradiction.

(c) True, by contraposition. Suppose $1/x$ is rational, $1/x = a/b$. Since $1/x \neq 0$ we have $a \neq 0$, so $x = b/a$ is rational. Hence if x is irrational, $1/x$ is irrational. (Note $x \neq 0$ because 0 is rational.)

PRACTICE • Rosen §1.7 Ex. 15–16 • contraposition

Prove by contraposition: (a) if $x + y \geq 2$ for real x, y , then $x \geq 1$ or $y \geq 1$; (b) if m, n are integers and mn is even, then m is even or n is even.

Solution. (a) The contrapositive of $p \rightarrow (q \vee r)$ is $(\neg q \wedge \neg r) \rightarrow \neg p$: if $x < 1$ and $y < 1$, then $x + y < 2$. This is true by adding the two inequalities.

(b) Contrapositive: if m and n are both odd, then mn is odd. With $m = 2a + 1$, $n = 2b + 1$: $mn = 4ab + 2a + 2b + 1 = 2(2ab + a + b) + 1$, odd. ■

PRACTICE • Rosen §1.7 Ex. 17 • contraposition and contradiction, same statement

Show that if n is an integer and $n^3 + 5$ is odd, then n is even, using (a) a proof by contraposition, (b) a proof by contradiction.

Solution. (a) Contrapositive: if n is odd then $n^3 + 5$ is even. If $n = 2k + 1$, then $n^3 = 8k^3 + 12k^2 + 6k + 1 = 2(4k^3 + 6k^2 + 3k) + 1$ is odd, so $n^3 + 5 = 2(4k^3 + 6k^2 + 3k + 3) + 1$ is even.

(b) Assume $n^3 + 5$ is odd and n is odd. Then n^3 is odd, so $n^3 + 5$ is odd+odd=even. This contradicts “ $n^3 + 5$ is odd”. ■

(Compare: contraposition proves $\neg q \rightarrow \neg p$ directly; contradiction assumes $p \wedge \neg q$ and reaches any contradiction.)

PRACTICE • Rosen §1.7 Ex. 23 • contradiction (pigeonhole)

Show that at least ten of any 64 days chosen must fall on the same day of the week.

Solution. Assume the opposite: each of the 7 days of the week receives at most 9 of the chosen days. Then there are at most $7 \cdot 9 = 63$ chosen days, but we chose 64: contradiction. Hence some weekday has at least 10 of them. (In general n objects in k boxes force some box to hold at least $\lceil n/k \rceil$; here $\lceil 64/7 \rceil = 10$.) ■

PRACTICE • Rosen §1.7 Ex. 25 • contradiction with parity cases

Use a proof by contradiction to show that there is no rational number r with $r^3 + r + 1 = 0$.

Solution. Assume $r = a/b$ is a root, with integers $a, b, b \neq 0$, in lowest terms (so a, b are not both even). Multiplying $r^3 + r + 1 = 0$ by b^3 :

$$a^3 + ab^2 + b^3 = 0.$$

Check the parity of the left side in the three possible cases:

	a^3	ab^2	b^3	sum
a even, b odd	even	even	odd	odd
a odd, b even	odd	even	even	odd
a odd, b odd	odd	odd	odd	odd

In every case the sum is odd, but it equals 0, which is even: contradiction. So no rational root exists. ■

PRACTICE • Rosen §1.7 Ex. 31 • several equivalent statements

Show that these statements about the integer x are equivalent: (i) $3x + 2$ is even, (ii) $x + 5$ is odd, (iii) x^2 is even.

Solution. Prove the cycle (i) $\rightarrow$ (ii) $\rightarrow$ (iii) $\rightarrow$ (i).

(i) $\rightarrow$ (ii): $3x + 2$ even means $3x$ is even. If x were odd, $3x$ would be odd; so x is even, and $x + 5$ is odd.

(ii) $\rightarrow$ (iii): $x + 5$ odd means x is even, $x = 2k$, so $x^2 = 4k^2$ is even.

(iii) $\rightarrow$ (i): if x were odd then x^2 would be odd; so x is even, $x = 2k$, and $3x + 2 = 2(3k + 1)$ is even. ■

PRACTICE • Rosen §1.7 Ex. 34–35 • is this reasoning correct?

(a) Is this reasoning for solving $\sqrt{2x^2 - 1} = x$ correct? (1) $\sqrt{2x^2 - 1} = x$; (2) $2x^2 - 1 = x^2$; (3) $x^2 - 1 = 0$; (4) $(x - 1)(x + 1) = 0$; (5) $x = 1$ or $x = -1$. (b) The same question for $\sqrt{x + 3} = 3 - x$: (2) $x + 3 = x^2 - 6x + 9$; (3) $0 = x^2 - 7x + 6$; (4) $0 = (x - 1)(x - 6)$; (5) $x = 1$ or $x = 6$.

Solution. Every step is a correct *implication*, but squaring ($a = b \Rightarrow a^2 = b^2$) is not reversible, so the reasoning only shows that any solution must be one of the listed values. Each candidate must be **checked in the original equation**; otherwise extraneous solutions creep in.

(a) $x = 1$: $\sqrt{1} = 1 \checkmark$. $x = -1$: $\sqrt{1} = 1 \neq -1$ (a square root is never negative) $\times$. Only $x = 1$ is a solution.

(b) $x = 1$: $\sqrt{4} = 2 = 3 - 1 \checkmark$. $x = 6$: $\sqrt{9} = 3 \neq 3 - 6 = -3 \times$. Only $x = 1$ is a solution.

PRACTICE • Rosen §1.7 Ex. 38 • counterexample

Find a counterexample to the statement that every positive integer can be written as the sum of the squares of three integers.

Solution. Take $n = 7$. Only the squares 0, 1, 4 do not exceed 7. Sums of three of them are 0, 1, 2, 3, 4, 5, 6, 8, 9, 12; the value 7 never occurs. So 7 is a counterexample. ■

PRACTICE • Rosen §1.8 Ex. 27 • conjecture, then proof by cases

Formulate a conjecture about the final decimal digit of the fourth power of an integer and prove it by cases.

Solution. Small cases: $1^4 = 1$, $2^4 = 16$, $3^4 = 81$, $4^4 = 256$, $5^4 = 625$, $6^4 = 1296$, $7^4 = 2401$, $8^4 = 4096$, $9^4 = 6561$, $10^4 = 10000$. **Conjecture:** the last digit of n^4 is 0, 1, 5 or 6.

Proof. Write $n = 10q + d$ with d the last digit. Every term of $(10q + d)^4$ other than d^4 contains a factor 10, so n^4 and d^4 have the same last digit. Check the ten cases $d = 0, \dots, 9$:

d	0	1	2	3	4	5	6	7	8	9
d^4	0	1	16	81	256	625	1296	2401	4096	6561
last digit	0	1	6	1	6	5	6	1	6	1

Every case ends in 0, 1, 5 or 6. ■

PRACTICE • Rosen §1.8 Ex. 29–30 • exhaustive proofs

Prove: (a) there is no positive integer n with $n^2 + n^3 = 100$; (b) there are no integer solutions of $2x^2 + 5y^2 = 14$.

Solution. (a) $n^2 + n^3$ increases with n : $n = 1, 2, 3, 4$ give 2, 12, 36, 80, all below 100, and $n = 5$ gives $150 > 100$. For $n \geq 5$ the value only grows. So 100 is never attained.

(b) $5y^2 \leq 14$ gives $y^2 \leq 2$, so $y \in \{-1, 0, 1\}$. If $y = 0$: $2x^2 = 14$, $x^2 = 7$, not a square. If $y = \pm 1$: $2x^2 = 9$, impossible (odd). No case works. ■

PRACTICE • Rosen §1.8 Ex. 22 • forward reasoning

Use forward reasoning to show that if x is a non-zero real number, then $x^2 + 1/x^2 \geq 2$.

Solution. Start from a fact that is always true: the square of a real number is non-negative, so $(x - \frac{1}{x})^2 \geq 0$. Expanding, $x^2 - 2 + \frac{1}{x^2} \geq 0$, hence $x^2 + \frac{1}{x^2} \geq 2$. ■ (Equality holds exactly when $x = \pm 1$.)

PRACTICE • Rosen §1.8 Ex. 9–10 • constructive and non-constructive existence

(a) Prove that there are 100 consecutive positive integers that are not perfect squares. Is your proof constructive? (b) Prove that either $2 \cdot 10^{500} + 15$ or $2 \cdot 10^{500} + 16$ is not a perfect square. Is your proof constructive?

Solution. (a) Constructive. $50^2 = 2500$ and $51^2 = 2601$. No integer lies strictly between 50 and 51, so none of 2501, 2502, ..., 2600 is a perfect square. That is $2600 - 2501 + 1 = 100$ consecutive integers.

(b) Non-constructive. Let $N = 2 \cdot 10^{500} + 15$. If both $N = a^2$ and $N + 1 = b^2$ with positive integers a, b , then $b^2 - a^2 = 1$, so $(b - a)(b + a) = 1$; but $b + a \geq 2$, impossible. So at least one of $N, N + 1$ is not a perfect square. The proof does not say *which*.

PRACTICE • Rosen §1.8 Ex. 17 • existence and uniqueness

Suppose that a and b are odd integers with $a \neq b$. Show there is a unique integer c such that $|a - c| = |b - c|$.

Solution. Existence. Take $c = \frac{a+b}{2}$. Since a and b are both odd, $a + b$ is even, so c is an integer, and $|a - c| = |\frac{a-b}{2}| = |b - c|$.

Uniqueness. Suppose $|a - c| = |b - c|$. Then $a - c = b - c$ or $a - c = -(b - c)$. The first gives $a = b$, excluded. The second gives $2c = a + b$, so $c = \frac{a+b}{2}$. Hence c must be this number. ■

Sets

Sets are the language in which relations and functions are defined, so it pays to have these definitions exact.

5.1 Sets and elements

DEFINITION — Set

A *set* is a well-defined collection of distinct objects, called its *elements* or *members*. We write $a \in A$ if a belongs to A and $a \notin A$ otherwise. Order and repetition do not matter. $|A|$ denotes the number of elements (the *cardinality*) of a finite set A .

EXAMPLE

Roster form: $A = \{1, 2, 3\}$. Set-builder form: $B = \{x \in \mathbb{Z} \mid 0 < x < 4\} = \{1, 2, 3\}$. Here $2 \in A$, $5 \notin A$, $|A| = 3$, and $\{1, 2, 2, 3\} = \{3, 1, 2\} = A$.

REMEMBER — standard number sets

$\mathbb{N} = \{0, 1, 2, \dots\}$ (natural numbers), $\mathbb{Z}$ (integers), $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$, $\mathbb{Q}$ (rationals), $\mathbb{R}$ (reals). A set may contain other sets: $\{\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}\}$ has four elements, each itself a set.

DEFINITION — Universal set and Venn diagram

†All sets in a discussion are subsets of one *universal set* U , drawn as a rectangle. In a *Venn diagram* each set is a region (usually a circle) inside the rectangle.

EXAMPLE

With $U = \{a, b, \dots, z\}$ (the 26 letters), the set $V = \{a, e, i, o, u\}$ of vowels is a circle inside the rectangle U ; the 21 consonants lie outside the circle.

5.2 Singleton set

DEFINITION — Singleton

A set with exactly one element is a *singleton set*.

EXAMPLE

$\{7\}$ and $\{\emptyset\}$ are singletons. Note $\{7\} \neq 7$: the first is a set containing the number 7.

5.3 Subset and proper subset

DEFINITION — Subset

B is a *subset* of A , written $B \subseteq A$, if every element of B is an element of A :

$$B \subseteq A \iff \forall x (x \in B \rightarrow x \in A).$$

EXAMPLE

$\{1, 2\} \subseteq \{1, 2, 3\}$, and $\{1, 2, 3\} \subseteq \{1, 2, 3\}$ (every set is a subset of itself), and $\emptyset \subseteq A$ for every set A (the condition holds vacuously).

DEFINITION — Proper subset

B is a *proper subset* of A , written $B \subset A$, if $B \subseteq A$ and $B \neq A$; that is,

$$\forall x (x \in B \rightarrow x \in A) \wedge \exists x (x \in A \wedge x \notin B).$$

EXAMPLE

$\{1, 2\} \subset \{1, 2, 3\}$ because $3 \in \{1, 2, 3\}$ but $3 \notin \{1, 2\}$. But $\{1, 2, 3\}$ is not a proper subset of itself.

WATCH OUT — $\in$ is not $\subseteq$

Let $S = \{2, \{2\}\}$. Then $2 \in S$ and $\{2\} \in S$ (both are elements), and $\{2\} \subseteq S$ (because $2 \in S$), and $\{\{2\}\} \subseteq S$ (because $\{2\} \in S$). But $\{2\}$ is *not* an element of $\{2, 3\}$, and 2 is not an element of $\{\{2\}\}$. Always ask: *is it listed as a member* (use $\in$), or *are all its members listed* (use $\subseteq$)?

5.4 Empty (null) set

DEFINITION — Empty set

The *empty set* (or *null set*), written $\emptyset$ or $\{ \}$, is the set with no elements; $|\emptyset| = 0$.

WATCH OUT

$\emptyset \neq \{\emptyset\}$. The set $\{\emptyset\}$ has one element (namely $\emptyset$), so $|\{\emptyset\}| = 1$.

5.5 Equal sets

DEFINITION — Equality of sets

$A = B$ if they have exactly the same elements:

$$A = B \iff \forall x (x \in B \rightarrow x \in A) \wedge \forall x (x \in A \rightarrow x \in B),$$

i.e. $A \subseteq B$ and $B \subseteq A$. (To prove two sets equal, prove both inclusions.)

EXAMPLE

$\{x \mid x^2 = 1\} = \{-1, 1\}$ and $\{1, 2, 3\} = \{3, 2, 1\}$, but $\{1, 2\} \neq \{1, 2, 3\}$.

5.6 Cardinality[†]

DEFINITION — Finite and infinite sets

A set is *finite* if it has n elements for some non-negative integer n (its *cardinality* $|A| = n$); otherwise it is *infinite*.

EXAMPLE

$|\{1, 3, 5, 7, 9\}| = 5$; $|\emptyset| = 0$; $|\{\emptyset\}| = 1$; the set of positive integers is infinite.

REMEMBER — union of two finite sets

$|A \cup B| = |A| + |B| - |A \cap B|$: adding $|A| + |B|$ counts the elements of $A \cap B$ twice, so subtract them once. If $A \cap B = \emptyset$, $|A \cup B| = |A| + |B|$.

5.7 Power set

DEFINITION — Power set

The *power set* $\mathcal{P}(A)$ is the set of *all subsets* of A (including $\emptyset$ and A itself). If $|A| = n$ then $|\mathcal{P}(A)| = 2^n$, because each element is either in or out of a subset.

EXAMPLE

For $A = \{1, 2, 3\}$:

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}, \quad |\mathcal{P}(A)| = 2^3 = 8.$$

Also $\mathcal{P}(\emptyset) = \{\emptyset\}$ (one subset, $2^0 = 1$), and $\mathcal{P}(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}$.

5.8 Cartesian product

DEFINITION — Ordered pair and Cartesian product

An *ordered pair* (a, b) is written with round brackets; order matters: $(a, b) \neq (b, a)$ unless $a = b$. The *Cartesian product* of A and B is the set of all ordered pairs

$$A \times B = \{(a, b) \mid a \in A, b \in B\}, \quad |A \times B| = |A| \cdot |B|.$$

More generally $A_1 \times \cdots \times A_n$ is the set of ordered n -tuples $(a_1, \dots, a_n)$ with $a_i \in A_i$, and $A^n = A \times \cdots \times A$ (n factors)[†].

EXAMPLE

$A = \{a, b\}$, $B = \{1, 2, 3\}$:

$$A \times B = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}, \quad |A \times B| = 6.$$

$B \times A = \{(1, a), (1, b), (2, a), \dots\} \neq A \times B$: the Cartesian product is **not commutative**. For $A = \{1, 2\}$: $A^2 = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$ and $|A^3| = 2^3 = 8$.

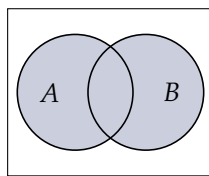
5.9 Set operations

DEFINITION — Union, intersection, difference, complement

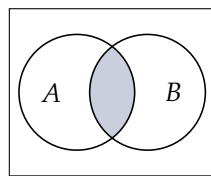
For sets A, B in a universal set U :

union	$A \cup B = \{x \mid x \in A \vee x \in B\}$
intersection	$A \cap B = \{x \mid x \in A \wedge x \in B\}$
difference	$A - B = \{x \mid x \in A \wedge x \notin B\}$
complement	$\overline{A} = U - A$

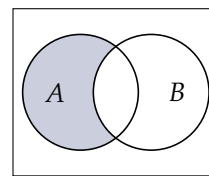
A and B are *disjoint* if $A \cap B = \emptyset$.



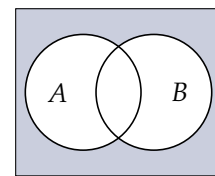
$A \cup B$



$A \cap B$



$A - B$



$\overline{A}$

EXAMPLE

$A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$: $A \cup B = \{1, 2, 3, 4\}$, $A \cap B = \{2, 3\}$, $A - B = \{1\}$, $B - A = \{4\}$. Since a relation is a set of ordered pairs, these operations apply to relations too (used in Chapter 6).

EXAMPLE — with a universal set

$U = \{1, 2, \dots, 10\}$ and $A = \{1, 2, \dots, 5\}$ (the positive integers up to 5): $\overline{A} = \{6, 7, 8, 9, 10\}$. If instead $U = \mathbb{Z}^+$ and $A = \{n \in \mathbb{Z}^+ \mid n > 10\}$, then $\overline{A} = \{1, 2, \dots, 10\}$. The complement depends on the universal set.

5.10 Set identities[†]

Set operations obey laws that mirror the laws of logic in Chapter 1 ($\cup \leftrightarrow \vee$, $\cap \leftrightarrow \wedge$, complement $\leftrightarrow \neg$, $U \leftrightarrow \text{T}$, $\emptyset \leftrightarrow \text{F}$).

Law	Identities	
Identity	$A \cup \emptyset = A$	$A \cap U = A$
Domination	$A \cup U = U$	$A \cap \emptyset = \emptyset$
Idempotent	$A \cup A = A$	$A \cap A = A$
Complementation	$\overline{\overline{A}} = A$	
Commutative	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Associative	$A \cup (B \cup C) = (A \cup B) \cup C$	$A \cap (B \cap C) = (A \cap B) \cap C$
Distributive	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
De Morgan	$\overline{A \cap B} = \overline{A} \cup \overline{B}$	$\overline{A \cup B} = \overline{A} \cap \overline{B}$
Absorption	$A \cup (A \cap B) = A$	$A \cap (A \cup B) = A$
Complement	$A \cup \overline{A} = U$	$A \cap \overline{A} = \emptyset$
Difference	$A - B = A \cap \overline{B}$	

Three ways to prove an identity, shown for De Morgan $\overline{A \cap B} = \overline{A} \cup \overline{B}$:

EXAMPLE — 1. show each side is a subset of the other

If $x \in \overline{A \cap B}$ then $x \notin A \cap B$, so $x \notin A$ or $x \notin B$; that is, $x \in \overline{A}$ or $x \in \overline{B}$, so $x \in \overline{A} \cup \overline{B}$. Conversely, if $x \in \overline{A} \cup \overline{B}$ then $x \notin A$ or $x \notin B$, so x is not in both, $x \notin A \cap B$, so $x \in \overline{A \cap B}$. Both inclusions hold, so the sets are equal.

EXAMPLE — 2. set-builder notation and logical equivalences

$$\begin{aligned}\overline{A \cap B} &= \{x \mid \neg(x \in A \wedge x \in B)\} = \{x \mid \neg(x \in A) \vee \neg(x \in B)\} \quad (\text{De Morgan for logic}) \\ &= \{x \mid x \in \overline{A} \vee x \in \overline{B}\} = \overline{A} \cup \overline{B}.\end{aligned}$$

EXAMPLE — 3. membership table: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Write 1 if x is in the set and 0 if not; one row per combination of memberships.

A	B	C	$B \cup C$	$A \cap (B \cup C)$	$A \cap B$	$A \cap C$	$(A \cap B) \cup (A \cap C)$
1	1	1	1	1	1	1	1
1	1	0	1	1	1	0	1
1	0	1	1	1	0	1	1
1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0

The two highlighted columns (5th and 8th) agree in every row, so the sets are equal.

5.11 Generalised union and intersection[†]

DEFINITION — Union and intersection of a family

For sets $A_1, A_2, \dots, A_n$:

$$\bigcup_{i=1}^n A_i = A_1 \cup \dots \cup A_n = \{x \mid x \in A_i \text{ for at least one } i\}, \quad \bigcap_{i=1}^n A_i = \{x \mid x \in A_i \text{ for every } i\}.$$

The same notation with $i = 1$ to ∞ is used for infinite families.

EXAMPLE

Let $A_i = \{i, i+1, i+2, \dots\}$. Then $\bigcup_{i=1}^n A_i = A_1 = \{1, 2, 3, \dots\}$ and $\bigcap_{i=1}^n A_i = A_n = \{n, n+1, \dots\}$ (the sets shrink as i grows).

Let $A_i = \{1, 2, \dots, i\}$. Then $\bigcup_{i=1}^{\infty} A_i = \mathbb{Z}^+$ and $\bigcap_{i=1}^{\infty} A_i = \{1\}$ (the sets grow; only 1 is in all of them).

5.12 Symmetric difference[†]

DEFINITION — Symmetric difference

$A \oplus B$ is the set of elements in *exactly one* of A and B :

$$A \oplus B = (A - B) \cup (B - A) = (A \cup B) - (A \cap B).$$

Logically, $x \in A \oplus B$ iff $(x \in A) \oplus (x \in B)$ (exclusive or).

EXAMPLE

$\{1, 3, 5\} \oplus \{1, 2, 3\} = \{2, 5\}$: 1 and 3 are in both and are removed. Also $A \oplus A = \emptyset$ and $A \oplus \emptyset = A$.

5.13 Sets as bit strings[†]

DEFINITION — Bit-string representation

For a finite universal set $U = \{a_1, \dots, a_n\}$ in a fixed order, a subset S is represented by the bit string whose i th bit is 1 if $a_i \in S$ and 0 otherwise. Then union is bitwise OR, intersection is bitwise AND, complement flips every bit, difference $A - B$ is A AND (NOT B), and symmetric difference is bitwise XOR.

EXAMPLE

$U = \{1, 2, \dots, 10\}$: the odd integers $\{1, 3, 5, 7, 9\}$ are 10 1010 1010, the integers not exceeding 5 are 11 1110 0000. Then

$$\begin{array}{ll} \text{union (OR)} & 11 \ 1110 \ 1010 = \{1, 2, 3, 4, 5, 7, 9\} \\ \text{intersection (AND)} & 10 \ 1010 \ 0000 = \{1, 3, 5\} \end{array}$$

5.14 Practice questions from Rosen

These are chosen from Rosen §2.1 and §2.2. Routine list-the-elements exercises and the basic identity proofs (idempotent, domination, and so on) are left out because the chapter above already shows them; what remains are questions where a student can slip, in the style of the PYQs.

PRACTICE • Rosen §2.1 Ex. 9–10 • true or false with $\emptyset$

Determine whether each statement is true or false.

- (a) $0 \in \emptyset$ (b) $\emptyset \in \{0\}$ (c) $\{0\} \subset \emptyset$ (d) $\emptyset \subset \{0\}$ (e) $\{0\} \in \{0\}$ (f) $\{0\} \subset \{0\}$ (g) $\{\emptyset\} \subseteq \{\emptyset\}$
 (h) $\emptyset \in \{\emptyset\}$ (i) $\emptyset \in \{\emptyset, \{\emptyset\}\}$ (j) $\{\emptyset\} \in \{\emptyset\}$ (k) $\{\emptyset\} \in \{\{\emptyset\}\}$ (l) $\{\emptyset\} \subset \{\emptyset, \{\emptyset\}\}$
 (m) $\{\{\emptyset\}\} \subset \{\emptyset, \{\emptyset\}\}$ (n) $\{\{\emptyset\}\} \subset \{\{\emptyset\}, \{\emptyset\}\}$

Solution. Here $\subset$ means *proper* subset. Ask two separate questions: “is it an *element*?” ($\in$: look at the members listed inside the outer braces) and “is every member of the left set a member of the right?” ($\subseteq$).

Reason		Reason	
(a)	F $\emptyset$ has no members	(h)	T $\emptyset$ is the member of $\{\emptyset\}$
(b)	F the only member of $\{0\}$ is 0	(i)	T $\emptyset$ is listed
(c)	F nothing is a proper subset of $\emptyset$	(j)	F the only member of $\{\emptyset\}$ is $\emptyset$, not $\{\emptyset\}$
(d)	T $\emptyset \subseteq$ every set, and $\emptyset \neq \{0\}$	(k)	T the only member of $\{\{\emptyset\}\}$ is $\{\emptyset\}$
(e)	F the only member is 0, not $\{0\}$	(l)	T $\emptyset$ is a member of the right side, which also has the extra member $\{\emptyset\}$
(f)	F a set is not a <i>proper</i> subset of itself	(m)	T $\{\emptyset\}$ is a member of the right side; proper because $\emptyset$ is also there
(g)	T every set is a subset of itself	(n)	F $\{\{\emptyset\}, \{\emptyset\}\} = \{\{\emptyset\}\}$, so the sets are equal

PRACTICE • Rosen §2.1 Ex. 7, 8 and 18 • $\in$ versus $\subseteq$

- (a) For each set, say whether 2 is an element, and whether $\{2\}$ is an element: (i) $\{x \in \mathbb{R} \mid x \text{ is an integer greater than } 1\}$; (ii) $\{x \in \mathbb{R} \mid x \text{ is the square of an integer}\}$; (iii) $\{2, \{2\}\}$; (iv) $\{\{2\}, \{\{2\}\}\}$; (v) $\{\{2\}, \{2, \{2\}\}\}$; (vi) $\{\{\{2\}\}\}$.
 (b) Find two sets A and B such that $A \in B$ and $A \subseteq B$.

Solution. (a) List the members of each set and compare.

Set	$2 \in \text{set?}$	$\{2\} \in \text{set?}$
(i) all reals that are integers > 1	T	F (members are numbers)
(ii) squares of integers $(0, 1, 4, 9, \dots)$	F (2 is not a square)	F
(iii) $\{2, \{2\}\}$	T	T
(iv) $\{\{2\}, \{\{2\}\}\}$	F	T
(v) $\{\{2\}, \{2, \{2\}\}\}$	F	T
(vi) $\{\{\{2\}\}\}$	F	F (the member is $\{\{2\}\}$)

(b) We need A to be both a *member* of B and a *subset* of B , so every member of A must also be in B . Take $A = \emptyset$ and $B = \{\emptyset\}$: $\emptyset \in B$, and $\emptyset \subseteq B$. Another: $A = \{a\}$, $B = \{a, \{a\}\}$ (then $a \in B$ gives $A \subseteq B$, and $\{a\}$ is listed in B , so $A \in B$).

PRACTICE • Rosen §2.1 Ex. 20, 21, 23, 24 • cardinality and power sets

- (a) Find the cardinality of $\emptyset$, $\{\emptyset\}$, $\{\emptyset, \{\emptyset\}\}$ and $\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$.
(b) Find $\mathcal{P}(\{\emptyset, \{\emptyset\}\})$.
(c) How many elements do $\mathcal{P}(\{a, b, \{a, b\}\})$, $\mathcal{P}(\{\emptyset, a, \{a\}, \{\{a\}\}\})$ and $\mathcal{P}(\mathcal{P}(\emptyset))$ have?
(d) Is each of these the power set of some set? (i) $\emptyset$ (ii) $\{\emptyset, \{a\}\}$ (iii) $\{\emptyset, \{a\}, \{\emptyset, a\}\}$ (iv) $\{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

Solution. (a) $|\emptyset| = 0$, $|\{\emptyset\}| = 1$, $|\{\emptyset, \{\emptyset\}\}| = 2$, and the last set has 3 members ($\emptyset$, $\{\emptyset\}$, $\{\emptyset, \{\emptyset\}\}$), so 3.

(b) The set $S = \{\emptyset, \{\emptyset\}\}$ has 2 members, so $\mathcal{P}(S)$ has 4: $\mathcal{P}(S) = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$.

(c) Count the members of the inner set first, then use $|\mathcal{P}(S)| = 2^{|S|}$: $\{a, b, \{a, b\}\}$ has 3 members, so $2^3 = 8$. $\{\emptyset, a, \{a\}, \{\{a\}\}\}$ has 4 members, so $2^4 = 16$. $\mathcal{P}(\emptyset) = \{\emptyset\}$ has 1 member, so $\mathcal{P}(\mathcal{P}(\emptyset)) = \{\emptyset, \{\emptyset\}\}$ has $2^1 = 2$.

(d) A power set always contains $\emptyset$ and has 2^n members. (i) No: $\emptyset$ has no members, but $\mathcal{P}(S) \ni \emptyset$ always. (ii) Yes: $\mathcal{P}(\{a\}) = \{\emptyset, \{a\}\}$. (iii) No: it has 3 members and 3 is not a power of 2. (iv) Yes: $\mathcal{P}(\{a, b\})$.

PRACTICE • Rosen §2.1 Ex. 22 and 25 • proofs about power sets

- (a) Prove that $\mathcal{P}(A) \subseteq \mathcal{P}(B)$ if and only if $A \subseteq B$.
(b) If A and B have the same power set, can you conclude $A = B$?

Solution. (a) ($\Leftarrow$) Let $A \subseteq B$ and take any $X \in \mathcal{P}(A)$, so $X \subseteq A$. Then $X \subseteq A \subseteq B$, so $X \subseteq B$, i.e. $X \in \mathcal{P}(B)$. Hence $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

($\Rightarrow$) Let $\mathcal{P}(A) \subseteq \mathcal{P}(B)$. Since $A \subseteq A$, we have $A \in \mathcal{P}(A)$, so $A \in \mathcal{P}(B)$, which means $A \subseteq B$. ■

(b) Yes. If $\mathcal{P}(A) = \mathcal{P}(B)$ then $\mathcal{P}(A) \subseteq \mathcal{P}(B)$ gives $A \subseteq B$ and $\mathcal{P}(B) \subseteq \mathcal{P}(A)$ gives $B \subseteq A$ by part (a). So $A = B$.

PRACTICE • Rosen §2.1 Ex. 26, 31–38 • Cartesian products

Let $A = \{a, b, c\}$, $B = \{x, y\}$, $C = \{0, 1\}$.

- (a) List $A \times B \times C$ and $B \times B \times B$.
(b) How many elements do $A \times B$, $A \times B \times C$ and A^n have if $|A| = m$, $|B| = n$, $|C| = p$?
(c) Prove: (i) $\emptyset \times A = A \times \emptyset = \emptyset$; (ii) if $A \subseteq C$ and $B \subseteq D$ then $A \times B \subseteq C \times D$; (iii) for non-empty A, B : $A \times B = B \times A$ only if $A = B$; (iv) if $A \times B = \emptyset$ then $A = \emptyset$ or $B = \emptyset$.

Solution. (a) $A \times B \times C$ consists of all triples (letter from A , letter from B , digit); there are $3 \cdot 2 \cdot 2 = 12$:

$$(a, x, 0), (a, x, 1), (a, y, 0), (a, y, 1), (b, x, 0), (b, x, 1), \\ (b, y, 0), (b, y, 1), (c, x, 0), (c, x, 1), (c, y, 0), (c, y, 1).$$

$B \times B \times B$ has $2^3 = 8$ triples:

$$(x, x, x), (x, x, y), (x, y, x), (x, y, y), (y, x, x), (y, x, y), (y, y, x), (y, y, y).$$

(b) One choice for the first coordinate per element of the first set and one for the second, so the counts multiply: $|A \times B| = mn$, $|A \times B \times C| = mnp$, $|A^n| = m^n$. (Check: $3 \cdot 2 \cdot 2 = 12$.)

(c)(i) $A \times \emptyset$ would need a pair (a, b) with $b \in \emptyset$; there is none, so it is empty. Same for $\emptyset \times A$.

(ii) Take $(x, y) \in A \times B$. Then $x \in A \subseteq C$ and $y \in B \subseteq D$, so $(x, y) \in C \times D$.

(iii) If $A = B$ then trivially $A \times B = B \times A$. Conversely suppose $A \times B = B \times A$ with $A, B \neq \emptyset$. Take $a \in A$; pick any $b \in B$. Then $(a, b) \in A \times B = B \times A$, so $a \in B$. Hence $A \subseteq B$; the same argument with the roles swapped gives $B \subseteq A$. So $A = B$. (Example of failure: $A = \{1\}$, $B = \{2\}$: $(1, 2) \neq (2, 1)$.)

(iv) Contrapositive: if $A \neq \emptyset$ and $B \neq \emptyset$, pick $a \in A, b \in B$; then $(a, b) \in A \times B$, so $A \times B \neq \emptyset$.

PRACTICE • Rosen §2.2 Ex. 14 and 25 • compute set expressions; recover the sets

(a) Let $A = \{0, 2, 4, 6, 8, 10\}$, $B = \{0, 1, 2, 3, 4, 5, 6\}$, $C = \{4, 5, 6, 7, 8, 9, 10\}$. Find $A \cap B \cap C$, $A \cup B \cup C$, $(A \cup B) \cap C$ and $(A \cap B) \cup C$.

(b) Find A and B if $A - B = \{1, 5, 7, 8\}$, $B - A = \{2, 10\}$ and $A \cap B = \{3, 6, 9\}$.

Solution. (a) $A \cap B = \{0, 2, 4, 6\}$ and $A \cup B = \{0, 1, 2, 3, 4, 5, 6, 8, 10\}$. Then

- $A \cap B \cap C = \{0, 2, 4, 6\} \cap C = \{4, 6\}$;
- $A \cup B \cup C = \{0, 1, 2, \dots, 10\}$;
- $(A \cup B) \cap C = \{4, 5, 6, 8, 10\}$;
- $(A \cap B) \cup C = \{0, 2\} \cup C = \{0, 2, 4, 5, 6, 7, 8, 9, 10\}$.

(b) Every element of A is either in B or not, so $A = (A - B) \cup (A \cap B) = \{1, 5, 7, 8\} \cup \{3, 6, 9\} = \{1, 3, 5, 6, 7, 8, 9\}$. Likewise $B = (B - A) \cup (A \cap B) = \{2, 10\} \cup \{3, 6, 9\} = \{2, 3, 6, 9, 10\}$. Check: $A - B = \{1, 5, 7, 8\}$ ✓.

PRACTICE • Rosen §2.2 Ex. 16, 18, 24 • prove a set identity

Let A, B, C be sets. Prove:

- (a) $A \cup (B - A) = A \cup B$; (b) $(B - A) \cup (C - A) = (B \cup C) - A$; (c) $(A - B) - C = (A - C) - (B - C)$.

Solution. Use $X - Y = X \cap \bar{Y}$ and the identities in the table.

(a) $A \cup (B - A) = A \cup (B \cap \bar{A}) = (A \cup B) \cap (A \cup \bar{A}) = (A \cup B) \cap U = A \cup B$. (Distributive, complement, identity laws.)

(b) $(B - A) \cup (C - A) = (B \cap \bar{A}) \cup (C \cap \bar{A}) = (B \cup C) \cap \bar{A} = (B \cup C) - A$ (distributive law, backwards).

(c) Left side: $(A - B) - C = A \cap \bar{B} \cap \bar{C}$. Right side:

$$\begin{aligned} (A - C) - (B - C) &= (A \cap \bar{C}) \cap \overline{(B \cap \bar{C})} = (A \cap \bar{C}) \cap (\bar{B} \cup C) \quad (\text{De Morgan, } \bar{\bar{C}} = C) \\ &= (A \cap \bar{C} \cap \bar{B}) \cup (A \cap \bar{C} \cap C) = (A \cap \bar{B} \cap \bar{C}) \cup \emptyset = A \cap \bar{B} \cap \bar{C}. \end{aligned}$$

Both sides equal $A \cap \bar{B} \cap \bar{C}$. ■

PRACTICE • Rosen §2.2 Ex. 17 • membership table

Show that $\overline{A \cap B \cap C} = \bar{A} \cup \bar{B} \cup \bar{C}$ (a) by showing each side is a subset of the other, (b) using a membership table.

Solution. (a) If $x \in \overline{A \cap B \cap C}$, then $x \notin A \cap B \cap C$, so x fails to be in at least one of A, B, C ; that is, $x \in \bar{A}$ or $x \in \bar{B}$ or $x \in \bar{C}$, so $x \in \bar{A} \cup \bar{B} \cup \bar{C}$. Reversing each step gives the other inclusion.

(b) Eight rows, one for each way an element can lie in A, B, C (1 = in, 0 = not in):

A	B	C	$A \cap B \cap C$	$\overline{A \cap B \cap C}$	$\bar{A}$	$\bar{B}$	$\bar{C}$	$\bar{A} \cup \bar{B} \cup \bar{C}$
1	1	1	1	0	0	0	0	0
1	1	0	0	1	0	0	1	1
1	0	1	0	1	0	1	0	1
1	0	0	0	1	0	1	1	1
0	1	1	0	1	1	0	0	1
0	1	0	0	1	1	0	1	1
0	0	1	0	1	1	1	0	1
0	0	0	0	1	1	1	1	1

The two bold columns are identical, so the sets are equal.

PRACTICE • Rosen §2.2 Ex. 29 and 30 • what can you conclude?

(a) What can you say about A and B if (i) $A \cup B = A$; (ii) $A \cap B = A$; (iii) $A - B = A$; (iv) $A \cap B = B \cap A$; (v) $A - B = B - A$?

(b) Can you conclude $A = B$ if (i) $A \cup C = B \cup C$; (ii) $A \cap C = B \cap C$; (iii) both (i) and (ii) hold?

Solution. (a) (i) $A \cup B = A$ means every element of B is already in A : $B \subseteq A$. (ii) $A \cap B = A$ means every element of A is in B : $A \subseteq B$. (iii) $A - B = A$ means no element of A is removed: $A \cap B = \emptyset$ (A and B are disjoint). (iv) always true (commutativity), so nothing can be said. (v) $A - B$ and $B - A$ are disjoint, so if they are equal both are empty, giving $A \subseteq B$ and $B \subseteq A$: $A = B$.

(b) (i) No: $A = \{1\}$, $B = \emptyset$, $C = \{1\}$ give $A \cup C = \{1\} = B \cup C$ but $A \neq B$. (ii) No: $A = \{1\}$,

$B = \{2\}$, $C = \emptyset$ give $A \cap C = \emptyset = B \cap C$. (iii) Yes. Let $x \in A$. If $x \in C$ then $x \in A \cap C = B \cap C$, so $x \in B$. If $x \notin C$ then $x \in A \cup C = B \cup C$ and $x \notin C$, so $x \in B$. Hence $A \subseteq B$, and by symmetry $B \subseteq A$. ■

PRACTICE • Rosen §2.2 Ex. 32, 35–41 • symmetric difference

- (a) Find $\{1, 3, 5\} \oplus \{1, 2, 3\}$.
- (b) Show that $A \oplus B = (A \cup B) - (A \cap B)$.
- (c) For $A \subseteq U$ show that $A \oplus A = \emptyset$, $A \oplus \emptyset = A$, $A \oplus U = \bar{A}$ and $A \oplus \bar{A} = U$.
- (d) Show that $A \oplus B = B \oplus A$, that $(A \oplus B) \oplus B = A$, and that $A \oplus B = A$ implies $B = \emptyset$.
- (e) If $A \oplus C = B \oplus C$, must $A = B$?

Solution. (a) Elements in exactly one set: $\{2, 5\}$.

(b) $x \in A \oplus B$ iff x is in exactly one of A, B . That means $x \in A \cup B$ (in at least one) but $x \notin A \cap B$ (not in both), i.e. $x \in (A \cup B) - (A \cap B)$. ■

(c) Write p for “ $x \in A$ ”. Then “ $x \in A \oplus A$ ” is $p \oplus p$, which is always false, so $A \oplus A = \emptyset$. Similarly $p \oplus \text{F} \equiv p$ gives $A \oplus \emptyset = A$; $p \oplus \text{T} \equiv \neg p$ gives $A \oplus U = \bar{A}$; and $p \oplus \neg p \equiv \text{T}$ gives $A \oplus \bar{A} = U$.

(d) Exclusive or is commutative and associative, so $\oplus$ on sets is too: $A \oplus B = B \oplus A$ and $(A \oplus B) \oplus B = A \oplus (B \oplus B) = A \oplus \emptyset = A$. If $A \oplus B = A$, then

$$B = \emptyset \oplus B = (A \oplus A) \oplus B = A \oplus (A \oplus B) = A \oplus A = \emptyset.$$

(e) Yes. $A = (A \oplus C) \oplus C = (B \oplus C) \oplus C = B$, using $(X \oplus C) \oplus C = X$ from (d).

PRACTICE • Rosen §2.2 Ex. 46 • inclusion–exclusion for three sets

Show that $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$. Use it to find how many students take at least one subject if 30 take Computer Science, 25 Mathematics, 20 Physics, 10 take CS and Maths, 8 CS and Physics, 5 Maths and Physics, and 3 take all three.

Solution. Proof. For two sets, $|X \cup Y| = |X| + |Y| - |X \cap Y|$. Apply it with $X = A \cup B$ and $Y = C$:

$$|A \cup B \cup C| = |A \cup B| + |C| - |(A \cup B) \cap C|.$$

Expand $|A \cup B| = |A| + |B| - |A \cap B|$. By the distributive law $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$, and applying the two-set formula again,

$$|(A \cap C) \cup (B \cap C)| = |A \cap C| + |B \cap C| - |A \cap B \cap C|.$$

Substituting both gives the three-set formula. ■

Numbers. $30 + 25 + 20 - 10 - 8 - 5 + 3 = 55$ students take at least one subject. (Adding the sizes counts a student in two sets twice, so pairs are subtracted; subtracting pairs removes the triple-overlap students altogether, so they are added back once.)

PRACTICE • Rosen §2.2 Ex. 50 and 51 • indexed unions and intersections

Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$ for every positive integer i if

- (a) $A_i = \{i, i+1, i+2, \dots\}$; (b) $A_i = \{0, i\}$; (c) $A_i = (0, i)$, the reals x with $0 < x < i$; (d) $A_i = (i, \infty)$; (e) $A_i = \{-i, i\}$; (f) $A_i = [-i, i]$.

Solution. Ask: which x lie in *some* A_i (union) and which lie in *every* A_i (intersection)?

	A_i	$\cup A_i$	$\cap A_i$
(a)	$\{i, i+1, \dots\}$ (shrinks)	$\{1, 2, 3, \dots\} = \mathbb{Z}^+$	$\emptyset$ (no x is $\geq i$ for every i)
(b)	$\{0, i\}$	$\{0, 1, 2, 3, \dots\}$	$\{0\}$
(c)	$(0, i)$ (grows)	$(0, \infty)$	$(0, 1)$
(d)	(i, ∞) (shrinks)	$(1, \infty)$	$\emptyset$
(e)	$\{-i, i\}$	$\mathbb{Z} - \{0\}$	$\emptyset$
(f)	$[-i, i]$ (grows)	$\mathbb{R}$	$[-1, 1]$

For a growing family the union is “the limit” and the intersection is A_1 ; for a shrinking family the intersection is the “limit” and the union is A_1 . In (b) the number 0 is in every A_i , and in (e) $A_1 = \{-1, 1\}$ but $1 \notin A_2 = \{-2, 2\}$, so nothing survives.

PRACTICE • Rosen §2.2 Ex. 52–56 • sets as bit strings

Let $U = \{1, 2, \dots, 10\}$ and represent a subset by a 10-bit string whose i th bit is 1 iff i is in the set.

- Write bit strings for $\{3, 4, 5\}$, $\{1, 3, 6, 10\}$ and $\{2, 3, 4, 7, 8, 9\}$.
- Find the sets given by 11 1100 1111, 01 0111 1000 and 10 0000 0001.
- What are the bit strings of $\emptyset$ and of U ? What operation on bit strings gives $A - B$ and $A \oplus B$?

Solution. (a) Put 1 in each listed position: $\{3, 4, 5\}$ is 00 1110 0000; $\{1, 3, 6, 10\}$ is 10 1001 0001; $\{2, 3, 4, 7, 8, 9\}$ is 01 1100 1110.

(b) Read the positions of the 1s: 11 1100 1111 = $\{1, 2, 3, 4, 7, 8, 9, 10\}$ (positions 5, 6 are 0); 01 0111 1000 = $\{2, 4, 5, 6, 7\}$; 10 0000 0001 = $\{1, 10\}$.

(c) $\emptyset$ is all zeros 00 0000 0000; U is all ones. $A - B = A \cap \overline{B}$ is **A AND (NOT B)**: keep a bit of A only where B has 0. $A \oplus B$ is bitwise **XOR**. Example with $A = \{1, 3, 5\} = 10 1010 0000$ and $B = \{1, 2, 3\} = 11 1000 0000$: $A - B$: 10 1010 0000 AND 00 0111 1111 = 00 0010 0000 = $\{5\}$; $A \oplus B = 01 0010 0000 = \{2, 5\}$, agreeing with the answer to the symmetric-difference question.

Relations

6.1 Relations and ordered pairs

DEFINITION — Relation

A *relation from A to B* is a subset $R \subseteq A \times B$. A *relation on A* is a subset of $A \times A$. We write aRb (or $(a, b) \in R$) to say a is related to b . Because the elements are *ordered* pairs, (a, b) and (b, a) are different, so aRb does not automatically give bRa .

EXAMPLE

$A = \{1, 2, 3\}$ and $R = \{(a, b) \in A \times A \mid a < b\} = \{(1, 2), (1, 3), (2, 3)\}$. Here $1R3$ but not $3R1$. A set with $|A| = n$ has 2^{n^2} relations on it (each of the n^2 pairs is in or out).

6.2 Properties of relations

Let R be a relation on a set A .

DEFINITION — Reflexive

R is *reflexive* if every element is related to itself: $\forall a \in A, (a, a) \in R$.

EXAMPLE

$A = \{1, 2, 3, 4\}$.

$R_1 = \{(1, 2), (2, 3), (1, 1)\}$ is **not** reflexive: $(2, 2), (3, 3), (4, 4)$ are missing.

$R_2 = \{(1, 1), (2, 2), (1, 2), (3, 4), (3, 3), (4, 4)\}$ is reflexive: $(1, 1), (2, 2), (3, 3), (4, 4)$ are all present.

“ $a \mid b$ ” on the positive integers is reflexive since $a \mid a$. (On all integers $\mathbb{Z}$ it fails: the usual textbook definition of $a \mid b$ requires $a \neq 0$, so $0 \nmid 0$.)

DEFINITION — Symmetric

R is *symmetric* if whenever aRb then bRa : $\forall a, b \in A, (a, b) \in R \rightarrow (b, a) \in R$.

EXAMPLE

$R_1 = \{(1, 2), (2, 1), (3, 4)\}$ is **not** symmetric: $(3, 4) \in R_1$ but $(4, 3) \notin R_1$.

$R_2 = \{(1, 2), (2, 1), (4, 4)\}$ is symmetric (each pair has its reverse; $(4, 4)$ is its own reverse).

“ $a \mid b$ ” is not symmetric: $2 \mid 4$ but $4 \nmid 2$.

DEFINITION — Antisymmetric

R is *antisymmetric* if the only way to have both aRb and bRa is $a = b$: $\forall a, b, (aRb \wedge bRa) \rightarrow a = b$. (Equivalently: no pair of *distinct* elements is related in both directions.)

EXAMPLE

$R_1 = \{(1,2), (3,4)\}$ and $R_2 = \{(1,2), (1,1), (3,3)\}$ are antisymmetric. $R_3 = \{(3,4), (4,3), (1,1)\}$ is **not**: $3R4$ and $4R3$ with $3 \neq 4$. “ $a \mid b$ ” on the positive integers is antisymmetric: $a \mid b$ and $b \mid a$ force $a = b$.

WATCH OUT

“Antisymmetric” is not the opposite of “symmetric”. A relation can be both (e.g. $\{(1,1), (2,2)\}$), or neither (e.g. $\{(1,2), (2,1), (2,3)\}$).

DEFINITION — Transitive

R is *transitive* if aRb and bRc always give aRc : $\forall a, b, c \in A, (aRb \wedge bRc) \rightarrow aRc$.

EXAMPLE

$R_1 = \{(1,2), (2,3), (3,4), (1,3)\}$ is **not** transitive: $(2,3), (3,4) \in R_1$ but $(2,4) \notin R_1$.
 $R_2 = \{(1,2), (3,4)\}$ is transitive (no chain aRb, bRc exists, so the condition holds vacuously).
 $R_3 = \{(1,2), (2,3), (1,3)\}$ is transitive: the only chain $(1,2), (2,3)$ has $(1,3) \in R_3$.

DEFINITION — Irreflexive and asymmetric[†]

R is *irreflexive* if no element is related to itself: $\forall a \in A, (a,a) \notin R$. R is *asymmetric* if aRb always rules out bRa : $\forall a, b \in A, (a,b) \in R \rightarrow (b,a) \notin R$. (Taking $b = a$ shows that an asymmetric relation has no loops.)

EXAMPLE

On $A = \{1,2,3\}$: $R_1 = \{(1,2), (2,3)\}$ is irreflexive and asymmetric. $R_2 = \{(1,2), (2,1)\}$ is irreflexive but **not** asymmetric ($(1,2)$ and $(2,1)$ are both present). $R_3 = \{(1,1), (1,2)\}$ is not irreflexive (it has the loop $(1,1)$) and also not reflexive ($(2,2) \notin R_3$). The relation “ $<$ ” on $\mathbb{Z}$ is asymmetric; “ $\leq$ ” is not.

WATCH OUT

“Not reflexive” does not mean “irreflexive”: R_3 above is neither. And **asymmetric = antisymmetric + irreflexive**: an asymmetric relation is always antisymmetric, but an antisymmetric one need not be asymmetric (for example $\{(1,1)\}$).

Relation	Reflexive	Irreflexive	Symmetric	Antisymmetric	Asymmetric	Transitive
$\leq$ on $\mathbb{Z}$	T	F	F	T	F	T
$<$ on $\mathbb{Z}$	F	T	F	T	T	T
$=$ on a non-empty set	T	F	T	T	F	T
$\mid$ on $\mathbb{Z}^+$	T	F	F	T	F	T
$\neq$ on $\{1,2,3\}$	F	T	T	F	F	F
congruence mod 3 on $\mathbb{Z}$	T	F	T	F	F	T

6.3 Representing relations

DEFINITION — Matrix representation

For a relation R on $A = \{a_1, \dots, a_n\}$, the *relation matrix* M is the $n \times n$ matrix with $M_{ij} = 1$ if $(a_i, a_j) \in R$ and 0 otherwise.

EXAMPLE

$A = \{1, 2, 3\}$, $R = \{(1, 1), (3, 1), (3, 2)\}$:

$$M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}$$

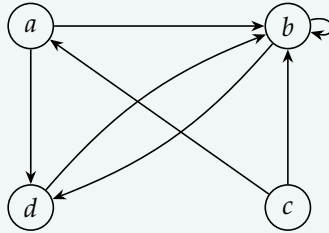
Row i , column j is 1 exactly when iRj ; e.g. row 3 has 1s in columns 1 and 2 for $(3, 1), (3, 2)$.

DEFINITION — Digraph representation

The *directed graph* of R has a vertex for each element of A and an arrow $a \rightarrow b$ for each $(a, b) \in R$; a pair (a, a) is drawn as a loop.

EXAMPLE

$R = \{(a, b), (a, d), (b, b), (c, a), (c, b), (d, b), (b, d)\}$ on $\{a, b, c, d\}$:



REMEMBER — reading properties off a matrix or digraph

- **Reflexive:** all diagonal entries are 1 / every vertex has a loop.
- **Symmetric:** M equals its transpose / every arrow has a reverse arrow.
- **Antisymmetric:** no off-diagonal pair with $M_{ij} = M_{ji} = 1$ / no two-way arrows between different vertices.
- **Transitive:** whenever there is a path $a \rightarrow b \rightarrow c$ there is also a direct arrow $a \rightarrow c$.

DEFINITION — Join, meet and Boolean product of 0–1 matrices[†]

For 0–1 matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ of the same size, the *join* $A \vee B$ has entries $a_{ij} \vee b_{ij}$ and the *meet* $A \wedge B$ has entries $a_{ij} \wedge b_{ij}$. If A is $m \times k$ and B is $k \times n$, the *Boolean product* $A \odot B$ is the $m \times n$ matrix with

$$(A \odot B)_{ij} = (a_{i1} \wedge b_{1j}) \vee (a_{i2} \wedge b_{2j}) \vee \dots \vee (a_{ik} \wedge b_{kj}),$$

that is, ordinary matrix multiplication with $\times$ replaced by $\wedge$ and $+$ by $\vee$. The Boolean power $M^{[n]}$ is $M \odot M \odot \dots \odot M$ (n factors).

REMEMBER — what the matrix operations mean

- $M_{R_1 \cup R_2} = M_{R_1} \vee M_{R_2}$ and $M_{R_1 \cap R_2} = M_{R_1} \wedge M_{R_2}$.
- $M_{R^{-1}}$ is the **transpose** of M_R ; $M_{\bar{R}}$ is M_R with every bit flipped.
- $M_{S \circ R} = M_R \odot M_S$ (composition is defined in the next section) and $M_{R^n} = M_R^{[n]}$.

EXAMPLE

On $\{1, 2, 3\}$ let $R = \{(1, 2), (2, 3)\}$ and $S = \{(2, 1), (3, 3)\}$. Then

$$M_R = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad M_S = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad M_R \vee M_S = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} (= R \cup S),$$

$$M_R \wedge M_S = \mathbf{0} \quad (R \cap S = \emptyset), \quad M_R \odot M_S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

For the product, row 1 of M_R is $(0, 1, 0)$, so row 1 of the result is row 2 of M_S , namely $(1, 0, 0)$; row 2 of M_R is $(0, 0, 1)$, giving row 3 of M_S , $(0, 0, 1)$; row 3 is zero. The result is $\{(1, 1), (2, 3)\}$, which is the composition $S \circ R$.

6.4 Combining relations[†]

Relations are sets of ordered pairs, so union, intersection, difference and symmetric difference apply directly. Three further constructions are specific to relations.

DEFINITION — Complement, inverse, composition, powers

Let R be a relation from A to B .

- **Complement:** $\bar{R} = (A \times B) - R = \{(a, b) \mid (a, b) \notin R\}$.
- **Inverse:** $R^{-1} = \{(b, a) \mid (a, b) \in R\}$ (reverse every pair). R is symmetric exactly when $R = R^{-1}$.
- **Composition:** if S is a relation from B to C , then $S \circ R = \{(a, c) \mid \exists b \in B ((a, b) \in R \wedge (b, c) \in S)\}$: first R , then S . In the digraph, $(a, c) \in S \circ R$ when there is a two-step path $a \rightarrow b \rightarrow c$ whose first arrow is in R and second arrow is in S .
- **Powers** of a relation R on A : $R^1 = R$ and $R^{n+1} = R^n \circ R$.

EXAMPLE — complement and inverse

On $A = \{1, 2, 3\}$ let $R = \{(1, 2), (2, 3)\}$. Then $R^{-1} = \{(2, 1), (3, 2)\}$ and $\bar{R}$ consists of the other $9 - 2 = 7$ pairs: $\{(1, 1), (1, 3), (2, 1), (2, 2), (3, 1), (3, 2), (3, 3)\}$.

EXAMPLE — composition

$R = \{(1,1), (1,4), (2,3), (3,1), (3,4)\}$ from $A = \{1,2,3\}$ to $B = \{1,2,3,4\}$ and $S = \{(1,0), (2,0), (3,1), (3,2), (4,1)\}$ from B to $C = \{0,1,2\}$. For each $(a,b) \in R$, look up every $(b,c) \in S$:

$(1,1) \in R, (1,0) \in S$	gives $(1,0)$
$(1,4) \in R, (4,1) \in S$	gives $(1,1)$
$(2,3) \in R, (3,1), (3,2) \in S$	gives $(2,1), (2,2)$
$(3,1) \in R, (1,0) \in S$	gives $(3,0)$
$(3,4) \in R, (4,1) \in S$	gives $(3,1)$

So $S \circ R = \{(1,0), (1,1), (2,1), (2,2), (3,0), (3,1)\}$.

EXAMPLE — powers

$R = \{(1,2), (2,3), (3,4)\}$ on $\{1,2,3,4\}$ (a path $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$). R^2 joins pairs two steps apart: $R^2 = \{(1,3), (2,4)\}$; $R^3 = \{(1,4)\}$; $R^4 = \emptyset$.

REMEMBER — a test for transitivity

R is transitive $\iff R \circ R \subseteq R \iff R^n \subseteq R$ for every $n \geq 1$. (In the path example $R^2 \not\subseteq R$, so R is not transitive.)

WATCH OUT

Books differ on the order in $S \circ R$. Here (as in Rosen) $S \circ R$ means “first R , then S ”, the same convention as for functions ($f \circ g$ means g first). Always check which convention a question uses.

6.5 Closures[†]

DEFINITION — Closure of a relation

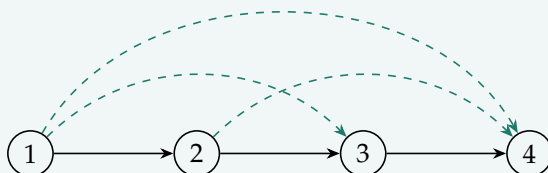
The *closure* of R with respect to a property P is the **smallest** relation that contains R and has property P (smallest means it is contained in every relation with P that contains R), when such a relation exists.

Closure	Formula	How to build it
reflexive	$R \cup \Delta$, where $\Delta = \{(a,a) \mid a \in A\}$	add the missing loops
symmetric $\sigma(R)$	$R \cup R^{-1}$	add the reverse of every pair
transitive R^*	$R \cup R^2 \cup \dots \cup R^n, n = A $	add (a,c) whenever a path $a \rightarrow \dots \rightarrow c$ exists

EXAMPLE

$R = \{(1,2), (2,3), (3,4)\}$ on $A = \{1,2,3,4\}$.

- Reflexive closure: $R \cup \{(1,1), (2,2), (3,3), (4,4)\}$.
- Symmetric closure: $R \cup \{(2,1), (3,2), (4,3)\}$.
- Transitive closure: $R \cup R^2 \cup R^3 = \{(1,2), (2,3), (3,4), (1,3), (2,4), (1,4)\}$. The three new pairs are dashed below.



EXAMPLE — symmetric closure

$R = \{(1,2), (1,3), (2,3)\}$ on $\{1,2,3\}$. Then $R^{-1} = \{(2,1), (3,1), (3,2)\}$ and $\sigma(R) = \{(1,2), (2,1), (1,3), (3,1), (2,3), (3,2)\}$, which is symmetric.

REMEMBER — equivalence relation from any relation

The smallest equivalence relation containing R is obtained by taking the reflexive closure, then the symmetric closure, then the transitive closure. Do the **transitive closure last**: the transitive closure of a symmetric relation is still symmetric, but making a transitive relation symmetric can destroy transitivity.

6.6 Equivalence relations

DEFINITION — Equivalence relation

A relation that is **reflexive, symmetric and transitive**.

EXAMPLE — congruence modulo 3

On $\mathbb{Z}$, let $a \sim b$ iff $3 \mid (a - b)$. Reflexive: $a - a = 0$ is divisible by 3. Symmetric: if $3 \mid (a - b)$ then $3 \mid (b - a)$. Transitive: if $3 \mid (a - b)$ and $3 \mid (b - c)$ then $3 \mid (a - b) + (b - c) = a - c$. So $\sim$ is an equivalence relation.

It splits $\mathbb{Z}$ into three *equivalence classes*[†] $[0] = \{\dots, -3, 0, 3, 6, \dots\}$, $[1] = \{\dots, 1, 4, 7, \dots\}$, $[2] = \{\dots, 2, 5, 8, \dots\}$, which are disjoint and cover $\mathbb{Z}$.

EXAMPLE — not an equivalence relation

$(\mathbb{N}, \leq)$: reflexive ($a \leq a$) and transitive, but not symmetric ($1 \leq 2$ yet $2 \not\leq 1$). So $\leq$ is not an equivalence relation.

DEFINITION — Equivalence class and partition[†]

Let R be an equivalence relation on A . The *equivalence class* of a is $[a] = \{x \in A \mid aRx\}$. A *partition* of A is a collection of non-empty, pairwise disjoint subsets of A whose union is A .

REMEMBER — equivalence relations $\leftrightarrow$ partitions

For an equivalence relation R on A the following are equivalent: aRb ; $[a] = [b]$; $[a] \cap [b] \neq \emptyset$. Hence the classes form a partition of A . Conversely, each partition of A gives an equivalence relation: aRb iff a and b lie in the same block. So equivalence relations on A and partitions of A correspond one-to-one.

EXAMPLE

Partition $\{a, b, c, d\}$ into blocks $\{a, b\}$ and $\{c, d\}$. The relation “in the same block” is

$$R = \{(a, a), (a, b), (b, a), (b, b), (c, c), (c, d), (d, c), (d, d)\},$$

which has $2^2 + 2^2 = 8$ pairs (a block of size k contributes k^2 pairs). Here $(a, b) \in R$ but $(b, c) \notin R$, and $[a] = [b] = \{a, b\}$, $[c] = [d] = \{c, d\}$ are disjoint with union $\{a, b, c, d\}$.

EXAMPLE — congruence classes

For congruence modulo m there are exactly m classes $[0], [1], \dots, [m-1]$, where $[r] = \{\dots, r - m, r, r + m, r + 2m, \dots\}$. Modulo 5: $[2] = \{\dots, -8, -3, 2, 7, 12, \dots\}$, and $[7] = [2]$ because $7 \equiv 2 \pmod{5}$.

6.7 Partial orders

DEFINITION — Partial order

A relation that is **reflexive**, **antisymmetric** and **transitive**. A set A together with a partial order $\preceq$ is a *partially ordered set* (poset) $(A, \preceq)$. Examples: $(\mathbb{N}, \leq)$ and $(\mathbb{Z}^+, |)$. Posets are studied in Chapter 7.

6.8 n -ary relations and databases[†]

DEFINITION — n -ary relation

An n -ary relation on sets $A_1, \dots, A_n$ is a subset of $A_1 \times \dots \times A_n$; n is its *degree* and the A_i are its *domains*. (A binary relation has degree 2.) In a relational database an n -ary relation is a table: each row (*record*) is an n -tuple and each column a *field*. A domain is a *primary key* if its value identifies every record uniquely. Three operations build new tables:

- **selection** s_C : keep the records that satisfy condition C ;
- **projection** $P_{i_1, \dots, i_m}$: keep only the listed fields (and remove duplicate rows);
- **join** J_p : combine records of two tables whose last p fields of the first equal the first p fields of the second.

EXAMPLE

Students = {(101, Asha, CS, 1), (102, Ravi, Math, 2), (103, Mei, CS, 2), (104, Omar, CS, 1)} has degree 4 (ID, Name, Major, Year). ID is a primary key; Major is not (CS repeats).

Selection with C: Major = CS $\wedge$ Year = 1 gives {(101, Asha, CS, 1), (104, Omar, CS, 1)}.

Projection $P_{3,4}$ gives {(CS,1), (Math,2), (CS,2)}: the rows (CS, 1) from records 101 and 104 collapse into one.

Join J_1 of {(Asha, CS), (Ravi, Math)} and {(CS, Engineering), (Math, Science)} gives {(Asha, CS, Engineering), (Ravi, Math, Science)}.

6.9 Solved previous-year questions

PYQ • Mid-Sem Oct 2023 Q2(c) • 3 marks

Let $X = \{1, 2, 3\}$ and R a relation on X defined by $xRy \iff x \neq y$ for all $x, y \in X$. Find the elements of the relation R and R^{-1} . Check whether R is transitive.

Solution. xRy holds exactly for pairs of *different* elements, so all $3 \times 3 = 9$ pairs except the three diagonal ones:

$$R = \{(1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2)\}.$$

R^{-1} reverses each pair: $(2, 1), (3, 1), (1, 2), (3, 2), (1, 3), (2, 3)$, so

$$R^{-1} = \{(1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2)\} = R$$

(as expected: R is symmetric).

Transitivity. $(1, 2) \in R$ and $(2, 1) \in R$, so transitivity would need $(1, 1) \in R$. But $1 \neq 1$ is false, so $(1, 1) \notin R$. Hence R is **not transitive**.

PYQ • Mid-Sem Oct 2023 Q3(a) • 4 marks

Let R and S be relations on X . Determine whether each statement is true or false; prove it if true, otherwise give a counterexample.

- (i) If R is transitive then R^{-1} is transitive.
- (ii) $\sigma(R_1 \cap R_2) = \sigma(R_1) \cap \sigma(R_2)$, where $\sigma(R) = R \cup R^{-1}$.
- (iii) A relation can simultaneously be symmetric and antisymmetric.
- (iv) Suppose that R is a relation on X that is symmetric and transitive but not reflexive.

Solution. (i) True. Let R be transitive and let $(a, b), (b, c) \in R^{-1}$. By definition of the inverse, $(b, a) \in R$ and $(c, b) \in R$. Apply transitivity of R to the chain $(c, b), (b, a)$: it gives $(c, a) \in R$. Hence $(a, c) \in R^{-1}$. So R^{-1} is transitive. ■

(ii) False. Take $X = \{1, 2\}$, $R_1 = \{(1, 2)\}$, $R_2 = \{(2, 1)\}$.

- $R_1 \cap R_2 = \emptyset$, so $\sigma(R_1 \cap R_2) = \emptyset$.
- $\sigma(R_1) = \{(1, 2), (2, 1)\}$ and $\sigma(R_2) = \{(2, 1), (1, 2)\}$, so $\sigma(R_1) \cap \sigma(R_2) = \{(1, 2), (2, 1)\} \neq \emptyset$.

The two sides differ. (Only the inclusion $\sigma(R_1 \cap R_2) \subseteq \sigma(R_1) \cap \sigma(R_2)$ is always true.)

(iii) True. Symmetric says $(a, b) \in R \Rightarrow (b, a) \in R$; antisymmetric says $(a, b), (b, a) \in R \Rightarrow a = b$. Together, any pair in R must have $a = b$. So exactly the subsets of the identity relation are both. Example: $R = \{(1, 1), (2, 2)\}$ on $X = \{1, 2, 3\}$ (or $R = \emptyset$). Check: $(1, 1) \in R \Rightarrow (1, 1) \in R$ (symmetric) and the only two-way pairs are (a, a) (antisymmetric).

(iv) **Such a relation exists (true).** The statement as printed is incomplete; the natural readings are “a relation that is symmetric and transitive but not reflexive can exist” (true) or “symmetric and transitive imply reflexive” (false). One example settles both: $X = \{1, 2\}$, $R = \{(1, 1)\}$.

- Symmetric: the only pair $(1, 1)$ is its own reverse.
- Transitive: the only chain is $(1, 1), (1, 1)$, giving $(1, 1) \in R$.
- Not reflexive: $(2, 2) \notin R$.

The tempting “proof” of reflexivity — $aRb \Rightarrow bRa$, then $aRb \wedge bRa \Rightarrow aRa$ — fails because it needs a to be related to *something*; the element 2 is related to nothing. (Even $R = \emptyset$ on a non-empty set works.)

6.10 Practice questions from Rosen

Chosen from Rosen §9.1–9.6 in the style of the PYQs above: find the pairs, test the properties (with a proof or a counterexample), and prove or disprove a statement. Near-duplicates of the same skill have been left out.

PRACTICE • Rosen §9.1 Ex. 3 • test the six properties

For each relation on $\{1, 2, 3, 4\}$ decide whether it is reflexive, irreflexive, symmetric, antisymmetric, asymmetric and/or transitive.

- (a) $\{(2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (3, 4)\}$ (b) $\{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$
 (c) $\{(2, 4), (4, 2)\}$ (d) $\{(1, 2), (2, 3), (3, 4)\}$ (e) $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$
 (f) $\{(1, 3), (1, 4), (2, 3), (2, 4), (3, 1), (3, 4)\}$

Solution.

	Refl.	Irrefl.	Sym.	Antisym.	Asym.	Trans.	Decisive facts
(a)	F	F	F	F	F	T	$(1, 1) \notin R$; $(2, 4) \in R$, $(4, 2) \notin R$; $(2, 3), (3, 2)$ both in
(b)	T	F	T	F	F	T	classes $\{1, 2\}, \{3\}, \{4\}$; $(1, 2), (2, 1)$ both in
(c)	F	T	T	F	F	F	$(2, 4), (4, 2) \in R$ but $(2, 2) \notin R$
(d)	F	T	F	T	T	F	$(1, 2), (2, 3) \in R$ but $(1, 3) \notin R$
(e)	T	F	T	T	F	T	the identity relation
(f)	F	T	F	F	F	F	$(1, 4) \in R$, $(4, 1) \notin R$; $(1, 3), (3, 1) \in R$ but $(1, 1) \notin R$

Why (a) is transitive: $R = \{2, 3\} \times \{2, 3, 4\}$. In a chain aRb, bRc we have $a \in \{2, 3\}$ and $c \in \{2, 3, 4\}$, so $(a, c) \in R$. In (c) and (f) the failing chain uses the same pair twice in opposite directions: symmetric pairs $(x, y), (y, x)$ force the loop (x, x) if R is to be transitive.

PRACTICE • Rosen §9.1 Ex. 6 and 7 • relations on $\mathbb{R}$ and $\mathbb{Z}$

Determine whether the relation is reflexive, symmetric, antisymmetric and/or transitive. On $\mathbb{R}$: (a) $x + y = 0$; (b) $x = \pm y$; (c) $x = 2y$; (d) $xy \geq 0$; (e) $x = 1$ or $y = 1$. On $\mathbb{Z}$: (f) $x = y + 1$ or $x = y - 1$; (g) x is a multiple of y ; (h) $x = y^2$; (i) $x \geq y^2$.

Solution.

Relation	Refl.	Sym.	Antisym.	Trans.
(a) $x + y = 0$	F	T	F	F
(b) $x = \pm y$	T	T	F	T
(c) $x = 2y$	F	F	T	F
(d) $xy \geq 0$	T	T	F	F
(e) $x = 1$ or $y = 1$	F	T	F	F
(f) $x = y \pm 1$	F	T	F	F
(g) x a multiple of y	T	F	F	T
(h) $x = y^2$	F	F	T	F
(i) $x \geq y^2$	F	F	T	T

Justifications, in column order (reflexive, symmetric, antisymmetric, transitive). Each **F** needs one counterexample; each **T** needs a proof.

- (a) $1 + 1 \neq 0$. Symmetric: $x + y = 0 \Rightarrow y + x = 0$. $(1, -1), (-1, 1) \in R$ with $1 \neq -1$. $(1, -1), (-1, 1) \in R$ but $(1, 1) \notin R$.
- (b) $x = \pm x$. Symmetric: $x = \pm y \Rightarrow y = \pm x$. $(1, -1), (-1, 1)$. Transitive: $x = \pm y$ and $y = \pm z$ give $x = \pm z$.
- (c) $(1, 1)$: $1 \neq 2$. $(2, 1) \in R$, $(1, 2) \notin R$. Antisymmetric: $x = 2y$ and $y = 2x$ give $x = 4x$, so $x = y = 0$. $(4, 2), (2, 1) \in R$ but $4 \neq 2 \cdot 1$.
- (d) $x \cdot x \geq 0$. Symmetric by commutativity. $(1, 2), (2, 1)$. $(1, 0), (0, -1) \in R$ but $1 \cdot (-1) < 0$.
- (e) $(2, 2)$. $x = 1$ or $y = 1$ is symmetric in x, y . $(1, 2), (2, 1)$. $(2, 1), (1, 3) \in R$ but $(2, 3) \notin R$.
- (f) $(1, 1)$; symmetric; $(1, 2), (2, 1)$; $(1, 2), (2, 3) \in R$ but $(1, 3) \notin R$.
- (g) $x = 1 \cdot x$. $(4, 2) \in R$ but $(2, 4) \notin R$. $(2, -2), (-2, 2)$ both in R . Transitive: $x = ky, y = lz$ give $x = (kl)z$.
- (h) $(2, 2)$ since $2 \neq 4$. $(4, 2) \in R$, $(2, 4) \notin R$. Antisymmetric: $x = y^2$ and $y = x^2$ give $x = x^4$, so $x \in \{0, 1\}$, and then $y = x$. **Not** transitive: $(16, 4), (4, 2) \in R$ but $16 \neq 2^2$.
- (i) $(2, 2)$: $2 < 4$. $(4, 2) \in R$, $(2, 4) \notin R$. Antisymmetric: $x \geq y^2 \geq 0$ and $y \geq x^2 \geq 0$ give $x \geq y^2 \geq x^4$, so $x \in \{0, 1\}$; likewise $y \in \{0, 1\}$; the mixed cases $(0, 1), (1, 0)$ are not in R , so $x = y$. Transitive: if $x \geq y^2$ and $y \geq z^2 \geq 0$ then $y^2 \geq z^4$, so $x \geq z^4 \geq z^2$ (for an integer z , $z^4 \geq z^2$). Over $\mathbb{R}$ this would fail at $z = \frac{1}{2}$.

PRACTICE • Rosen §9.1 Ex. 8–10, 15, 22 • the empty relation; properties that co-occur

(a) Show that $R = \emptyset$ on a non-empty set S is symmetric and transitive but not reflexive. (b) Show that $R = \emptyset$ on $S = \emptyset$ is reflexive, symmetric and transitive. (c) Give an example of a relation that is (i) both symmetric and antisymmetric, (ii) neither. (d) Can a relation be neither reflexive nor irreflexive? (e) Must an asymmetric relation be antisymmetric? Must an antisymmetric relation be asymmetric?

Solution. (a) Symmetry says “if $(a, b) \in R$ then $(b, a) \in R$ ”; R has no pairs, so there is nothing to check and the condition holds vacuously. Transitivity is vacuous in the same way. Reflexivity needs $(s, s) \in R$ for each $s \in S$; pick any $s \in S$: $(s, s) \notin \emptyset$, so R is not reflexive.

(b) With $S = \emptyset$ there are no elements, so “for all $a \in S$, $(a, a) \in R$ ” is vacuously true, and so are the other two conditions.

(c) (i) $\{(1, 1), (2, 2)\}$ on $\{1, 2, 3\}$ (or $\emptyset$). (ii) $\{(1, 2), (2, 1), (2, 3)\}$: not symmetric ($(2, 3)$ has no $(3, 2)$), not antisymmetric ($(1, 2), (2, 1)$ with $1 \neq 2$).

(d) Yes: $\{(1, 1), (1, 2)\}$ on $\{1, 2\}$ has the loop $(1, 1)$ (so not irreflexive) but lacks $(2, 2)$ (so not reflexive).

(e) *Asymmetric* $\Rightarrow$ *antisymmetric*: yes. If aRb and bRa both held, that would contradict asymmetry, so the premise of antisymmetry never happens and the condition holds vacuously. *Antisymmetric* $\Rightarrow$ *asymmetric*: no. $R = \{(1, 1)\}$ is antisymmetric, but $(1, 1) \in R$ and asymmetry would need $(1, 1) \notin R$.

PRACTICE • Rosen §9.1 Ex. 26, 27, 40, 41 • inverse, complement, set operations on relations

- (a) Find R^{-1} and $\bar{R}$ for (i) $R = \{(a, b) \mid a < b\}$ on $\mathbb{Z}$, (ii) $R = \{(a, b) \mid a \text{ divides } b\}$ on $\mathbb{Z}^+$.
 (b) Let R_1 be “divides” and R_2 “is a multiple of” on $\mathbb{Z}^+$. Find $R_1 \cup R_2$, $R_1 \cap R_2$, $R_1 - R_2$, $R_2 - R_1$ and $R_1 \oplus R_2$.
 (c) Let R_1 and R_2 be congruence mod 3 and mod 4 on $\mathbb{Z}$. Find $R_1 \cap R_2$.

Solution. (a)(i) $R^{-1} = \{(a, b) \mid a > b\}$ (swap the roles) and $\bar{R} = \{(a, b) \mid a \not< b\} = \{(a, b) \mid a \geq b\}$.

(ii) $R^{-1} = \{(a, b) \mid b \text{ divides } a\}$, i.e. “ a is a multiple of b ”; $\bar{R} = \{(a, b) \mid a \nmid b\}$.

(b) $R_2 = R_1^{-1}$, so

- $R_1 \cup R_2 = \{(a, b) \mid a \mid b \text{ or } b \mid a\}$;
- $R_1 \cap R_2 = \{(a, a) \mid a \in \mathbb{Z}^+\}$ (if $a \mid b$ and $b \mid a$ then $a = b$, as both are positive);
- $R_1 - R_2 = \{(a, b) \mid a \mid b, a \neq b\}$ (a is a proper divisor of b);
- $R_2 - R_1 = \{(a, b) \mid b \mid a, a \neq b\}$;
- $R_1 \oplus R_2 = (R_1 - R_2) \cup (R_2 - R_1) = \{(a, b) \mid a \neq b \text{ and } (a \mid b \text{ or } b \mid a)\}$.

(c) $(a, b) \in R_1 \cap R_2$ iff $3 \mid (a - b)$ and $4 \mid (a - b)$, iff $12 \mid (a - b)$ (as $\text{lcm}(3, 4) = 12$): the intersection is congruence modulo 12.

PRACTICE • Rosen §9.1 Ex. 32, 33, 36 • composition and powers

- (a) $R = \{(1, 2), (1, 3), (2, 3), (2, 4), (3, 1)\}$ and $S = \{(2, 1), (3, 1), (3, 2), (4, 2)\}$. Find $S \circ R$.
 (b) R is “is a parent of” and S is “is a sibling of” on the set of people. What are $S \circ R$ and $R \circ S$?
 (c) For $R = \{(1, 1), (2, 1), (3, 2), (4, 3)\}$ find R^2 , R^3 and R^n for $n \geq 3$.
 (d) For $R = \{(a, b) \mid a > b\}$ find $R \circ R$ on $\mathbb{R}$, and on $\mathbb{Z}$.

Solution. (a) Take each $(a, b) \in R$ and every $(b, c) \in S$: $(1, 2) \rightarrow (2, 1)$ gives $(1, 1)$; $(1, 3) \rightarrow (3, 1)$, $(3, 2)$ give $(1, 1)$, $(1, 2)$; $(2, 3) \rightarrow (3, 1)$, $(3, 2)$ give $(2, 1)$, $(2, 2)$; $(2, 4) \rightarrow (4, 2)$ gives $(2, 2)$; $(3, 1)$: S has no pair starting at 1, so nothing. Hence $S \circ R = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$.

(b) $(a, c) \in S \circ R$ iff for some b , a is a parent of b and b is a sibling of c : a is a parent of c , and c has a sibling. $(a, c) \in R \circ S$ iff a is a sibling of some b who is a parent of c : a is an aunt or uncle of c . So $S \circ R \neq R \circ S$: composition is not commutative.

(c) $R^2 = R \circ R$: $(2, 1) \rightarrow (1, 1)$ gives $(2, 1)$; $(3, 2) \rightarrow (2, 1)$ gives $(3, 1)$; $(4, 3) \rightarrow (3, 2)$ gives $(4, 2)$; and $(1, 1)$. So $R^2 = \{(1, 1), (2, 1), (3, 1), (4, 2)\}$. Then $R^3 = R^2 \circ R = \{(1, 1), (2, 1), (3, 1), (4, 1)\}$ (now $(4, 3) \rightarrow (3, 1)$ gives $(4, 1)$), and $R^4 = R^3$; so $R^n = \{(1, 1), (2, 1), (3, 1), (4, 1)\}$ for all $n \geq 3$.

(d) $(a, c) \in R \circ R$ iff there is b with $a > b > c$. On $\mathbb{R}$ such a b exists whenever $a > c$ (take the midpoint), so $R \circ R = R$. On $\mathbb{Z}$ a b strictly between a and c exists iff $a - c \geq 2$, so $R \circ R = \{(a, c) \mid a - c \geq 2\} \subsetneq R$.

PRACTICE • Rosen §9.1 Ex. 25, 42–46 • counting relations

- (a) How many relations are there from an m -element set to an n -element set? (b) Of the 16 relations on $\{0, 1\}$, how many contain $(0, 1)$; how many are reflexive, irreflexive, symmetric, antisymmetric, asymmetric, transitive? (c) How many relations are there on $\{a, b, c, d\}$, and

how many of them contain (a, a) ? (d) For a set S with n elements and $a \in S$, how many relations on S contain no pair with a as first element? At least one such pair?

Solution. (a) A relation is a subset of $A \times B$, which has mn pairs, and each pair is in or out: 2^{mn} . (b) A relation on $\{0, 1\}$ is a subset of $\{(0, 0), (0, 1), (1, 0), (1, 1)\}$, so there are $2^4 = 16$. Fix $(0, 1)$ in: the other three pairs are free, so $2^3 = 8$ contain it. Now count by choosing the loops $(0, 0), (1, 1)$ (4 ways) and the off-diagonal pairs $(0, 1), (1, 0)$ (4 ways):

Property	Count	Reason
reflexive	4	both loops in; 2^2 choices for the rest
irreflexive	4	both loops out; 2^2 for the rest
symmetric	8	$(0, 1), (1, 0)$ both in or both out $(2) \times$ loops (4)
antisymmetric	12	not both $(0, 1), (1, 0)$ (3 ways) $\times$ loops (4)
asymmetric	3	no loops and not both: $\emptyset, \{(0, 1)\}, \{(1, 0)\}$
transitive	13	only $\{(0, 1), (1, 0)\}, \{(0, 0), (0, 1), (1, 0)\}$ and $\{(0, 1), (1, 0), (1, 1)\}$ fail

(A transitive relation containing both $(0, 1)$ and $(1, 0)$ must contain $(0, 0)$ and $(1, 1)$; every other relation on $\{0, 1\}$ is transitive.)

(c) $4 \times 4 = 16$ pairs, so $2^{16} = 65536$ relations; fixing (a, a) in leaves 15 free pairs: $2^{15} = 32768$.

(d) There are n^2 pairs, n of which have a first. None of these n : 2^{n^2-n} . At least one: total minus none, $2^{n^2} - 2^{n^2-n}$.

PRACTICE • Rosen §9.4 Ex. 1–3 • closures

(a) $R = \{(0, 1), (1, 1), (1, 2), (2, 0), (2, 2), (3, 0)\}$ on $\{0, 1, 2, 3\}$. Find its reflexive, symmetric and transitive closures.

(b) What is the reflexive closure of $\{(a, b) \mid a \neq b\}$ on $\mathbb{Z}$?

(c) What is the symmetric closure of $\{(a, b) \mid a \text{ divides } b\}$ on $\mathbb{Z}$?

Solution. (a) **Reflexive closure:** the loops present are $(1, 1), (2, 2)$; add $(0, 0), (3, 3)$. So it is $R \cup \{(0, 0), (3, 3)\}$.

Symmetric closure: $R^{-1} = \{(1, 0), (1, 1), (2, 1), (0, 2), (2, 2), (0, 3)\}$. Adding what is new: $(1, 0), (2, 1), (0, 2), (0, 3)$, giving $R \cup R^{-1}$ with 10 pairs.

Transitive closure: follow paths. $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$ is a cycle, so $0, 1, 2$ each reach every one of $0, 1, 2$ (that is 9 pairs). $3 \rightarrow 0$ gives $3 \rightarrow 0, 1, 2$. Nothing enters 3. So $R^* = \{0, 1, 2\} \times \{0, 1, 2\} \cup \{(3, 0), (3, 1), (3, 2)\}$, 12 pairs.

(b) Adding all missing loops (a, a) to “ $a \neq b$ ” gives every pair: the reflexive closure is $\mathbb{Z} \times \mathbb{Z}$ (the relation that is always true).

(c) $R \cup R^{-1} = \{(a, b) \mid a \mid b \text{ or } b \mid a\}$.

PRACTICE • Rosen §9.4 Ex. 29 • smallest relation with given properties

Find the smallest relation on $\{1, 2, 3, 4\}$ containing $\{(1, 2), (1, 4), (3, 3), (4, 1)\}$ that is (a) reflexive and transitive, (b) symmetric and transitive, (c) reflexive, symmetric and transitive.

Solution. Let R_0 be the given relation. **Transitive closure first:** $(1, 4), (4, 1)$ gives $(1, 1)$; $(4, 1), (1, 2)$ gives $(4, 2)$; $(4, 1), (1, 4)$ gives $(4, 4)$. So $R_0^* = R_0 \cup \{(1, 1), (4, 2), (4, 4)\}$.

(a) Add the only missing loop $(2, 2)$: $\{(1, 1), (1, 2), (1, 4), (2, 2), (3, 3), (4, 1), (4, 2), (4, 4)\}$ (8 pairs).

Adding a loop at 2 creates no new chains, since 2 has no other outgoing pair, so it is still transitive.

(b) *Symmetric closure first*: $\{(1,2), (2,1), (1,4), (4,1), (3,3)\}$. Then the transitive closure links 1, 2, 4 in both directions, so every pair of $\{1,2,4\} \times \{1,2,4\}$ appears (9 pairs) together with (3,3): 10 pairs in all.

(c) All four loops are already present in (b), so the answer is the same 10-pair relation: an equivalence relation with classes $\{1,2,4\}$ and $\{3\}$.

Order matters in (b): taking the transitive closure first and then symmetrising gives $R_0^* \cup (R_0^*)^{-1}$, which contains (2,1), (1,2) but not (2,2), so it is not transitive.

PRACTICE • Rosen §9.3 Ex. 3 and 7 • from a matrix to properties

List the ordered pairs of the relation on $\{1,2,3\}$ given by each matrix and decide whether it is reflexive, irreflexive, symmetric, antisymmetric, asymmetric and/or transitive.

$$(a) \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \quad (b) \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix} \quad (c) \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Solution. Row i , column j is 1 iff $(i,j) \in R$. Diagonal all 1 $\Rightarrow$ reflexive; diagonal all 0 $\Rightarrow$ irreflexive; $M = M^T \Rightarrow$ symmetric.

(a) $R = \{(1,1), (1,3), (2,2), (3,1), (3,3)\}$. Reflexive; symmetric ($M = M^T$); not antisymmetric ($(1,3), (3,1)$); transitive (e.g. $(1,3), (3,1) \Rightarrow (1,1)$ and $(3,1), (1,3) \Rightarrow (3,3)$ are present). An **equivalence relation** with classes $\{1,3\}, \{2\}$.

(b) $R = \{(1,2), (2,2), (3,2)\}$. Not reflexive ($(1,1) \notin R$), not irreflexive ($(2,2) \in R$), not symmetric ($(1,2) \in R, (2,1) \notin R$), antisymmetric (no two-way pair with distinct ends), not asymmetric (the loop $(2,2)$), transitive (every chain ends in 2, and $(a,2), (2,2) \Rightarrow (a,2)$). So it is antisymmetric and transitive only.

(c) $R = \{(1,1), (1,2), (1,3), (2,1), (2,3), (3,1), (3,2), (3,3)\}$. Not reflexive ($(2,2) \notin R$), not irreflexive, symmetric ($M = M^T$), not antisymmetric, not asymmetric, not transitive: $(2,1), (1,2) \in R$ but $(2,2) \notin R$. Only symmetric.

PRACTICE • Rosen §9.3 Ex. 14 and 15 • matrix operations and powers

Let $M_{R_1} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ and $M_{R_2} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$. Find the matrices of (a) $R_1 \cup R_2$, (b) $R_1 \cap R_2$,

(c) $R_2 \circ R_1$, (d) $R_1 \circ R_1$, (e) $R_1 \oplus R_2$. (f) For $M_R = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ find the matrices of R^2, R^3, R^4 .

Solution. (a) Join (entrywise OR): $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$. (b) Meet (entrywise AND): $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$.

(e) Entrywise XOR: $\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$.

(c) $M_{R_2 \circ R_1} = M_{R_1} \odot M_{R_2}$. Row i of the product is the OR of those rows of M_{R_2} selected by the 1s

in row i of M_{R_1} . Row 1 of M_{R_1} is $(0, 1, 0)$: take row 2 of M_{R_2} , $(0, 1, 1)$. Row 2 is $(1, 1, 1)$: OR of all three rows = $(1, 1, 1)$. Row 3 is $(1, 0, 0)$: row 1 of M_{R_2} , $(0, 1, 0)$. Result $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$.

(d) $M_{R_1} \odot M_{R_1}$ in the same way: row 1 picks row 2 of M_{R_1} , $(1, 1, 1)$; row 2 picks all rows, $(1, 1, 1)$; row 3 picks row 1, $(0, 1, 0)$. Result $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$.

(f) Repeat the row-selection rule: $M^{[2]} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$, $M^{[3]} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$, $M^{[4]} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$.

PRACTICE • Rosen §9.5 Ex. 1, 24 and 26 • which are equivalence relations?

(a) Which of these relations on $\{0, 1, 2, 3\}$ are equivalence relations? For the others, name a property that fails. For each equivalence relation give the classes.

- (i) $\{(0, 0), (1, 1), (2, 2), (3, 3)\}$
- (ii) $\{(0, 0), (0, 2), (2, 0), (2, 2), (2, 3), (3, 2), (3, 3)\}$
- (iii) $\{(0, 0), (1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$
- (iv) $\{(0, 0), (1, 1), (1, 3), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$
- (v) $\{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 2), (3, 3)\}$

(b) Which of these matrices represent equivalence relations?

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Solution. Check reflexive (all four loops), symmetric (each pair has its reverse), transitive (each chain closes).

(a)(i) Yes. Classes $\{0\}, \{1\}, \{2\}, \{3\}$.

(ii) No: $(1, 1) \notin R$ (not reflexive); also $(0, 2), (2, 3) \in R$ but $(0, 3) \notin R$ (not transitive).

(iii) Yes. Classes $\{0\}, \{1, 2\}, \{3\}$.

(iv) No: reflexive and symmetric, but $(1, 3), (3, 2) \in R$ and $(1, 2) \notin R$ (not transitive).

(v) No: $(1, 2) \in R$ but $(2, 1) \notin R$ (not symmetric); also $(2, 0), (0, 1) \in R$ but $(2, 1) \notin R$.

(b) First matrix: no, since $M_{12} = 1$ but $M_{21} = 0$ (not symmetric). Second: yes; rows 1, 3 agree and rows 2, 4 agree, classes $\{1, 3\}, \{2, 4\}$. Third: yes; classes $\{1, 2, 3\}, \{4\}$. (Tip: a symmetric, reflexive 0–1 matrix is transitive exactly when its rows are either identical or have no 1 in common.)

PRACTICE • Rosen §9.5 Ex. 9, 15, 16, 25, 29, 39, 40 • $f(x) = f(y)$ gives an equivalence relation

(a) Let f be a function with domain A and $R = \{(x, y) \mid f(x) = f(y)\}$. Show R is an equivalence relation and describe its classes.

(b) Show that these are equivalence relations: on pairs of positive integers, $((a, b), (c, d)) \in R$ iff

(i) $a + d = b + c$, (ii) $ad = bc$; on bit strings, (iii) sRt iff s and t have the same number of 1s.

(c) Describe the class of $(1, 2)$ in (i) and in (ii), and the class of 011 in (iii).

Solution. (a) *Reflexive:* $f(x) = f(x)$. *Symmetric:* $f(x) = f(y) \Rightarrow f(y) = f(x)$. *Transitive:* $f(x) = f(y)$ and $f(y) = f(z)$ give $f(x) = f(z)$. The class of x is $[x] = \{y \mid f(y) = f(x)\}$: all elements with the same image as x . There is one class for each value in the range of f .

(b) Choose a function f and apply (a).

- (i) $a + d = b + c \iff a - b = c - d$, so it is " $f(a, b) = f(c, d)$ " with $f(a, b) = a - b$.
- (ii) $ad = bc \iff a/b = c/d$ (all positive), so $f(a, b) = a/b$.
- (iii) $f(s) =$ the number of 1s in s .

(c) (i) $[(1, 2)] = \{(c, d) \mid c - d = -1\} = \{(1, 2), (2, 3), (3, 4), \dots\}$; each class stands for one integer (the difference $a - b$). (ii) $[(1, 2)] = \{(c, d) \mid d = 2c\} = \{(1, 2), (2, 4), (3, 6), \dots\}$; each class stands for one positive rational (the fraction a/b). (iii) $[011]$ is the set of all bit strings with exactly two 1s: 011, 101, 110, 0011, 0101, $\dots$.

PRACTICE • Rosen §9.5 Ex. 35–37, 49 • congruence classes and refinement

(a) Find $[n]_5$ for $n = 2, 3, 6, -3$. (b) Find $[4]_m$ for $m = 2, 3, 6, 8$. (c) Describe the congruence classes modulo 6. (d) Show that the partition into classes mod 6 is a refinement of the partition into classes mod 3 (every mod-6 class lies inside a mod-3 class).

Solution. (a) $[n]_5 = \{5k + r \mid k \in \mathbb{Z}\}$ where r is the remainder of n mod 5. $[2]_5 = \{\dots, -8, -3, 2, 7, 12, \dots\}$; $[3]_5 = \{\dots, -7, -2, 3, 8, 13, \dots\}$; $[6]_5 = [1]_5 = \{\dots, -9, -4, 1, 6, 11, \dots\}$; $[-3]_5 = [2]_5$ because $-3 = 5(-1) + 2$.

(b) $[4]_2 = \{2k\} =$ all even integers; $[4]_3 = \{3k + 1\} = \{\dots, -5, -2, 1, 4, 7, \dots\}$; $[4]_6 = \{6k + 4\} = \{\dots, -8, -2, 4, 10, \dots\}$; $[4]_8 = \{8k + 4\} = \{\dots, -12, -4, 4, 12, \dots\}$.

(c) Six classes $[r]_6 = \{6k + r\}$ for $r = 0, 1, 2, 3, 4, 5$, for example $[0]_6 = \{\dots, -6, 0, 6, 12, \dots\}$.

(d) If $x \equiv y \pmod{6}$ then $6 \mid (x - y)$, so $3 \mid (x - y)$ and $x \equiv y \pmod{3}$: elements in one class mod 6 are always in one class mod 3. Concretely $[0]_6, [3]_6 \subseteq [0]_3$; $[1]_6, [4]_6 \subseteq [1]_3$; $[2]_6, [5]_6 \subseteq [2]_3$. Each mod-3 class is split into two mod-6 classes.

PRACTICE • Rosen §9.5 Ex. 41, 44, 47 • partitions and their relations

(a) Which are partitions of $\{1, \dots, 6\}$? (i) $\{1, 2\}, \{2, 3, 4\}, \{4, 5, 6\}$ (ii) $\{1\}, \{2, 3, 6\}, \{4\}, \{5\}$ (iii) $\{2, 4, 6\}, \{1, 3, 5\}$ (iv) $\{1, 4, 5\}, \{2, 6\}$.

(b) Which are partitions of $\mathbb{Z}$? (i) evens, odds (ii) positives, negatives (iii) classes of remainders 0, 1, 2 mod 3 (iv) $x < -100$; $|x| \leq 100$; $x > 100$ (v) not divisible by 3; even; remainder 3 mod 6.

(c) List the ordered pairs of the equivalence relation produced by the partition $\{0\}, \{1, 2\}, \{3, 4, 5\}$ of $\{0, \dots, 5\}$, and count the pairs for $\{0, 1, 2\}, \{3, 4, 5\}$.

Solution. A partition needs non-empty blocks that are **disjoint** and whose union is the **whole set**.

(a) (i) No: 2 lies in two blocks (and so does 4). (ii) Yes. (iii) Yes. (iv) No: 3 is in no block.

(b) (i) Yes. (ii) No: 0 is in no block. (iii) Yes. (iv) Yes. (v) No: 2 is both not divisible by 3 and even, so the first two blocks overlap.

(c) $(a, b) \in R$ exactly when a, b lie in the same block. The blocks contribute $1^2 + 2^2 + 3^2 = 14$ pairs:

$(0, 0); (1, 1), (1, 2), (2, 1), (2, 2); (3, 3), (3, 4), (3, 5), (4, 3), (4, 4), (4, 5), (5, 3), (5, 4), (5, 5)$.

The partition $\{0, 1, 2\}, \{3, 4, 5\}$ gives $3^2 + 3^2 = 18$ pairs: all (i, j) with $i, j \in \{0, 1, 2\}$ and all (i, j) with $i, j \in \{3, 4, 5\}$.

PRACTICE • Rosen §9.5 Ex. 55 and 56 • smallest equivalence relation; unions and intersections

- (a) Find the smallest equivalence relation on $\{a, b, c, d, e\}$ containing $\{(a, b), (a, c), (d, e)\}$.
 (b) If R_1, R_2 are equivalence relations on a non-empty set S , must (i) $R_1 \cup R_2$, (ii) $R_1 \cap R_2$, (iii) $R_1 \oplus R_2$ be an equivalence relation? Prove or give a counterexample.

Solution. (a) Take the reflexive closure (add $(a, a), (b, b), (c, c), (d, d), (e, e)$), then the symmetric closure (add $(b, a), (c, a), (e, d)$), then the transitive closure last: $(b, a), (a, c)$ give (b, c) and $(c, a), (a, b)$ give (c, b) . The result is every pair inside $\{a, b, c\}$ (9) and inside $\{d, e\}$ (4): 13 pairs, with classes $\{a, b, c\}$ and $\{d, e\}$.

(b)(i) **No.** On $S = \{1, 2, 3\}$ let R_1 have classes $\{1, 2\}, \{3\}$ and R_2 have classes $\{1\}, \{2, 3\}$. Then $(1, 2) \in R_1$ and $(2, 3) \in R_2$, so both are in $R_1 \cup R_2$, but $(1, 3)$ is in neither: not transitive.

(ii) **Yes.** Reflexive: every (a, a) is in both. Symmetric: if (a, b) is in both then so is (b, a) . Transitive: if $(a, b), (b, c)$ are in both, then (a, c) is in both.

(iii) **No.** Every (a, a) lies in both R_1 and R_2 , so it is removed from $R_1 \oplus R_2$: the result is not reflexive (for $S \neq \emptyset$).

PRACTICE • Rosen §9.5 Ex. 61–63 • how many equivalence relations?

- (a) List the equivalence relations on a 3-element set and find how many there are. (b) How many are there on a 4-element set? (c) Is the transitive closure of the symmetric closure of the reflexive closure of any relation always an equivalence relation?

Solution. Equivalence relations on S correspond to partitions of S , so count partitions.

(a) For $\{a, b, c\}$: $\{a, b, c\}$; $\{a\}, \{b, c\}$; $\{b\}, \{a, c\}$; $\{c\}, \{a, b\}$; $\{a\}, \{b\}, \{c\}$. That is **5**.

(b) Count by block sizes: 4 (one way); 3+1 (4 ways, choose the singleton); 2+2 (3 ways); 2+1+1 (6 ways, choose the pair); 1+1+1+1 (1 way). Total $1 + 4 + 3 + 6 + 1 = 15$. (These numbers 1, 2, 5, 15, 52, ... are the Bell numbers.)

(c) **Yes.** Let R_1 be the reflexive closure of R (reflexive), R_2 the symmetric closure of R_1 (symmetric, and still reflexive since $R_2 \supseteq R_1$), and R_3 the transitive closure of R_2 (transitive, and still reflexive as $R_3 \supseteq R_2$). R_3 is also symmetric: a path $x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_k$ of pairs of R_2 can be reversed, since each pair of R_2 has its reverse in R_2 . So R_3 is reflexive, symmetric and transitive. (The order matters: making a transitive relation symmetric last can destroy transitivity, as in Ex. 29 above.)

Partial Orders, Hasse Diagrams and Lattices

7.1 Partially ordered sets

DEFINITION — Poset

A *partial order* on a set S is a relation $\preceq$ that is **reflexive, antisymmetric and transitive**. The pair $(S, \preceq)$ is a *partially ordered set* or *poset*. (Notation: S is the partially ordered set, $\preceq$ the partial ordering relation.)

EXAMPLE

$(\mathbb{N}, \leq)$, $(\mathbb{Z}^+, |)$ (divisibility) and $(\mathcal{P}(A), \subseteq)$ are posets. For $(\mathbb{Z}^+, |)$: $a \mid a$; $a \mid b$ and $b \mid a$ give $a = b$; $a \mid b$ and $b \mid c$ give $a \mid c$.

DEFINITION — Comparable, total order

In a poset, a and b are *comparable* if $a \preceq b$ or $b \preceq a$; otherwise they are *incomparable*. If **every** two elements are comparable, $(S, \preceq)$ is a *totally ordered set* (a *chain*) and $\preceq$ is a *total order*.

EXAMPLE

In $(\mathbb{Z}^+, |)$: 3 and 15 are comparable ($3 \mid 15$), but 7 and 15 are incomparable ($7 \nmid 15$ and $15 \nmid 7$), so $(\mathbb{Z}^+, |)$ is a poset but not a total order. $(\mathbb{N}, \leq)$ is a total order: any two numbers satisfy $a \leq b$ or $b \leq a$.

7.2 Hasse diagrams

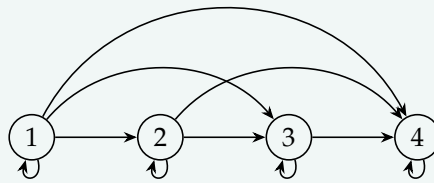
DEFINITION — Covering relation and Hasse diagram

In a finite poset, b *covers* a if $a \prec b$ (i.e. $a \preceq b$, $a \neq b$) and there is no c with $a \prec c \prec b$. The *Hasse diagram* draws each element as a dot/vertex and joins a to b by a line, with b placed **higher** than a , exactly when b covers a .

How to get the Hasse diagram from the digraph of the partial order:

1. remove all **self-loops** (reflexivity is understood);
2. remove all edges implied by **transitivity** ($a \rightarrow c$ whenever $a \rightarrow b \rightarrow c$);
3. drop the **arrowheads** (all edges point upward).

EXAMPLE — $(\{1, 2, 3, 4\}, \leq)$



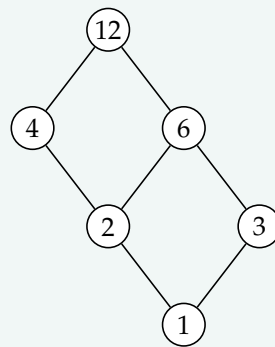
digraph of $\leq$ (loops, transitive edges)



Hasse diagram

EXAMPLE — divisors of 12 under divisibility

$(\{1, 2, 3, 4, 6, 12\}, |)$. Covers: $1 \rightarrow 2$, $1 \rightarrow 3$, $2 \rightarrow 4$, $2 \rightarrow 6$, $3 \rightarrow 6$, $4 \rightarrow 12$, $6 \rightarrow 12$. (No edge $1 \rightarrow 4$ because $1 \mid 2 \mid 4$; no edge $2 \rightarrow 12$ because $2 \mid 4 \mid 12$.)

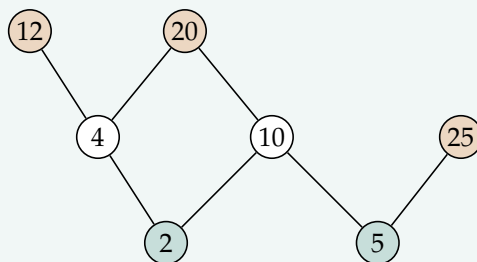


7.3 Maximal and minimal elements

DEFINITION — Maximal and minimal elements

Let $(S, \leq)$ be a poset. An element m is *maximal* if there is **no** element $x \in S$ with $m < x$ (nothing lies strictly above it). An element n is *minimal* if there is no $x \in S$ with $x < n$. In a Hasse diagram, maximal elements are the top vertices (no line going up) and minimal elements are the bottom vertices (no line going down). There may be several of each.

EXAMPLE — $(\{2, 4, 5, 10, 12, 20, 25\}, |)$



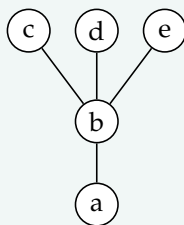
maximal: 12, 20, 25

minimal: 2, 5

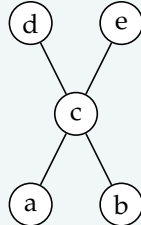
DEFINITION — Greatest and least elements

g is the *greatest element* if $x \preceq g$ for **every** $x \in S$ (it is above everything). ℓ is the *least element* if $\ell \preceq x$ for every $x \in S$. When they exist they are **unique**. A greatest element is maximal, but a maximal element need not be greatest (compare the poset above: 12, 20, 25 are all maximal, but none is above all others).

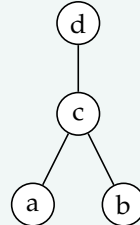
EXAMPLE — four small posets



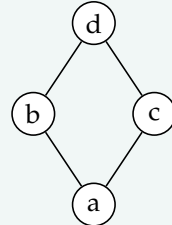
(1)



(2)



(3)



(4)

	maximal	minimal	greatest	least
(1)	c, d, e	a	none	a
(2)	d, e	a, b	none	none
(3)	d	a, b	d	none
(4)	d	a	d	a

7.4 Upper and lower bounds, LUB and GLB

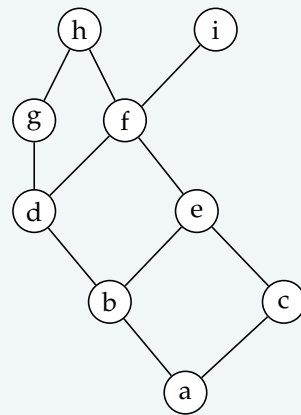
DEFINITION — Upper and lower bounds

Let $A \subseteq S$ be a subset of a poset $(S, \preceq)$. An element $u \in S$ is an *upper bound* of A if $a \preceq u$ for **all** $a \in A$. An element $\ell \in S$ is a *lower bound* of A if $\ell \preceq a$ for all $a \in A$. The bound need not belong to A .

DEFINITION — Least upper bound and greatest lower bound

The *least upper bound* (LUB, supremum, join) of A is the **least** element of the set of upper bounds of A . The *greatest lower bound* (GLB, infimum, meet) of A is the **greatest** element of the set of lower bounds. If they exist they are unique.

EXAMPLE — a nine-element poset



Subset	Upper bounds	Lower bounds	LUB	GLB
$\{a, b, c\}$	e, f, h, i	a	e	a
$\{h, i\}$	none	a, b, c, d, e, f	none	f
$\{a, c, d, f\}$	f, h, i	a	f	a

For $\{h, i\}$: nothing lies above both h and i , so there is no upper bound at all; below both lie f, d, e, b, c, a , and the greatest of these is f .

EXAMPLE — divisibility: $(\mathbb{Z}^+, |)$ and $A = \{3, 9, 12\}$

An upper bound is a multiple of 3, 9 and 12: the multiples of $\text{lcm}(3, 9, 12) = 36$. A lower bound is a common divisor: 1 and 3. So

$$\text{UB} = \{36, 72, 108, \dots\}, \quad \text{LB} = \{1, 3\}, \quad \text{LUB} = 36 = \text{lcm}, \quad \text{GLB} = 3 = \text{gcd}.$$

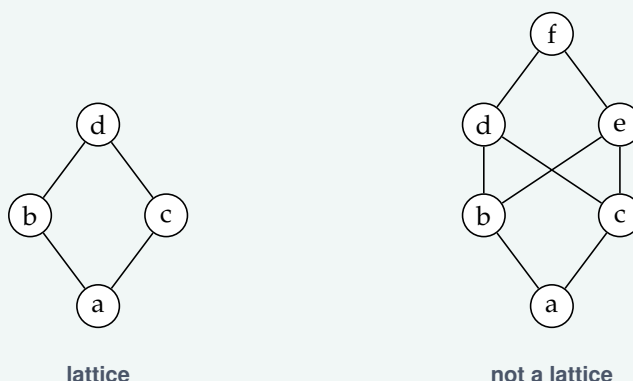
In $(\mathbb{Z}^+, |)$ the LUB of a set is its lcm and the GLB is its gcd.

7.5 Lattices

DEFINITION — Lattice

A poset is a *lattice* if **every pair** of elements a, b has a least upper bound (written $a \vee b$, the *join*) and a greatest lower bound ($a \wedge b$, the *meet*). Test: for each pair, collect the common upper bounds and check there is a unique least one; likewise for lower bounds.

EXAMPLE — lattice and non-lattice



Left: any two elements have a join and a meet (e.g. $b \vee c = d$, $b \wedge c = a$). Right: b and c have upper bounds d, e, f but d and e are incomparable, so there is *no least* upper bound; $\text{LUB}\{b, c\}$ does not exist.

EXAMPLE — more lattices and non-lattices

- Every chain such as $(\mathbb{N}, \leq)$ is a lattice: $a \vee b = \max(a, b)$, $a \wedge b = \min(a, b)$.
- $(\mathbb{Z}^+, |)$ is a lattice: $a \vee b = \text{lcm}(a, b)$, $a \wedge b = \text{gcd}(a, b)$; the divisors of 12 (previous section) form a lattice.
- $(\mathcal{P}(A), \subseteq)$ is a lattice: join = $\cup$, meet = $\cap$.
- $(\{1, 2, 3, 4, 6, 8, 12\}, |)$ is *not* a lattice: 8 and 12 have no common upper bound in the set ($\text{lcm} = 24$ is missing).

7.6 Dual of a poset[†]

DEFINITION — Dual poset

If $(S, \preceq)$ is a poset, its *dual* is $(S, \succeq)$, that is, the same set with the inverse relation $\preceq^{-1}$. The dual of a partial order is again a partial order (reflexivity, antisymmetry and transitivity all survive reversal). Its Hasse diagram is the original turned **upside down**, so maximal $\leftrightarrow$ minimal, greatest $\leftrightarrow$ least and LUB $\leftrightarrow$ GLB.

EXAMPLE

The dual of $(\{1, 2, 3\}, \leq)$ is $(\{1, 2, 3\}, \geq)$. The dual of $(\mathbb{Z}^+, |)$ is the poset “is a multiple of”. The dual of $(\mathcal{P}(A), \subseteq)$ is $(\mathcal{P}(A), \supseteq)$.

7.7 Lexicographic order[†]

DEFINITION — Lexicographic (dictionary) order

Let $(A, \preceq_1)$ and $(B, \preceq_2)$ be posets. On $A \times B$ define $(a_1, b_1) \prec (a_2, b_2)$ if $a_1 \prec_1 a_2$, or $a_1 = a_2$ and $b_1 \prec_2 b_2$. The first coordinate decides; the second is used only to break a tie. For strings over an ordered alphabet, compare letter by letter at the first place they differ; if one string is a proper prefix of the other, the shorter one comes first.

EXAMPLE

On $\mathbb{Z}^+ \times \mathbb{Z}^+$: $(2,9) \prec (3,1)$ because $2 < 3$ decides at once; $(3,1) \prec (3,4)$ because the first coordinates tie and $1 < 4$. For words: $\text{cat} \prec \text{cot}$ ($a < o$), $\text{art} \prec \text{arts}$ (prefix), $\text{discreet} \prec \text{discrete}$ (they first differ at the seventh letter, $e < t$). For bit strings with $0 < 1$: $0110 \prec 10$, and $01 \prec 010$.

7.8 Compatible total orders (topological sorting)[†]

DEFINITION — Compatible total order

A total order $\preceq_t$ on S is *compatible* with the partial order $\preceq$ if $a \preceq b$ implies $a \preceq_t b$. Every finite poset has one: *repeatedly pick a minimal element of what is left, list it next, and delete it*. This is how tasks with prerequisites are scheduled.

EXAMPLE

Divisors of 12 under divisibility (Hasse diagram earlier). Minimal: 1; remove it. Minimal now: 2, 3; take 2. Minimal now: 3, 4; take 3; then 4, 6 are minimal: take 4, then 6, then 12. Result: 1, 2, 3, 4, 6, 12. Other choices give other valid orders, such as 1, 3, 2, 6, 4, 12, but never 1, 4, ... before 2. For divisibility the ordinary order of size always works, since $a \mid b$ implies $a \leq b$.

7.9 Solved previous-year questions

PYQ • Mid-Sem Oct 2023 Q2(d) • 3 marks

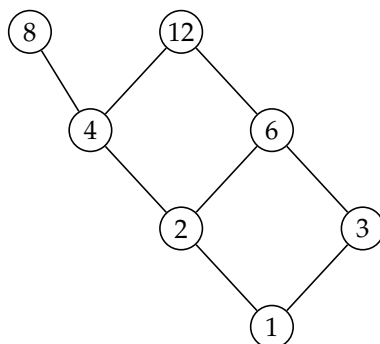
Let $A = \{1, 2, 3, 4, 6, 8, 12\}$ be the partial ordering on A defined by aRb if a divides b . Draw the Hasse diagram of the poset (A, R) .

Solution. The relation consists of the 7 reflexive pairs and the 15 pairs $(1, x)$ for $x = 2, 3, 4, 6, 8, 12$; $(2, 4), (2, 6), (2, 8), (2, 12)$; $(3, 6), (3, 12)$; $(4, 8), (4, 12)$; $(6, 12)$.

Step 1: delete the 7 self-loops. **Step 2:** delete edges implied by transitivity: $(1, 4)$ [via 2], $(1, 6)$ [via 2 or 3], $(1, 8)$ [via 2, 4], $(1, 12)$ [via 2, 3, 4, 6], $(2, 8)$ [via 4], $(2, 12)$ [via 4 or 6], $(3, 12)$ [via 6].

Step 3: draw what remains (8 covering edges) with arrows pointing up and then dropped:

1–2, 1–3, 2–4, 2–6, 3–6, 4–8, 4–12, 6–12.

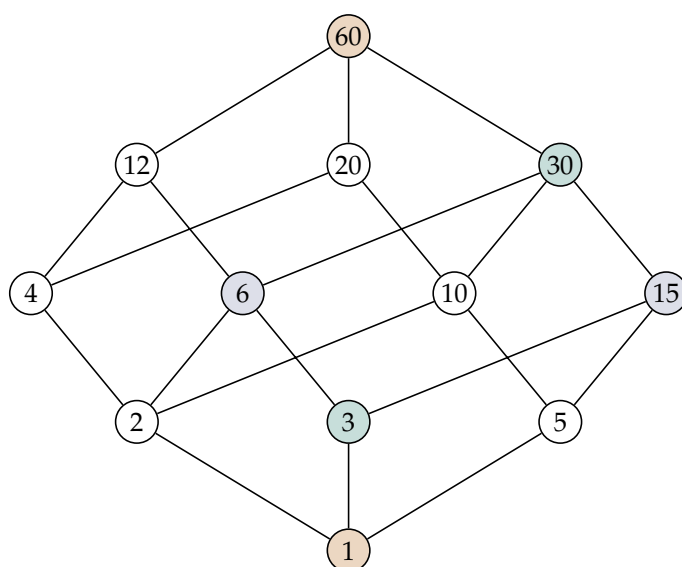


Levels: 1 | 2, 3 | 4, 6 | 8, 12. Elements at the same level need not be comparable (e.g. 2 and 3). Also: minimal = 1 (least element); maximal = 8, 12 (no greatest element).

PYQ • End-Sem Dec 2024 Q2(b) • 7 marks

Draw the Hasse diagram for the poset of divisors of 60, and answer: (I) identify the maximal and minimal elements; (II) find the least upper bound of 6 and 15; (III) find the greatest lower bound of 6 and 15.

Solution. Divisors of $60 = 2^2 \cdot 3 \cdot 5$: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60 ordered by divisibility. Covering pairs (prime multiples): 1–2, 3, 5; 2–4, 6, 10; 3–6, 15; 5–10, 15; 4–12, 20; 6–12, 30; 10–20, 30; 15–30; 12, 20, 30–60 (20 edges).



maximal 60, minimal 1

the pair 6, 15

LUB 30 and GLB 3

(I) Maximal and minimal elements. Nothing lies above 60 (every divisor divides it), so 60 is the only maximal element (it is also the greatest element). Nothing lies below 1, so 1 is the only minimal element (the least element).

(II) LUB of 6 and 15. Upper bounds are common multiples of 6 and 15 inside the set, i.e. multiples of 30: {30, 60}. The least is 30, and $30 \mid 60$. So $\text{LUB}(6, 15) = 30 = \text{lcm}(6, 15)$.

(III) GLB of 6 and 15. Lower bounds are common divisors: {1, 3}. The greatest is 3, so $\text{GLB}(6, 15) = 3 = \text{gcd}(6, 15)$.

7.10 Practice questions from Rosen

Chosen from Rosen §9.6 in the style of the PYQs above (draw a Hasse diagram; find maximal, minimal, greatest, least, bounds; decide whether something is a lattice). Routine repeats of these skills are left out.

PRACTICE • Rosen §9.6 Ex. 1 and 7 • which relations are partial orders?

(a) Which of these relations on $\{0, 1, 2, 3\}$ are partial orders? For the others, say which property fails.

(i) $\{(0, 0), (1, 1), (2, 2), (3, 3)\}$

(ii) $\{(0, 0), (1, 1), (2, 0), (2, 2), (2, 3), (3, 2), (3, 3)\}$

(iii) $\{(0, 0), (1, 1), (1, 2), (2, 2), (3, 3)\}$

(iv) $\{(0, 0), (1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$

(v) $\{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 2), (3, 3)\}$

(b) Which of these matrices represent partial orders?

$$(i) \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (ii) \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (iii) \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{pmatrix}$$

Solution. A partial order must be reflexive (all loops), antisymmetric and transitive. All five relations in (a) and all three matrices in (b) are reflexive, so check the other two properties.

(a)(i) Yes: the identity relation (no two different elements are comparable).

(ii) No. Not antisymmetric: $(2, 3), (3, 2) \in R$ with $2 \neq 3$. Also not transitive: $(3, 2), (2, 0) \in R$ but $(3, 0) \notin R$.

(iii) Yes: the only extra pair is $1 \preceq 2$.

(iv) Yes: $1 \preceq 2 \preceq 3$ and $1 \preceq 3$; 0 is comparable to nothing but itself.

(v) No. Not antisymmetric: $(0, 1), (1, 0) \in R$. Also not transitive: $(2, 0), (0, 1) \in R$ but $(2, 1) \notin R$.

(b)(i) No: $M_{12} = M_{21} = 1$, so $(1, 2)$ and $(2, 1)$ are both in R (not antisymmetric).

(ii) Yes: $R = \{(1, 1), (1, 2), (1, 3), (2, 2), (3, 3)\}$ is antisymmetric and transitive (no chain gives a pair that is missing): $1 \preceq 2, 1 \preceq 3$, with 2, 3 incomparable.

(iii) No: antisymmetric, but not transitive, since $(1, 3), (3, 4) \in R$ while $(1, 4) \notin R$.

PRACTICE • Rosen §9.6 Ex. 13–15 • duals, comparable elements

(a) Find the duals of $(\{0, 1, 2\}, \leq)$, $(\mathbb{Z}, \geq)$, $(\mathcal{P}(\mathbb{Z}), \supseteq)$ and $(\mathbb{Z}^+, |)$.

(b) Which of these pairs are comparable in $(\mathbb{Z}^+, |)$? (i) 5, 15 (ii) 6, 9 (iii) 8, 16 (iv) 7, 7.

(c) Find two incomparable elements in $(\mathcal{P}(\{0, 1, 2\}), \subseteq)$ and in $(\{1, 2, 4, 6, 8\}, |)$.

Solution. (a) The dual reverses the relation: $(\{0, 1, 2\}, \geq)$; $(\mathbb{Z}, \leq)$; $(\mathcal{P}(\mathbb{Z}), \subseteq)$; and $(\mathbb{Z}^+, \text{“is a multiple of”})$.

(b) (i) comparable, $5 \mid 15$. (ii) incomparable: $6 \nmid 9$ and $9 \nmid 6$. (iii) comparable, $8 \mid 16$. (iv) comparable: every element is comparable to itself ($7 \mid 7$).

(c) In $\mathcal{P}(\{0, 1, 2\})$: $\{0\}$ and $\{1\}$ (neither is a subset of the other); also $\{0, 1\}$ and $\{2\}$. In $(\{1, 2, 4, 6, 8\}, |)$: 4 and 6, since $4 \nmid 6$ and $6 \nmid 4$ (also 6, 8). A total order has no such pair.

PRACTICE • Rosen §9.6 Ex. 16–19 • lexicographic order

(a) With $S = \{1, 2, 3, 4\}$ and lexicographic order on $S \times S$, find all pairs less than $(2, 3)$ and all pairs greater than $(3, 1)$.

(b) Put in lexicographic order: (i) $(0, 1, 2, 3), (0, 1, 3, 2)$; (ii) $(1, 0, 1, 0, 1), (0, 1, 1, 1, 0)$.

(c) Put in dictionary order: quack, quick, quicksilver, quicksand, quacking; open, opener, opera, operand, opened; zoo, zero, zoom, zoology, zoological.

(d) Put in lexicographic order the bit strings 0, 01, 11, 001, 010, 011, 0001, 0101 (with $0 < 1$).

Solution. (a) $(a, b) \prec (2, 3)$ iff $a < 2$, or $a = 2$ and $b < 3$: $(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2)$ (6 pairs). $(a, b) \succ (3, 1)$ iff $a > 3$, or $a = 3$ and $b > 1$: $(3, 2), (3, 3), (3, 4), (4, 1), (4, 2), (4, 3), (4, 4)$ (7 pairs). (The lexicographic order on $S \times S$ is a total order, so its Hasse diagram is a single chain of 16 elements.)

(b) (i) They agree in the first two entries; at the third $2 < 3$, so $(0, 1, 2, 3) \prec (0, 1, 3, 2)$. (ii) The first entries decide: $0 < 1$, so $(0, 1, 1, 1, 0) \prec (1, 0, 1, 0, 1)$.

(c) quack $\prec$ quacking $\prec$ quick $\prec$ quicksand $\prec$ quicksilver (“quack” is a prefix of “quacking”;

then $a < i$ separates “quacking” from “quick”; “quick” is a prefix of “quicksand”; then $a < i$).
 open $\prec$ opened $\prec$ opener $\prec$ opera $\prec$ operand (prefix; “opene” then $d < r$; “ope” then $n < r$; prefix).
 zero $\prec$ zoo $\prec$ zoological $\prec$ zoology $\prec$ zoom ($e < o$; prefix; “zoolog” then $i < y$; “zoo” then $l < m$).

(d) A prefix comes first and otherwise the first differing bit decides: $0 \prec 0001 \prec 001 \prec 01 \prec 010 \prec 0101 \prec 011 \prec 11$.

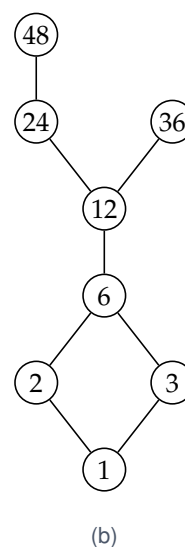
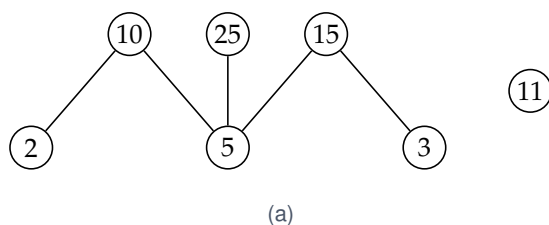
PRACTICE • Rosen §9.6 Ex. 22, 23, 29 • draw the Hasse diagram

Draw the Hasse diagram (a) for divisibility on $\{2, 3, 5, 10, 11, 15, 25\}$; (b) for divisibility on $\{1, 2, 3, 6, 12, 24, 36, 48\}$; (c) for inclusion on $\mathcal{P}(\{a, b, c\})$, and list its covering relation.

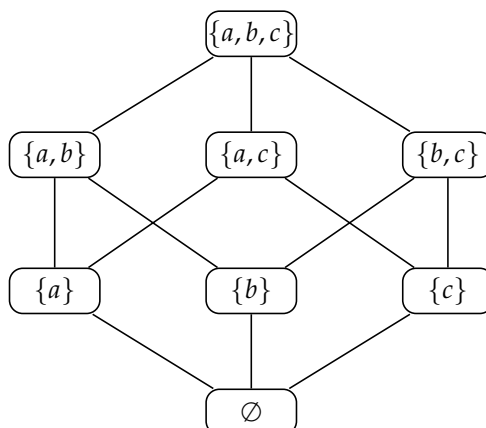
Solution. Find the covering pairs (b covers a when $a \mid b$ or $a \subset b$ with nothing strictly between) and draw them upward.

(a) Covers: $2 \rightarrow 10$, $5 \rightarrow 10$, $3 \rightarrow 15$, $5 \rightarrow 15$, $5 \rightarrow 25$. The number 11 divides nothing else and nothing else divides it, so it is an isolated vertex (both maximal and minimal). Note there is no edge $2 \rightarrow 5$ or $3 \rightarrow 25$.

(b) Covers: $1 \rightarrow 2$, $1 \rightarrow 3$, $2 \rightarrow 6$, $3 \rightarrow 6$, $6 \rightarrow 12$, $12 \rightarrow 24$, $12 \rightarrow 36$, $24 \rightarrow 48$. (No edge $6 \rightarrow 24$, since $6 \mid 12 \mid 24$; no edge $12 \rightarrow 48$, since $12 \mid 24 \mid 48$; $36 \nmid 48$.)



(c) A set covers another when it has exactly one more element. Covering relation: $\emptyset \rightarrow \{a\}, \{b\}, \{c\}$; $\{a\} \rightarrow \{a, b\}, \{a, c\}$; $\{b\} \rightarrow \{a, b\}, \{b, c\}$; $\{c\} \rightarrow \{a, c\}, \{b, c\}$; $\{a, b\}, \{a, c\}, \{b, c\} \rightarrow \{a, b, c\}$ ($3 + 6 + 3 = 12$ pairs). The Hasse diagram is a cube:



PRACTICE • Rosen §9.6 Ex. 33 and 35 • maximal, minimal, greatest, least, bounds

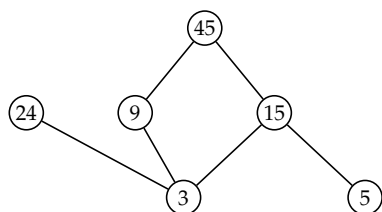
Answer (a)–(h) for each poset.

(a) maximal elements; (b) minimal elements; (c) is there a greatest element; (d) is there a least element; (e) all upper bounds of X ; (f) the LUB of X ; (g) all lower bounds of Y ; (h) the GLB of Y .

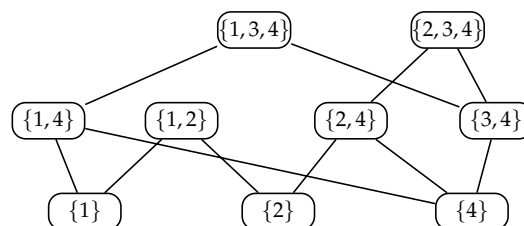
Poset 1: $(\{3, 5, 9, 15, 24, 45\}, |)$ with $X = \{3, 5\}$, $Y = \{15, 45\}$.

Poset 2: $(\{\{1\}, \{2\}, \{4\}, \{1, 2\}, \{1, 4\}, \{2, 4\}, \{3, 4\}, \{1, 3, 4\}, \{2, 3, 4\}\}, \subseteq)$ with $X = \{\{2\}, \{4\}\}$, $Y = \{\{1, 3, 4\}, \{2, 3, 4\}\}$.

Solution. Draw the Hasse diagrams first (edges are the covering pairs).



Poset 1



Poset 2

Poset 1. (a) maximal: 24, 45 (nothing in the set is a proper multiple of them). (b) minimal: 3, 5. (c) No greatest element ($24 \nmid 45$ and $45 \nmid 24$). (d) No least element ($3 \nmid 5$). (e) Upper bounds of $\{3, 5\}$ are the common multiples in the set: 15, 45. (f) LUB = 15 (the smallest, and $15 \mid 45$). (g) Lower bounds of $\{15, 45\}$ are the common divisors in the set: 3, 5, 15. (h) GLB = 15.

Poset 2. (a) maximal: $\{1, 2\}, \{1, 3, 4\}, \{2, 3, 4\}$. (b) minimal: $\{1\}, \{2\}, \{4\}$. (c) No greatest and (d) no least element. (e) Upper bounds of $\{\{2\}, \{4\}\}$ are the sets containing 2 and 4: $\{2, 4\}, \{2, 3, 4\}$. (f) LUB = $\{2, 4\}$. (g) Lower bounds of $\{\{1, 3, 4\}, \{2, 3, 4\}\}$ are the sets inside $\{3, 4\}$: $\{4\}, \{3, 4\}$. (h) GLB = $\{3, 4\}$.

PRACTICE • Rosen §9.6 Ex. 36, 40, 50, 60 • existence and uniqueness

(a) Give a poset that has (i) a minimal element but no maximal element, (ii) a maximal but no minimal element, (iii) neither. (b) Show that a poset has at most one greatest element. (c) Show that a finite non-empty poset has a maximal element. (d) Show that every totally ordered set is a lattice.

Solution. (a) (i) $(\mathbb{N}, \leq)$: minimal element 0, nothing is maximal. (ii) $(\{0, -1, -2, \dots\}, \leq)$: maximal 0, no minimal element. (iii) $(\mathbb{Z}, \leq)$.

(b) Suppose g and g' are both greatest. Since g' is greatest, $g \preceq g'$; since g is greatest, $g' \preceq g$. By antisymmetry $g = g'$. (The same argument shows the least element is unique.)

(c) Pick x_1 . If it is not maximal, some $x_2 \succ x_1$ exists; if x_2 is not maximal pick $x_3 \succ x_2$, and so on. By transitivity $x_1 \prec x_2 \prec x_3 \prec \dots$, so no element repeats (a repeat would give $x_i \prec x_i$). The set is finite, so the process must stop, at a maximal element.

(d) Let a, b be elements. Since any two elements are comparable, either $a \preceq b$ or $b \preceq a$. If $a \preceq b$, then b is the least upper bound of $\{a, b\}$ and a is the greatest lower bound; if $b \preceq a$ the roles swap. So every pair has a join and a meet: it is a lattice ($a \vee b = \max, a \wedge b = \min$).

PRACTICE • Rosen §9.6 Ex. 44 and 46 • is it a lattice?

Determine whether these posets are lattices: (a) $(\{1, 3, 6, 9, 12\}, |)$; (b) $(\{1, 5, 25, 125\}, |)$; (c) $(\mathbb{Z}, \geq)$; (d) $(\mathcal{P}(S), \supseteq)$. (e) Show that the dual of a lattice is a lattice.

Solution. A lattice needs a LUB and a GLB for **every pair**.

(a) Not a lattice. Take 6 and 9. Their upper bounds would be common multiples in the set, i.e. multiples of 18; none of 1, 3, 6, 9, 12 qualifies, so $\{6, 9\}$ has no upper bound at all (the same happens for 9, 12).

(b) Lattice. It is a chain $(1 \mid 5 \mid 25 \mid 125)$, and every chain is a lattice: $a \vee b = \text{lcm} = \max$, $a \wedge b = \text{gcd} = \min$.

(c) Lattice. $(\mathbb{Z}, \geq)$ is a total order. For a, b : the join (least upper bound in this order) is $\min(a, b)$ and the meet is $\max(a, b)$.

(d) Lattice. Upper bounds of A, B in the $\supseteq$ order are the sets X with $X \subseteq A$ and $X \subseteq B$, i.e. $X \subseteq A \cap B$; the “least” of them in this order is the largest such set, $A \cap B$. So join = $A \cap B$ and meet = $A \cup B$.

(e) In the dual, “upper bound” and “lower bound” swap roles. So the LUB of $\{a, b\}$ in the dual is the GLB in the original, and vice versa. Both exist in a lattice, so the dual is a lattice, with join and meet exchanged.

PRACTICE • Rosen §9.6 Ex. 62 and 63 • compatible total orders

(a) Find a compatible total order for divisibility on $\{1, 2, 3, 6, 8, 12, 24, 36\}$. (b) Find all compatible total orders for $(\{1, 2, 4, 5, 12, 20\}, \mid)$.

Solution. (a) If $a \mid b$ then $a \leq b$, so listing the elements by size works: 1, 2, 3, 6, 8, 12, 24, 36. (Check: $6 \mid 12, 24, 36$; $8 \mid 24$; $12 \mid 24, 36$; all appear later.)

(b) The Hasse diagram has covers $1 \rightarrow 2, 1 \rightarrow 5, 2 \rightarrow 4, 4 \rightarrow 12, 4 \rightarrow 20, 5 \rightarrow 20$. So 1 comes first; 2 before 4; 4 before 12 and 20; and 5 before 20. Count the orders of the last five elements: the chain 2, 4 is followed by 12 and 20 in either order.

- 2, 4, 12, 20: put 5 anywhere before 20: before 2, after 2, after 4, after 12: 4 ways.
- 2, 4, 20, 12: put 5 anywhere before 20: before 2, after 2, after 4: 3 ways.

Total 7 compatible total orders:

1, 2, 4, 5, 12, 20 1, 2, 4, 5, 20, 12 1, 2, 4, 12, 5, 20 1, 2, 5, 4, 12, 20
1, 2, 5, 4, 20, 12 1, 5, 2, 4, 12, 20 1, 5, 2, 4, 20, 12

Functions

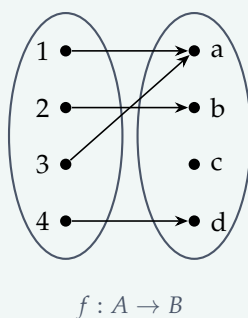
8.1 Functions: domain, codomain, image, range

DEFINITION — Function

A function $f : A \rightarrow B$ assigns to **every** element $a \in A$ **exactly one** element $f(a) \in B$. A is the domain, B the codomain. If $f(a) = b$ we call b the *image* of a and a a *pre-image* of b . The *range* of f is the set of all images, $\{f(a) \mid a \in A\} \subseteq B$.

EXAMPLE

$A = \{1, 2, 3, 4\}$, $B = \{a, b, c, d\}$, with $f(1) = a$, $f(2) = b$, $f(3) = a$, $f(4) = d$.



Domain = A ; codomain = B ; the image of 4 is d ; the pre-images of a are $\{1, 3\}$; the range is $\{a, b, d\}$, which is smaller than the codomain because c has no pre-image.

WATCH OUT

A rule is *not* a function if some element of A has *no* arrow or *two* arrows (see panel (4) below). It is fine for two different elements to share one image ($f(1) = f(3) = a$ above).

8.2 One-to-one, onto, bijection

DEFINITION — One-to-one (injective)

f is *one-to-one* if different elements have different images:

$$f(a) = f(b) \Rightarrow a = b \quad (\text{equivalently } a \neq b \Rightarrow f(a) \neq f(b)).$$

To prove: assume $f(a) = f(b)$ and derive $a = b$. To disprove: exhibit $a \neq b$ with $f(a) = f(b)$.

EXAMPLE

$f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x + 1$ is one-to-one: $2a + 1 = 2b + 1 \Rightarrow a = b$. But $f(x) = x^2$ on $\mathbb{R}$ is not: $f(2) = f(-2) = 4$.

DEFINITION — Onto (surjective)

$f : A \rightarrow B$ is *onto* if every element of the codomain is an image: for every $y \in B$ there is an $x \in A$ with $f(x) = y$. Equivalently, $\text{range} = \text{codomain}$.

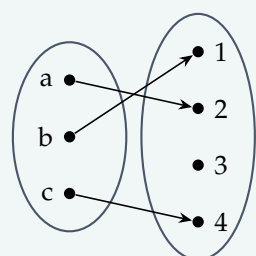
EXAMPLE

$f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^3$ is onto: given y , take $x = \sqrt[3]{y}$. But $f(x) = x^2$ from $\mathbb{R}$ to $\mathbb{R}$ is not onto: -1 has no pre-image. (As a map $\mathbb{R} \rightarrow [0, \infty)$ it *is* onto.)

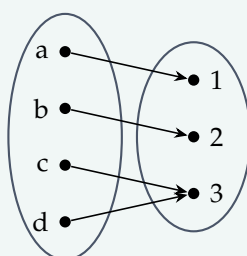
DEFINITION — Bijection / one-to-one correspondence

f is a *bijection* if it is both one-to-one and onto: every element of B is hit by exactly one element of A .

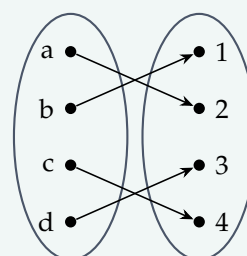
EXAMPLE — the five cases



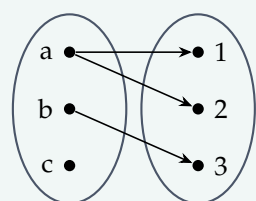
(1) one-to-one,
not onto



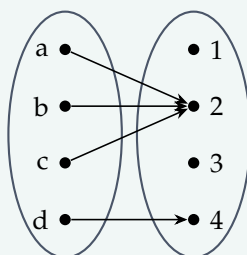
(2) onto,
not one-to-one



(3) one-to-one
and onto (bijection)



(4) **not a function**
(a has two images, c none)



(5) neither one-to-one
nor onto

REMEMBER — counting for finite sets

[†]If $|A| = m$ and $|B| = n$: a one-to-one function needs $m \leq n$; an onto function needs $m \geq n$; a bijection needs $m = n$. The number of functions $A \rightarrow B$ is n^m .

8.3 Inverse function

DEFINITION — Inverse function

If $f : A \rightarrow B$ is a **bijection**, its *inverse* $f^{-1} : B \rightarrow A$ is defined by $f^{-1}(b) = a \iff f(a) = b$. Only bijections have inverses (one-to-one so each b has *at most* one pre-image; onto so it has *at least* one). Method: write $y = f(x)$, solve for x in terms of y , then $f^{-1}(y) = x$.

EXAMPLE

$f(x) = x + 2$: $y = x + 2 \Rightarrow x = y - 2$, so $f^{-1}(y) = y - 2$; check $f(f^{-1}(y)) = (y - 2) + 2 = y$.

$f(x) = 3x - 1$: $y = 3x - 1 \Rightarrow x = \frac{y+1}{3}$, so $f^{-1}(y) = \frac{y+1}{3}$.

$f(x) = x^2$ on $\mathbb{R}$ has no inverse (not one-to-one). Restricted to $[0, \infty) \rightarrow [0, \infty)$ it is a bijection with inverse $\sqrt{y}$.

8.4 Composition of functions

DEFINITION — Composition

For $g : A \rightarrow B$ and $f : B \rightarrow C$, the *composition* $f \circ g : A \rightarrow C$ is

$$(f \circ g)(x) = f(g(x)) \quad (\text{apply } g \text{ first, then } f).$$

Likewise $(g \circ f)(x) = g(f(x))$. Composition is defined only if the range of the inner function lies in the domain of the outer one. In general $f \circ g \neq g \circ f$ — composition is **not commutative**.

EXAMPLE

$f(x) = 2x + 3$, $g(x) = 3x + 2$:

$$(f \circ g)(x) = f(3x + 2) = 2(3x + 2) + 3 = 6x + 7,$$

$$(g \circ f)(x) = g(2x + 3) = 3(2x + 3) + 2 = 6x + 11.$$

Since $6x + 7 \neq 6x + 11$, $f \circ g \neq g \circ f$.

8.5 Floor and ceiling functions

DEFINITION — Floor and ceiling

$\lfloor x \rfloor$ is the *greatest integer* $\leq x$; $\lceil x \rceil$ is the *least integer* $\geq x$. Every real x can be written $x = n + f$ with $n = \lfloor x \rfloor \in \mathbb{Z}$ and fractional part $0 \leq f < 1$.

EXAMPLE

$\lfloor 3.7 \rfloor = 3$, $\lceil 3.2 \rceil = 4$, $\lfloor -3.7 \rfloor = -4$ (the greatest integer not exceeding -3.7), $\lceil -3.2 \rceil = -3$, $\lfloor 5 \rfloor = \lceil 5 \rceil = 5$.

EXAMPLE — Prove $\lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + \frac{1}{2} \rfloor$

Write $x = n + f$, $n = \lfloor x \rfloor$, $0 \leq f < 1$. Split into two cases.

Case 1: $0 \leq f < \frac{1}{2}$. Then $2x = 2n + 2f$ with $0 \leq 2f < 1$, so $\lfloor 2x \rfloor = 2n$. Also $x + \frac{1}{2} = n + (f + \frac{1}{2})$ with $f + \frac{1}{2} < 1$, so $\lfloor x + \frac{1}{2} \rfloor = n$. RHS = $n + n = 2n$ ✓.

Case 2: $\frac{1}{2} \leq f < 1$. Then $1 \leq 2f < 2$, so $\lfloor 2x \rfloor = 2n + 1$. Also $1 \leq f + \frac{1}{2} < \frac{3}{2}$, so $\lfloor x + \frac{1}{2} \rfloor = n + 1$. RHS = $n + (n + 1) = 2n + 1$ ✓.

Both cases agree, so the identity holds for all real x . ■

8.6 Is a rule a function?†

A rule $f : A \rightarrow B$ is a function only if **every** element of A gets **exactly one** value, and that value lies in B . Three things can go wrong:

Problem	Example	Why
undefined at some a	$f(x) = 1/x$ from $\mathbb{R}$ to $\mathbb{R}$	no value at $x = 0$
two values at some a	$f(x) = \pm\sqrt{x^2 + 1}$ from $\mathbb{R}$ to $\mathbb{R}$	each x gets two values
value outside the codomain	$f(n) = n/2$ from $\mathbb{Z}$ to $\mathbb{Z}$; $f(x) = \sqrt{x}$ from $\mathbb{R}$ to $\mathbb{R}$	$f(1) = \frac{1}{2} \notin \mathbb{Z}$; $\sqrt{-1} \notin \mathbb{R}$

By contrast $f(n) = \sqrt{n^2 + 1}$ from $\mathbb{Z}$ to $\mathbb{R}$ is a function.

8.7 Image of a set and inverse image†

DEFINITION — Image and inverse image

Let $f : A \rightarrow B$. For $S \subseteq A$ the *image* of S is $f(S) = \{f(s) \mid s \in S\} \subseteq B$. For $T \subseteq B$ the *inverse image* of T is $f^{-1}(T) = \{a \in A \mid f(a) \in T\} \subseteq A$: all the pre-images of all elements of T .

EXAMPLE

$f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$. Then $f(\{-2, -1, 0, 1\}) = \{4, 1, 0, 1\} = \{0, 1, 4\}$; $f^{-1}(\{4\}) = \{-2, 2\}$; $f^{-1}(\{-1\}) = \emptyset$; $f^{-1}(\{x \mid 0 < x < 1\}) = (-1, 0) \cup (0, 1)$.

WATCH OUT

$f^{-1}(T)$ makes sense for *every* function, invertible or not; it is a set. The inverse *function* $f^{-1}(y)$ exists only for bijections. The notation is the same, the meaning is not.

8.8 Old functions, new functions†

DEFINITION — Identity, sum, product, monotone, characteristic function

- The *identity* on A is $\iota_A(x) = x$; it is a bijection.
- For $f, g : A \rightarrow \mathbb{R}$: $(f + g)(x) = f(x) + g(x)$ and $(fg)(x) = f(x)g(x)$.
- $f : \mathbb{R} \rightarrow \mathbb{R}$ is *strictly increasing* if $x < y \Rightarrow f(x) < f(y)$, and *strictly decreasing* if $x < y \Rightarrow f(x) > f(y)$.
- For $S \subseteq U$ the *characteristic function* $f_S : U \rightarrow \{0, 1\}$ has $f_S(x) = 1$ if $x \in S$ and 0 if $x \notin S$.

EXAMPLE

A strictly increasing (or decreasing) function is one-to-one: if $a \neq b$, say $a < b$, then $f(a) < f(b)$ (or $>$), so $f(a) \neq f(b)$. Hence $f(x) = 2x + 1$ and $f(x) = -x^3$ are one-to-one. With $f(x) = x^2$ and $g(x) = x + 1$: $(f + g)(x) = x^2 + x + 1$ and $(fg)(x) = x^3 + x^2$. For $U = \{1, \dots, 5\}$ and $S = \{2, 5\}$ the values $f_S(1), \dots, f_S(5)$ are 0, 1, 0, 0, 1: the bit string 01001 of Chapter 5.

8.9 Composition, inverses and one-to-one/onto[†]

REMEMBER — facts to memorise

Let $g : A \rightarrow B$ and $f : B \rightarrow C$.

- If f and g are both one-to-one, then $f \circ g$ is one-to-one. If both are onto, then $f \circ g$ is onto. Hence a composition of bijections is a bijection.
- If $f \circ g$ is one-to-one then g is one-to-one; if $f \circ g$ is onto then f is onto.
- For bijections, $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$ (undo the last step first, like taking off shoes before socks), and $f \circ f^{-1} = \iota$, $f^{-1} \circ f = \iota$.

EXAMPLE

$g(x) = 2x$ and $f(x) = x + 1$. Then $(f \circ g)(x) = 2x + 1$, whose inverse is $y \mapsto \frac{y-1}{2}$. And $g^{-1}(y) = \frac{y}{2}$, $f^{-1}(y) = y - 1$, so $(g^{-1} \circ f^{-1})(y) = g^{-1}(y - 1) = \frac{y-1}{2}$, the same function. The other order, $f^{-1} \circ g^{-1}$, gives $\frac{y}{2} - 1 \neq \frac{y-1}{2}$.

8.10 Floor and ceiling: more properties[†]

REMEMBER — rules for $\lfloor \cdot \rfloor$ and $\lceil \cdot \rceil$

For real x and integer n :

$$x - 1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x + 1, \quad \lfloor x + n \rfloor = \lfloor x \rfloor + n, \quad \lceil x + n \rceil = \lceil x \rceil + n,$$

$$\lceil x \rceil = \lfloor x \rfloor \text{ if } x \text{ is an integer, else } \lceil x \rceil = \lfloor x \rfloor + 1; \quad \lceil -x \rceil = -\lfloor x \rfloor, \quad \lfloor -x \rfloor = -\lceil x \rceil.$$

EXAMPLE — the ceiling counts “how many blocks”

Data of n bits fill $\lceil n/8 \rceil$ bytes: $n = 4$ needs 1 byte, $n = 8$ needs 1, $n = 9$ needs 2. The number of integers m with $a \leq m \leq b$ is $\lfloor b \rfloor - \lceil a \rceil + 1$: for $a = 1.5, b = 4.2$ it is $4 - 2 + 1 = 3$ ($m = 2, 3, 4$).

8.11 Partial functions[†]

DEFINITION — Partial function

A *partial function* $f : A \rightarrow B$ assigns to each element of its *domain of definition* $D \subseteq A$ exactly one element of B , and is undefined outside D . If $D = A$ it is a *total* function (an ordinary function).

EXAMPLE

$f : \mathbb{Z} \rightarrow \mathbb{R}, f(n) = 1/n$ is a partial function: domain $\mathbb{Z}$, codomain $\mathbb{R}$, domain of definition $\mathbb{Z} - \{0\}$, undefined at 0. It becomes a total function on $\mathbb{Z} - \{0\}$.

8.12 Solved previous-year questions

PYQ • Mid-Sem Oct 2023 Q2(a) • 2 marks

What do you mean by composition of functions? Let $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = 4x + 3$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ defined as $g(x) = x/3$. Find $f \circ g(x)$.

Solution. Composition. If $g : A \rightarrow B$ and $f : B \rightarrow C$, the composite $f \circ g : A \rightarrow C$ is the function $(f \circ g)(x) = f(g(x))$: first apply g , then apply f to the result.

Computation.

$$(f \circ g)(x) = f(g(x)) = f\left(\frac{x}{3}\right) = 4 \cdot \frac{x}{3} + 3 = \frac{4x}{3} + 3 = \frac{4x + 9}{3}.$$

(For contrast, $(g \circ f)(x) = \frac{4x+3}{3} = \frac{4x}{3} + 1 \neq (f \circ g)(x)$.)

PYQ • Mid-Sem Oct 2023 Q2(b) • 2 marks

Give an example of a function from $\mathbb{N}$ to $\mathbb{N}$ that is onto but not one-to-one.

Solution. Take $\mathbb{N} = \{1, 2, 3, \dots\}$ and $f(n) = \lceil n/2 \rceil$, so $f(1) = 1$, $f(2) = 1$, $f(3) = 2$, $f(4) = 2$, $f(5) = 3, \dots$

Not one-to-one: $f(1) = f(2) = 1$ although $1 \neq 2$.

Onto: for any $m \in \mathbb{N}$, $f(2m) = \lceil m \rceil = m$, so every m has a pre-image.

(If $\mathbb{N}$ includes 0, use $f(n) = \lfloor n/2 \rfloor$: $f(0) = f(1) = 0$, and $f(2m) = m$. Another example: $f(1) = 1$ and $f(n) = n - 1$ for $n \geq 2$.)

PYQ • End-Sem Dec 2024 Q1(b) • 5 marks

Let $f(x) = 3x + 2$ and $g(x) = x^2 - 4$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$.

Solution.

$$(f \circ g)(x) = f(g(x)) = f(x^2 - 4) = 3(x^2 - 4) + 2 = 3x^2 - 12 + 2 = \boxed{3x^2 - 10},$$

$$(g \circ f)(x) = g(f(x)) = g(3x + 2) = (3x + 2)^2 - 4 = 9x^2 + 12x + 4 - 4 = \boxed{9x^2 + 12x}.$$

They are different functions (e.g. at $x = 1$: $f \circ g$ gives -7 but $g \circ f$ gives 21), which shows that composition is not commutative.

PYQ • End-Sem Dec 2024 Q4 • 10 marks

Let A and B be finite sets with $|A| = m$ and $|B| = n$. Derive a formula for the number of onto functions from A to B . Use it to find the number of onto functions from $A = \{1, 2, 3, 4\}$ to $B = \{A, B, C\}$.

Solution. Derivation (inclusion–exclusion). Let $B = \{b_1, \dots, b_n\}$. The total number of functions $A \rightarrow B$ is n^m (each of the m elements independently picks one of n images).

A function is onto exactly when *no* element of B is missed. For each i let S_i be the set of functions that miss b_i (never map to b_i). Then the number of onto functions is $n^m - |S_1 \cup \dots \cup S_n|$.

- A function that misses a specified set of k elements of B maps A into the remaining $n - k$

elements, so there are $(n - k)^m$ of them: $|S_{i_1} \cap \cdots \cap S_{i_k}| = (n - k)^m$.

- There are $\binom{n}{k}$ ways to choose which k elements are missed.

By the inclusion–exclusion principle,

$$|S_1 \cup \cdots \cup S_n| = \binom{n}{1}(n - 1)^m - \binom{n}{2}(n - 2)^m + \binom{n}{3}(n - 3)^m - \cdots$$

Therefore

$$\#_{\text{onto}}(m, n) = \sum_{k=0}^n (-1)^k \binom{n}{k} (n - k)^m$$

which expands to $n^m - \binom{n}{1}(n - 1)^m + \binom{n}{2}(n - 2)^m - \cdots + (-1)^{n-1} \binom{n}{n-1} 1^m$. (This is 0 if $m < n$, and equals $n!$ when $m = n$.)

Application: $m = 4, n = 3$.

$$3^4 - \binom{3}{1}2^4 + \binom{3}{2}1^4 - \binom{3}{3}0^4 = 81 - 3(16) + 3(1) - 0 = 81 - 48 + 3 = \boxed{36}.$$

Check by direct counting. An onto function from a 4-element set to a 3-element set must send exactly two elements to one image and one element each to the other two. Choose which element of B receives two elements: 3 ways; choose the two elements of A mapped to it: $\binom{4}{2} = 6$ ways; send the remaining two elements of A to the remaining two elements of B one-to-one: $2! = 2$ ways. Total $3 \cdot 6 \cdot 2 = 36$ ✓.

8.13 Practice questions from Rosen

Chosen from Rosen §2.3 in the style of the PYQs above (composition, one-to-one and onto, examples and counterexamples). Near-repeats, graph-drawing and calculator-style exercises are left out.

PRACTICE • Rosen §2.3 Ex. 1–3 • is it a function?

- (a) Why is f not a function from $\mathbb{R}$ to $\mathbb{R}$ if (i) $f(x) = 1/x$, (ii) $f(x) = \sqrt{x}$, (iii) $f(x) = \pm\sqrt{x^2 + 1}$?
 (b) Is f a function from $\mathbb{Z}$ to $\mathbb{R}$ if (i) $f(n) = \pm n$, (ii) $f(n) = \sqrt{n^2 + 1}$, (iii) $f(n) = 1/(n^2 - 4)$?
 (c) Is f a function from the set of bit strings to the integers if (i) $f(S)$ is the position of a 0 bit in S , (ii) $f(S)$ is the number of 1 bits in S , (iii) $f(S)$ is the smallest i such that the i th bit of S is 1, and $f(S) = 0$ for the empty string?

Solution. A function needs **exactly one** value in the codomain for **every** element of the domain.

(a) (i) No value at $x = 0$. (ii) No real value for negative x . (iii) Two values for every x (both signs).

(b) (i) No: $n = 1$ would get both 1 and -1 . (ii) Yes: $n^2 + 1 > 0$, so there is exactly one non-negative real square root. (iii) No: undefined at $n = \pm 2$.

(c) (i) No: the string 010 has two 0 bits (two possible values) and 111 has none (no value). (ii) Yes. (iii) No: the non-empty string 000 has no 1 bit, so nothing is assigned to it; the rule only covers the empty string as a special case.

PRACTICE • Rosen §2.3 Ex. 12–14 • one-to-one and onto, with counterexamples

- (a) Which of these functions from $\mathbb{Z}$ to $\mathbb{Z}$ are one-to-one? Which are onto? (i) $f(n) = n - 1$ (ii) $f(n) = n^2 + 1$ (iii) $f(n) = n^3$ (iv) $f(n) = \lceil n/2 \rceil$.

(b) Which of these functions $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ are onto? (i) $2m - n$ (ii) $m^2 - n^2$ (iii) $m + n + 1$ (iv) $|m| - |n|$ (v) $m^2 - 4$.

Solution. (a)

	$f(n)$	One-to-one?	Onto?	Reason
(i)	$n - 1$	T	T	$n - 1 = m - 1 \Rightarrow n = m; y = f(y + 1)$
(ii)	$n^2 + 1$	F	F	$f(1) = f(-1) = 2$; values are ≥ 1 , so 0 has no pre-image
(iii)	n^3	T	F	strictly increasing; 2 is not a cube
(iv)	$\lceil n/2 \rceil$	F	T	$f(1) = f(2) = 1; f(2m) = m$ for every m

(b) (i) Onto: given k , take $(m, n) = (0, -k)$, then $2 \cdot 0 - (-k) = k$. **(ii) Not onto:** $m^2 - n^2 = (m - n)(m + n)$, and $m - n, m + n$ have the same parity, so the product is odd or a multiple of 4; hence 2 (or any number $\equiv 2 \pmod{4}$) is never a value. **(iii) Onto:** take $(m, n) = (k - 1, 0)$. **(iv) Onto:** for $k \geq 0$ take $(k, 0)$; for $k < 0$ take $(0, -k)$, giving $0 - |-k| = k$. **(v) Not onto:** $m^2 - 4 \geq -4$ for every m , so -5 is never a value.

PRACTICE • Rosen §2.3 Ex. 20 • example functions $\mathbb{N} \rightarrow \mathbb{N}$

Let $\mathbb{N} = \{1, 2, 3, \dots\}$. Give an example of a function from $\mathbb{N}$ to $\mathbb{N}$ that is (a) one-to-one but not onto, (b) onto but not one-to-one, (c) both one-to-one and onto but not the identity function, (d) neither.

Solution. (a) $f(n) = n + 1$. One-to-one ($n + 1 = m + 1 \Rightarrow n = m$), but 1 has no pre-image.

(b) $f(n) = \lceil n/2 \rceil$ (PYQ M23-2b): $f(1) = f(2) = 1$, and $f(2m) = m$.

(c) Swap neighbours: $f(n) = n + 1$ if n is odd, $f(n) = n - 1$ if n is even, i.e. $1 \leftrightarrow 2, 3 \leftrightarrow 4, \dots$. Then $f(f(n)) = n$, so f is a bijection (it is its own inverse), and $f(1) = 2 \neq 1$.

(d) $f(n) = 1$ for all n : not one-to-one ($f(1) = f(2)$) and not onto (2 has no pre-image).

PRACTICE • Rosen §2.3 Ex. 22, 23, 39 • bijections $\mathbb{R} \rightarrow \mathbb{R}$ and inverses

Determine whether each is a bijection from $\mathbb{R}$ to $\mathbb{R}$, and find the inverse of each bijection. (a) $f(x) = -3x + 4$ (b) $f(x) = -3x^2 + 7$ (c) $f(x) = \frac{x+1}{x+2}$ (d) $f(x) = x^5 + 1$ (e) $f(x) = \frac{x^2+1}{x^2+2}$. (f) Show that $f(x) = ax + b$ with $a \neq 0$ is invertible and find f^{-1} .

Solution. (a) Bijection. One-to-one: $-3a + 4 = -3b + 4 \Rightarrow a = b$. Onto: $y = -3x + 4$ gives $x = \frac{4-y}{3} \in \mathbb{R}$. Inverse $f^{-1}(y) = \frac{4-y}{3}$.

(b) No. Not one-to-one: $f(1) = f(-1) = 4$ (and the range is $(-\infty, 7]$, so it is not onto).

(c) No, it is not even a function from $\mathbb{R}$ to $\mathbb{R}$: it is undefined at $x = -2$. (On $\mathbb{R} - \{-2\}$ it is one-to-one, but 1 is never a value, so it is not onto $\mathbb{R}$ either.)

(d) Bijection. x^5 is strictly increasing, so one-to-one; given y , $x = \sqrt[5]{y-1}$ (real fifth roots exist for negative numbers too). Inverse $f^{-1}(y) = \sqrt[5]{y-1}$.

(e) No. $f(1) = f(-1) = \frac{2}{3}$, so not one-to-one; also $f(x) = 1 - \frac{1}{x^2+2} \in [\frac{1}{2}, 1)$, so it is not onto.

(f) One-to-one: $ax_1 + b = ax_2 + b \Rightarrow x_1 = x_2$ since $a \neq 0$. Onto: $y = ax + b$ is solved by $x = \frac{y-b}{a}$.

So f is a bijection and $f^{-1}(y) = \frac{y-b}{a}$.

PRACTICE • Rosen §2.3 Ex. 33–35 • composition and one-to-one, onto

Let $g : A \rightarrow B$ and $f : B \rightarrow C$. (a) Show that if f and g are one-to-one, then $f \circ g$ is one-to-one. (b) Show that if f and g are onto, then $f \circ g$ is onto. (c) If f and $f \circ g$ are one-to-one, does it follow that g is one-to-one? (d) If f and $f \circ g$ are onto, does it follow that g is onto?

Solution. (a) Suppose $(f \circ g)(a_1) = (f \circ g)(a_2)$, i.e. $f(g(a_1)) = f(g(a_2))$. Since f is one-to-one, $g(a_1) = g(a_2)$; since g is one-to-one, $a_1 = a_2$. ■

(b) Let $c \in C$. As f is onto there is $b \in B$ with $f(b) = c$; as g is onto there is $a \in A$ with $g(a) = b$. Then $(f \circ g)(a) = f(b) = c$. ■

(c) Yes. If $g(a_1) = g(a_2)$ then $f(g(a_1)) = f(g(a_2))$, i.e. $(f \circ g)(a_1) = (f \circ g)(a_2)$, and since $f \circ g$ is one-to-one, $a_1 = a_2$. (In fact the hypothesis on f is not needed: if $f \circ g$ is one-to-one then g is.)

(d) No. Counterexample: $A = \{1\}$, $B = \{1, 2\}$, $C = \{1\}$, with $g(1) = 1$, $f(1) = f(2) = 1$. Then f is onto C and $f \circ g : A \rightarrow C$ is onto, but g is not onto B (2 has no pre-image). What is true: if $f \circ g$ is onto then f is onto.

PRACTICE • Rosen §2.3 Ex. 38 and 70 • when do $f \circ g$ and $g \circ f$ agree; inverse of a composition

(a) Let $f(x) = ax + b$ and $g(x) = cx + d$. Find necessary and sufficient conditions on a, b, c, d so that $f \circ g = g \circ f$. (b) Show that if $f : Y \rightarrow Z$ and $g : X \rightarrow Y$ are invertible, then $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$.

Solution. (a) $(f \circ g)(x) = a(cx + d) + b = acx + ad + b$ and $(g \circ f)(x) = c(ax + b) + d = acx + bc + d$. The x -terms already agree, so the functions are equal iff $ad + b = bc + d$, i.e.

$$d(a - 1) = b(c - 1).$$

Check with the chapter example $f = 2x + 3$, $g = 3x + 2$: $d(a - 1) = 2$, $b(c - 1) = 6$, unequal, so they do not commute. Examples that do: $f = 2x$, $g = 3x$ ($b = d = 0$), or $f = x + 1$, $g = x + 5$ ($a = c = 1$).

(b) Compute the composition in both orders:

$$(f \circ g) \circ (g^{-1} \circ f^{-1}) = f \circ (g \circ g^{-1}) \circ f^{-1} = f \circ \iota_Y \circ f^{-1} = f \circ f^{-1} = \iota_Z,$$

$$(g^{-1} \circ f^{-1}) \circ (f \circ g) = g^{-1} \circ (f^{-1} \circ f) \circ g = g^{-1} \circ g = \iota_X.$$

So $g^{-1} \circ f^{-1}$ undoes $f \circ g$; it is the inverse. ■

PRACTICE • Rosen §2.3 Ex. 40 and 41 • image of a union and intersection

Let $f : A \rightarrow B$ and $S, T \subseteq A$. (a) Show $f(S \cup T) = f(S) \cup f(T)$. (b) Show $f(S \cap T) \subseteq f(S) \cap f(T)$. (c) Give an example where the inclusion in (b) is proper. (d) Show that equality holds in (b) if f is one-to-one.

Solution. (a) $y \in f(S \cup T)$ iff $y = f(x)$ for some $x \in S \cup T$, iff $y = f(x)$ for some $x \in S$ or for some $x \in T$, iff $y \in f(S) \cup f(T)$. ■

(b) Let $y \in f(S \cap T)$, so $y = f(x)$ with $x \in S$ and $x \in T$. Then $y \in f(S)$ and $y \in f(T)$. ■

(c) $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$, $S = \{-1\}$, $T = \{1\}$. Then $S \cap T = \emptyset$, so $f(S \cap T) = \emptyset$, but

$$f(S) \cap f(T) = \{1\} \cap \{1\} = \{1\}.$$

(d) Let $y \in f(S) \cap f(T)$: $y = f(s) = f(t)$ with $s \in S, t \in T$. Since f is one-to-one, $s = t$, so this element lies in $S \cap T$ and $y \in f(S \cap T)$. ■

PRACTICE • Rosen §2.3 Ex. 42–45 • inverse images

(a) For $f(x) = x^2$ on $\mathbb{R}$ find $f^{-1}(\{1\})$, $f^{-1}(\{x \mid 0 < x < 1\})$ and $f^{-1}(\{x \mid x > 4\})$. (b) For $g(x) = \lfloor x \rfloor$ find $g^{-1}(\{0\})$, $g^{-1}(\{-1, 0, 1\})$ and $g^{-1}(\{x \mid 0 < x < 1\})$. (c) For $f : A \rightarrow B$ and $S, T \subseteq B$ show that $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$ and $f^{-1}(\overline{S}) = \overline{f^{-1}(S)}$.

Solution. Recall $f^{-1}(T) = \{a \mid f(a) \in T\}$.

(a) $x^2 = 1$: $\{-1, 1\}$. $0 < x^2 < 1$: $-1 < x < 1, x \neq 0$, i.e. $(-1, 0) \cup (0, 1)$. $x^2 > 4$: $|x| > 2$, i.e. $(-\infty, -2) \cup (2, \infty)$.

(b) $\lfloor x \rfloor = 0$ iff $0 \leq x < 1$: $[0, 1)$. $\lfloor x \rfloor \in \{-1, 0, 1\}$ iff $-1 \leq x < 2$: $[-1, 2)$. $\lfloor x \rfloor$ is always an integer and no integer lies strictly between 0 and 1, so the inverse image of $\{x \mid 0 < x < 1\}$ is $\emptyset$.

(c) $a \in f^{-1}(S \cup T)$ iff $f(a) \in S \cup T$ iff $f(a) \in S$ or $f(a) \in T$ iff $a \in f^{-1}(S) \cup f^{-1}(T)$. Also $a \in f^{-1}(\overline{S})$ iff $f(a) \notin S$ iff $a \notin f^{-1}(S)$. ■

(Unlike images, inverse images respect unions, intersections and complements exactly. Compare ros:2.3-40.)

PRACTICE • Rosen §2.3 Ex. 54 and 73 • prove or disprove: floor and ceiling

Prove or disprove for all real x, y : (a) $\lceil -x \rceil = -\lfloor x \rfloor$; (b) $\lfloor 2x \rfloor = 2\lfloor x \rfloor$; (c) $\lceil x \rceil + \lceil y \rceil - \lceil x + y \rceil$ is 0 or 1; (d) $\lceil xy \rceil = \lceil x \rceil \lceil y \rceil$.

Solution. (a) **True.** Let $n = \lfloor x \rfloor$, so $n \leq x < n + 1$. Multiplying by -1 : $-n - 1 < -x \leq -n$. The least integer that is $\geq -x$ is $-n$, so $\lceil -x \rceil = -n = -\lfloor x \rfloor$.

(b) **False.** $x = \frac{1}{2}$: $\lfloor 2x \rfloor = \lfloor 1 \rfloor = 1$ but $2\lfloor \frac{1}{2} \rfloor = 0$. (The correct statement is $\lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + \frac{1}{2} \rfloor$, proved earlier in this chapter.)

(c) **True.** Write $x = \lceil x \rceil - a$ and $y = \lceil y \rceil - b$ with $0 \leq a, b < 1$ (because $\lceil x \rceil - 1 < x \leq \lceil x \rceil$). Put $N = \lceil x \rceil + \lceil y \rceil$, an integer; then $x + y = N - (a + b)$ with $0 \leq a + b < 2$.

• If $0 \leq a + b < 1$: $N - 1 < x + y \leq N$, so $\lceil x + y \rceil = N$ and the expression is 0.

• If $1 \leq a + b < 2$: $N - 2 < x + y \leq N - 1$, so $\lceil x + y \rceil = N - 1$ and the expression is 1.

(d) **False.** $x = y = \frac{3}{2}$: $\lceil xy \rceil = \lceil \frac{9}{4} \rceil = 3$ but $\lceil x \rceil \lceil y \rceil = 2 \cdot 2 = 4$.

PRACTICE • Rosen §2.3 Ex. 56–59, 61 • counting with floor and ceiling

(a) For real $a < b$, use floor and/or ceiling to express the number of integers n with (i) $a \leq n \leq b$, (ii) $a < n < b$. (b) How many bytes are needed to encode n bits when $n = 4, 10, 500, 3000, 7, 17, 1001, 28800$? (c) Data go out in blocks of 1500 bytes. How many blocks are needed for 150 kilobytes, 384 kilobytes, 1.544 megabytes and 45.3 megabytes (1 kilobyte = 1000 bytes, 1 megabyte = 1 000 000 bytes)?

Solution. (a) (i) The integers are $\lceil a \rceil, \lceil a \rceil + 1, \dots, \lfloor b \rfloor$, so there are $\lfloor b \rfloor - \lceil a \rceil + 1$. (ii) The integers are $\lfloor a \rfloor + 1, \dots, \lceil b \rceil - 1$, so there are $\lceil b \rceil - \lfloor a \rfloor - 1$. Check with $a = 1.5, b = 4.2$: (i) $4 - 2 + 1 = 3$ (2, 3, 4); (ii) $5 - 1 - 1 = 3$. With $a = 1.2, b = 1.8$ both give 0.

(b) A byte holds 8 bits, so n bits need $\lceil n/8 \rceil$ bytes (a partly filled byte still counts): $n = 4, 10, 500, 3000, 7, 17, 1001, 28800$ give 1, 2, 63, 375, 1, 3, 126, 3600.

(c) Blocks = $\lceil \text{bytes}/1500 \rceil$: $150\,000/1500 = 100$ gives 100; $384\,000/1500 = 256$ gives 256;

$1\,544\,000/1500 \approx 1029.3$ gives 1030; $45\,300\,000/1500 = 30\,200$ gives 30 200.

PRACTICE • Rosen §2.3 Ex. 71 • characteristic functions

Let f_S be the characteristic function of $S \subseteq U$. Show that for all $x \in U$: (a) $f_{A \cap B}(x) = f_A(x)f_B(x)$; (b) $f_{A \cup B}(x) = f_A(x) + f_B(x) - f_A(x)f_B(x)$; (c) $f_{\overline{A}}(x) = 1 - f_A(x)$; (d) $f_{A \oplus B}(x) = f_A(x) + f_B(x) - 2f_A(x)f_B(x)$.

Solution. (c) If $x \in A$ then $f_A(x) = 1$ and $x \notin \overline{A}$, so $f_{\overline{A}}(x) = 0 = 1 - 1$; if $x \notin A$ then $f_A(x) = 0$ and $f_{\overline{A}}(x) = 1 = 1 - 0$.

(a), (b), (d) $f_A(x)$ and $f_B(x)$ can be only 0 or 1, so check all four combinations:

f_A	f_B	$f_{A \cap B}$	$f_A f_B$	$f_{A \cup B}$	$f_A + f_B - f_A f_B$	$f_{A \oplus B}$	$f_A + f_B - 2f_A f_B$
0	0	0	0	0	0	0	0
0	1	0	0	1	1	1	1
1	0	0	0	1	1	1	1
1	1	1	1	1	1	0	0

In every row the columns match, which proves (a), (b) and (d). (The formulas are the arithmetic versions of AND, OR and XOR from Chapter 1, and the string of values $f_S(1), \dots, f_S(n)$ is the bit string of Chapter 5.) ■

PRACTICE • Rosen §2.3 Ex. 72 • finite sets of equal size

Let $f : A \rightarrow B$ where A and B are finite and $|A| = |B|$. Show that f is one-to-one if and only if it is onto. Does this hold for infinite sets?

Solution. ($\Rightarrow$) If f is one-to-one, distinct elements have distinct images, so $|f(A)| = |A| = |B|$. Since $f(A) \subseteq B$ and both are finite of the same size, $f(A) = B$: f is onto.

($\Leftarrow$) Let f be onto and suppose it is not one-to-one. Two different elements of A share an image, so $|f(A)| \leq |A| - 1 < |B|$; but f onto means $f(A) = B$, a contradiction. So f is one-to-one. ■

Infinite sets: no. On $\mathbb{N}$, $f(n) = n + 1$ is one-to-one but not onto, and $f(n) = \lceil n/2 \rceil$ is onto but not one-to-one.

PRACTICE • Rosen §2.3 Ex. 77 • partial functions

For each partial function give its domain, codomain, domain of definition and the set where it is undefined, and say whether it is total. (a) $f : \mathbb{Z} \rightarrow \mathbb{R}$, $f(n) = 1/n$. (b) $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Q}$, $f(m, n) = m/n$. (c) $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$, $f(m, n) = mn$. (d) $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$, $f(m, n) = m - n$ if $m > n$.

Solution.

	Domain	Codomain	Domain of definition	Undefined for	Total?
(a)	$\mathbb{Z}$	$\mathbb{R}$	$\mathbb{Z} - \{0\}$	$n = 0$	no
(b)	$\mathbb{Z} \times \mathbb{Z}$	$\mathbb{Q}$	$\{(m, n) \mid n \neq 0\}$	$n = 0$	no
(c)	$\mathbb{Z} \times \mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z} \times \mathbb{Z}$	nowhere	yes
(d)	$\mathbb{Z} \times \mathbb{Z}$	$\mathbb{Z}$	$\{(m, n) \mid m > n\}$	$m \leq n$	no

Corrections to the class notes

While building this guide I compared your handwritten pages with the standard definitions. Most of your notes are right. The items below are the places where the notes are incomplete, or where a fix will save you marks in an exam. Page numbers refer to the pages of your scanned PDF (the dated pages, 18 Aug – 15 Sep 2026).

Page	In your notes	Correct version
5	$\exists$: “true for some but not true for all”	$\exists x P(x)$ means <i>at least one</i> element satisfies P . It is still true when <i>all</i> elements satisfy P ; it is false only when <i>none</i> does. (Your own second note, “some / at least 1”, is the right one.) See Ch. 2.
8	Q2 English reading: “a student who is friends with everyone but his friends are not friends with each other”	The formula $\exists x \forall y \forall z ((F(x, y) \wedge F(x, z) \wedge y \neq z) \rightarrow \neg F(y, z))$ says only that <i>any two distinct friends of x are not friends</i> . “Friends with everyone” is not in the formula. Read it as: “there is a student none of whose friends are friends with each other.” See Ch. 2, nested quantifiers.
9	Modus tollens: premises $\neg q$ and $p \rightarrow q$ written, no conclusion	Conclusion is $\neg p$: $p \rightarrow q, \neg q \therefore \neg p$. See Ch. 3.
12	Book-and-exam proof: $R(c)$ and $\neg Q(c)$ combined using “Addition”; then $R(c) \wedge \neg Q(x)$	Joining two known statements is Conjunction , not Addition (Addition gives $p \vee q$ from p). Also write $\neg Q(c)$, not $\neg Q(x)$, before applying existential generalisation to reach $\exists x (R(x) \wedge \neg Q(x))$. See Ch. 3.
16	Exhaustive proof, $n = 4$: line reads like $3^3 \geq 3^4$	The left side is $(n + 1)^3 = 5^3 = 125$, so the line is $5^3 \geq 3^4$, i.e. $125 \geq 81$. The numbers you wrote (125, 81) are right; only the base “5” looks like a 3. See Ch. 4.
18	Odd-number sum first written $1 + 3 + 5 + \dots + (2k + 1) = n^2$	The last term of the sum of the first n odd numbers is $2n - 1$: $1 + 3 + \dots + (2n - 1) = n^2$. The later lines of your page already use this form. See Ch. 4.
20	“ $a \mid b$ ” is antisymmetric (no domain stated)	True on the <i>positive</i> integers. On all of $\mathbb{Z}$ it fails: $2 \mid -2$ and $-2 \mid 2$ but $2 \neq -2$. Reflexivity on $\mathbb{Z}$ fails only because the usual definition of $a \mid b$ requires $a \neq 0$ (so $0 \nmid 0$), as your notes say. See Ch. 6.
3	Laws table: the absorption-law formulas are missing (the name is written without them)	$p \vee (p \wedge q) \equiv p$ and $p \wedge (p \vee q) \equiv p$. Added in Ch. 1.

REMEMBER

Everything else on those pages, including all truth tables, the translations on pages 2 and 8, the rules-of-inference examples on pages 9–12, the $1 = 2$ fallacy, the induction proofs, and the relation examples, was checked and is correct.

One-page revision sheet

Logic

Conditional	$p \rightarrow q \equiv \neg q \rightarrow \neg p \equiv \neg p \vee q$; $\neg(p \rightarrow q) \equiv p \wedge \neg q$; converse and inverse are <i>not</i> equivalent to $p \rightarrow q$
Biconditional	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$; true when p, q have the same value
De Morgan	$\neg(p \wedge q) \equiv \neg p \vee \neg q$, $\neg(p \vee q) \equiv \neg p \wedge \neg q$
Types	tautology (always T), contradiction (always F), contingency (depends); satisfiable = some row makes it T
Quantifiers	$\neg \forall x P \equiv \exists x \neg P$, $\neg \exists x P \equiv \forall x \neg P$; use $\forall \rightarrow$ and $\exists \wedge$; $\exists$ = at least one
Inference rules	MP $p \rightarrow q, p \therefore q$; MT $p \rightarrow q, \neg q \therefore \neg p$; HS $p \rightarrow q, q \rightarrow r \therefore p \rightarrow r$; Addition $p \therefore p \vee q$; Simplification $p \wedge q \therefore p$; Conjunction $p, q \therefore p \wedge q$; DS $p \vee q, \neg p \therefore q$; Resolution $p \vee q, \neg p \vee r \therefore q \vee r$
Quantified rules	UI (instantiate $\forall$ to any c); UG; EI (name a <i>new</i> c); EG
Fallacies	affirming the conclusion ($q, p \rightarrow q \therefore p$); denying the hypothesis ($\neg p, p \rightarrow q \therefore \neg q$)

Proofs

Direct ($p \rightarrow q$); contraposition ($\neg q \rightarrow \neg p$); contradiction (assume the negation, reach a false statement); vacuous (hypothesis false); trivial (conclusion true); cases / exhaustive; **induction** = basis step + inductive step ($P(k) \Rightarrow P(k+1)$). Cancelling $(a - b)$ when $a = b$ is dividing by 0.

Sets

Counting	$ \mathcal{P}(A) = 2^{ A }$; $ A \times B = A B $; $ A \cup B \cup C = \sum A - \sum A \cap B + A \cap B \cap C $
Identities	De Morgan, distributive, absorption; $A - B = A \cap \bar{B}$; $A \oplus B = (A - B) \cup (B - A) = (A \cup B) - (A \cap B)$; prove by two inclusions, by set-builder and logic, or by a membership table
Watch out	$\emptyset \neq \{\emptyset\}$; $x \in A$ (element) is different from $x \subseteq A$ (subset); bit strings: $\cup$ = OR, $\cap$ = AND, $\oplus$ = XOR

Relations, posets

Relation types	reflexive $(a, a) \in R$; irreflexive; symmetric $aRb \Rightarrow bRa$; antisymmetric $aRb \wedge bRa \Rightarrow a = b$; asymmetric $aRb \Rightarrow b \not R a$; transitive $aRb \wedge bRc \Rightarrow aRc$; number of relations on A : $2^{ A ^2}$
Operations	R^{-1} swaps pairs; $S \circ R$ = first R , then S ; $M_{S \circ R} = M_R \odot M_S$; transitive iff $R^n \subseteq R$
Closures	reflexive $R \cup \Delta$; symmetric $R \cup R^{-1}$; transitive $R \cup R^2 \cup \dots$; smallest equivalence containing R : reflexive, then symmetric, then transitive <i>last</i>
Equivalence	reflexive + symmetric + transitive; classes are disjoint and partition A ; a block of size k gives k^2 pairs; congruence mod m has m classes
Partial orders	reflexive + antisymmetric + transitive; Hasse diagram: draw upward, omit loops and transitive edges; maximal = nothing above, minimal = nothing below; greatest/least must be comparable to <i>all</i>
Bounds, lattice	upper bound $\geq$ all of S ; LUB = least upper bound; GLB = greatest lower bound; lattice = every pair has a LUB and a GLB (divisibility: lcm and gcd)

Functions

Function types	one-to-one: $f(a) = f(b) \Rightarrow a = b$; onto: every y in the codomain is hit; bijection = both; inverse exists iff bijection
Composition	$(f \circ g)(x) = f(g(x))$: apply g first; in general $f \circ g \neq g \circ f$; $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$; one-to-one and onto both survive composition
Images	$f(S)$ = images of S ; $f^{-1}(T)$ = all pre-images of T (exists for every f); $f(S \cap T) \subseteq f(S) \cap f(T)$, but f^{-1} respects $\cup, \cap$ exactly
Floor, ceiling	$\lfloor x \rfloor \leq x \leq \lceil x \rceil$; $\lceil -x \rceil = -\lfloor x \rfloor$; $\lceil n/8 \rceil$ bytes for n bits
Onto count	onto functions $m \rightarrow n$: $\sum_{k=0}^n (-1)^k \binom{n}{k} (n-k)^m$; $4 \rightarrow 3$ gives $81 - 48 + 3 = 36$